L3ORON Institute

CERTNEXUS®

CERTIFIED ARTIFICIAL INTELLIGENCE PRACTITIONER AIP210

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CertNexus Authorized Instructor

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ABOUT THE TRAINER

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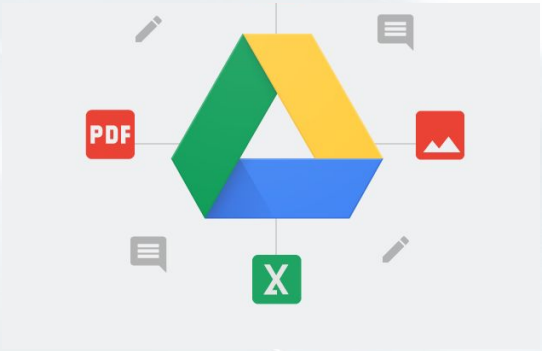
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WHAT'S THE PROBLEM THAT WE ARE FACING



CAREEM APPLICATION



GOOGLE DRIVE APPLICATION



1 NAME AN APPLICATION

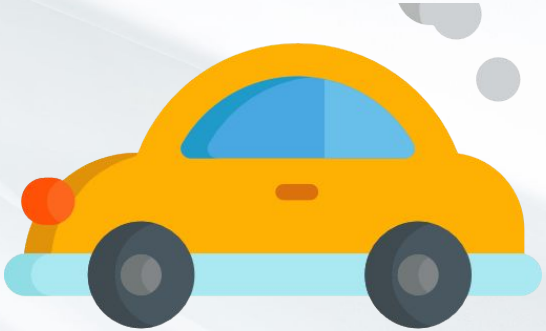
2 MENTION ITS PROBLEM

3 MENTION THE SOLUTION THAT THEY HAVE PROVIDED

HOW WE SHOULD THINK



BICYCLE



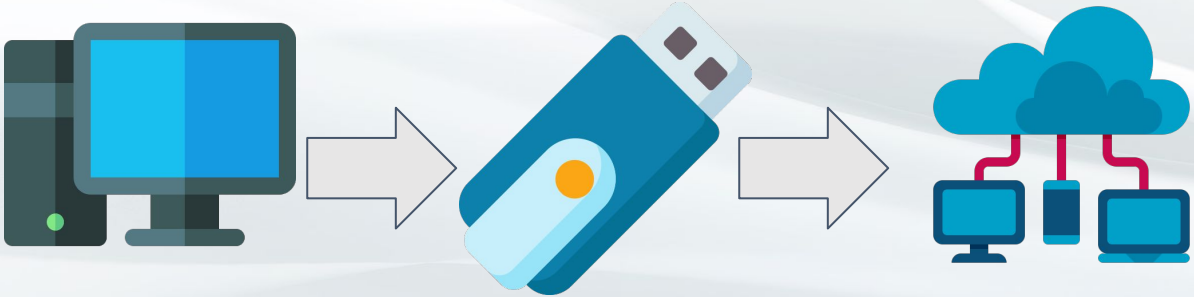
CAR



1- NAME 2 EXAMPLES

2- COMPARE THEM

3- MENTION HOW WE SHOULD THINK



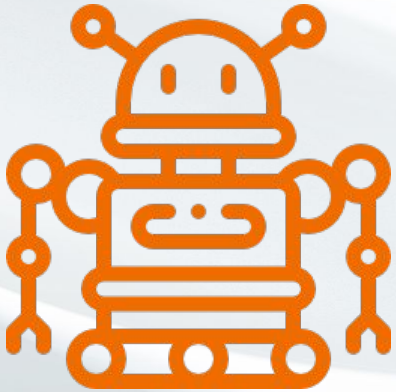
VARIOUS OPTIONS IN THE PROCESS



INTRODUCTION TO ARTIFICIAL INTELLIGENCE

HUMAN INTELLIGENCE	COMPUTER INTELLIGENCE





DEFINITION OF ARTIFICIAL INTELLIGENCE

It is the act of letting the computer

1- **think** as a human being

2- **take** decisions

3- and **predict** things!



DIFFERENCE BETWEEN A NORMAL MACHINE AND ARTIFICIAL INTELLIGENCE



Car



Air Conditioner



Curtains



Phone

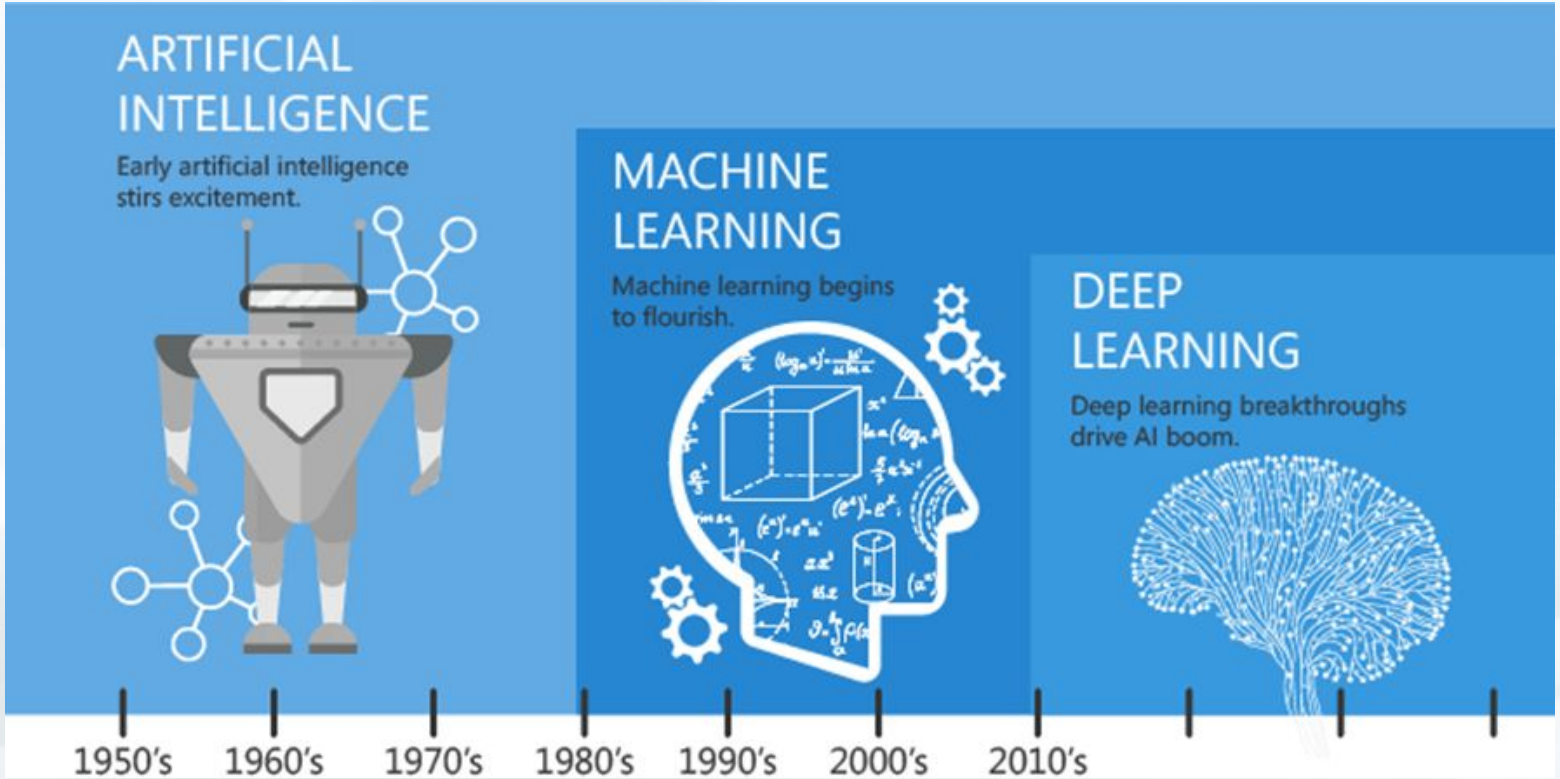


Television

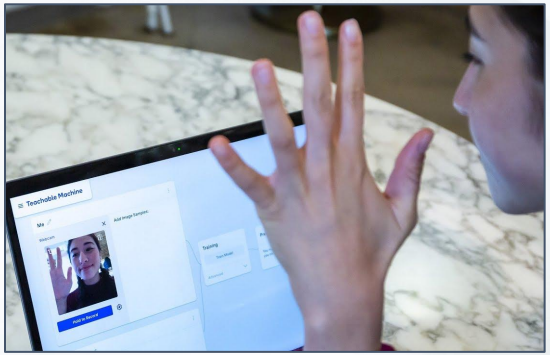


Computer

HAVE YOU EVER HEARD OF THESE TERMINOLOGIES?



QUICK ARTIFICIAL INTELLIGENCE EXERCISES



GAME



GAME



GAME

DATASET

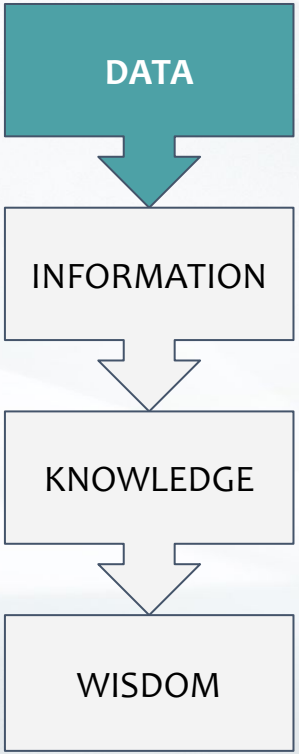
COURSE OUTLINES

LESSON 1	**Solving Business Problems Using AI and ML**
LESSON 2	Preparing Data
LESSON 3	Training, Evaluating, and Tuning a Machine Learning Model
LESSON 4	Building Linear Regression Models
LESSON 5	Building Forecasting Models
LESSON 6	Building Classification Models Using Regression and k-NN
LESSON 7	Building Clustering Models
LESSON 8	Building Decision Trees and Random Forests
LESSON 9	Building Support-Vector Machines
LESSON 10	Building Artificial Neural Networks
LESSON 11	Operationalizing Machine Learning Models
LESSON 12	Maintaining Machine Learning Operations

LESSON 1
Solving Business Problems Using AI and ML

Topic A	**Identify AI and ML Solutions for Business Problems**
Topic B	Formulate a Machine Learning Problem
Topic C	Select Approaches to Machine Learning

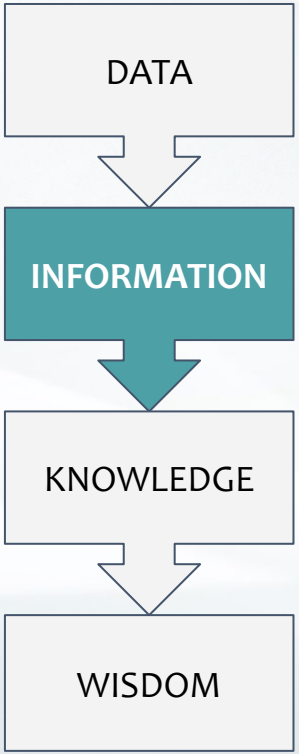
DIKW STRUCTURE: DATA | INFORMATION | KNOWLEDGE | WISDOM



COLUMN A	COLUMN B
50	20
50	30

This is called **RAW DATA**
It is **not really useful**

DIKW STRUCTURE: DATA | INFORMATION | KNOWLEDGE | WISDOM



COLUMN A	COLUMN B
50	20
50	30

DISTANCE	TIME	SPEED
50 KM	20 MIN	150 km/h
50 KM	30 MIN	100 km/h

DIKW STRUCTURE: DATA | INFORMATION | KNOWLEDGE | WISDOM



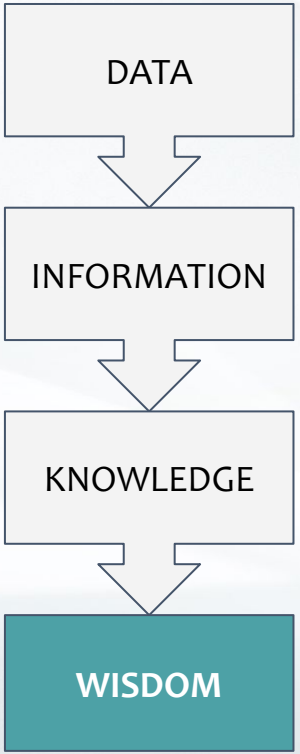
COLUMN A	COLUMN B
50	20
50	30

KNOWLEDGE is
ACTIONABLE INTELLIGENCE
(Information + Experience)

DISTANCE	TIME	SPEED
50 KM	20 MIN	150 km/h
50 KM	30 MIN	100 km/h

THIS DRIVER
WAS SPEEDING

DIKW STRUCTURE: DATA | INFORMATION | KNOWLEDGE | WISDOM



COLUMN A	COLUMN B
50	20
50	30

DISTANCE	TIME	SPEED
50 KM	20 MIN	150 km/h
50 KM	30 MIN	100 km/h

THIS DRIVER **WAS SPEEDING**

The insurance company should charge a **higher subscription fees** for the first driver as they are more likely to **get tickets and be in accidents**

WHAT'S THE DIFFERENCE



UNDERSTANDING BIG DATA

Collections of data that are **so large** and **complex** that they require advanced tools to process and analyze them.

VOLUME	VARIETY	VELOCITY	VERACITY	VALUE	VARIABILITY
Amount of data from different sources	types of data structured semi-structured unstructured	speed at which data processing must occur	accuracy or reliability of data	business value of collected data	how this data can be formatted

HANDLE LARGE DATASETS IN MACHINE LEARNING

PROBLEM

On projects with **massive datasets**, it can be **challenging** to store, move, and process the data. In other words, we don't have **unlimited resources**.

SOLUTION

For this reason, we will see 5 tips to handle large datasets in machine learning

HANDLE LARGE DATASETS IN MACHINE LEARNING

1	**WORK WITH LESS DATA WHEN POSSIBLE**
2	USE A MORE EFFICIENT DATA FORMAT
3	DON'T LOAD ALL DATA AT ONE TIME
4	RECONFIGURE YOUR SYSTEM MEMORY
5	USE A DIFFERENT SYSTEM WITH MORE MEMORY AND PROCESSING POWER

You can work with a **small sample** of the data pulled at random **from the larger set**. This will **speed up** your early work in the project.

As you progress toward a final model, you can **increase** the **size** of the **sample** to refine the solution.

NOTE: Some machine learning algorithms support various measures that enable you to **determine the minimal size of a sample set** that will produce good results.

HANDLE LARGE DATASETS IN MACHINE LEARNING

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3	DON'T LOAD ALL DATA AT ONE TIME
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5	USE A DIFFERENT SYSTEM WITH MORE MEMORY AND PROCESSING POWER

By using more efficient data formats, you may be able to use less storage, memory, and network bandwidth.

EXAMPLE: NetCDF or HDF **are better than** text files or CSV files

HANDLE LARGE DATASETS IN MACHINE LEARNING

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2	USE A MORE EFFICIENT DATA FORMAT
3	**DON'T LOAD ALL DATA AT ONE TIME**
4	RECONFIGURE YOUR SYSTEM MEMORY
5	USE A DIFFERENT SYSTEM WITH MORE MEMORY AND PROCESSING POWER

One possibility is using **relational databases**, which provide **querying** capabilities that enable you to **extract** and work with batches of data.

NOTE: Machine learning algorithms that use an **iterative approach** to cycle through data tend to **work well** with this approach.
However, **other algorithms** may **not be suitable** for this approach, since they require access to the entire dataset.

HANDLE LARGE DATASETS IN MACHINE LEARNING

1	WORK WITH LESS DATA WHEN POSSIBLE
2	USE A MORE EFFICIENT DATA FORMAT
3	DON'T LOAD ALL DATA AT ONE TIME
4	**RECONFIGURE YOUR SYSTEM MEMORY**
5	USE A DIFFERENT SYSTEM WITH MORE MEMORY AND PROCESSING POWER

If you can't **reduce** the amount of data to efficiently process it, then consider **increasing** the memory (RAM)

Check whether the machine learning tool is using **all of the memory** installed

If no, then you may be able to change the settings to use more memory.

If yes, then you may also need to consider installing more RAM.

HANDLE LARGE DATASETS IN MACHINE LEARNING

1	WORK WITH LESS DATA WHEN POSSIBLE
2	USE A MORE EFFICIENT DATA FORMAT
3	DON'T LOAD ALL DATA AT ONE TIME
4	RECONFIGURE YOUR SYSTEM MEMORY
5	**USE A DIFFERENT SYSTEM WITH MORE MEMORY AND PROCESSING POWER**

If you are **still facing problems** after you tried reconfiguring your system memory, then consider having a **new system**, or **renting cloud storage** from Amazon Web Services, Microsoft Azure, or Google Cloud Platform

In some situations, it may be best to use a platform designed for big data, such as **Apache Hadoop** with the Mahout machine learning library, or Apache Spark with the MLlib library.

DATA MINING

PROBLEM: I collected large amount of data and I need to convert them into actionable intelligence

SOLUTION: Data Mining: The process of using statistical analysis methods to find meaningful patterns in data.

CLUSTERS	Groupings of similar records that belong together.
DEPENDENCIES	Records that depend on each other for related information.
ANOMALIES	Records that don't seem to belong because they don't follow expected patterns

WHAT'S THE GOAL OF DATA MINING

Although the goal of "silver mining" is to extract silver, the goal of "data mining" is not to extract data.

Rather, the goal of data mining is to extract **knowledge**

Although Data Mining **can be performed without** Artificial Intelligence and Machine Learning, **yet**, AI and ML provide new ways to perform data mining tasks in a **much faster**, **more efficient**, and at a **lower cost** than other methods.

HOW BUSINESSES REVEAL KNOWLEDGE

- How certain environmental factors affect the failure rates of various types of machinery used in the business, providing knowledge that can improve predictive maintenance schedules.
- Factors that reveal a customer might be about to switch to a different service provider, indicating that it might be a good time to offer customer loyalty incentives, special offers, and so forth.
- How various attributes of customer data correlate with loan defaults.
- How customers' history of buying (how recent, how frequent, and purchase amounts) correlates with the types of offers they respond to.

EXAMPLES OF APPLIED AI AND ML IN BUSINESS

INFORM BUSINESS DECISIONS	**Predict** where and when office and call center staff, crews, fleet vehicles, and equipment are most likely to be needed to **maximize utilization and productivity**.
IMPROVE THE UTILIZATION OF STAFF AND EQUIPMENT	Make the **most effective use** of marketing and sales by focusing contact based on a customer's **interests, browsing, and buying history**
IMPROVE SALES AND CUSTOMER EXPERIENCE	Recommend additional/other products based on those the customer has **already shown interest in**.
AUTOMATED INSPECTION	Use **flying drones** to scan properties for forest fires, missing persons, wildlife counts, and so forth.

EXAMPLES OF APPLIED AI AND ML IN BUSINESS

FINANCE	Process **loan applications**, contracts, and other documents, providing recommendations such as **loan approval**, contract terms that should be reviewed, and so forth.
SECURITY	Use realtime security camera analysis to **track the movement** of people within a building to perform surveillance, identifying unusual traffic patterns or behavior, such as violent behavior, vandalism, and unaccompanied suspicious packages.
PROCESS GUIDANCE AND NAVIGATION	Scan live imaging during surgery to **guide and assist surgeons**
VISUAL INTERACTION	Provide a virtual "try it out" feature for online shopping, where customer can see what they will look like when wearing the product, see **what the product will look like** in their home, and so forth.

GUIDELINES TO SELECT APPROPRIATE BUSINESS APPLICATIONS FOR AI AND ML

Consider using an AI or ML solution when

- You currently have, or can obtain sufficient data
- The problem is sufficiently self-contained or isolated from outside influence
- It is acceptable for the machine to provide an answer without your being completely clear how it got that answer.
- It is acceptable if the solution is not always 100% right
- It is a learning problem, rather than just an automation problem

IDENTIFY CHALLENGES OF AI AND ML IN BUSINESS

COLLECTING AND STORING ESSENTIAL DATA	You have data, but that data might **not be on the right** systems, in the right location, or in the right structure to be useful. Or a solution may require data that you have **not collected yet**. Data might be fragmented across **multiple locations**, and may need to be **aggregated and processed** before it can be used.
CHANGING DATA LANDSCAPE	
MAKING INFORMED DECISIONS RELATED TO IMPLEMENTATION COSTS	
USING DATA LEGALLY AND ETHICALLY	
SELECTING AND IMPLEMENTING TECHNOLOGY	
IDENTIFYING STAFF WITH SPECIALIZED SKILLS	

IDENTIFY CHALLENGES OF AI AND ML IN BUSINESS

COLLECTING AND STORING ESSENTIAL DATA	If the structure of the data, or relationships and patterns within the data, **change over time**, the ML algorithms need to **re-learn** the new patterns of data so they will continue to perform well.
CHANGING DATA LANDSCAPE	
MAKING INFORMED DECISIONS RELATED TO IMPLEMENTATION COSTS	
USING DATA LEGALLY AND ETHICALLY	
SELECTING AND IMPLEMENTING TECHNOLOGY	
IDENTIFYING STAFF WITH SPECIALIZED SKILLS	

IDENTIFY CHALLENGES OF AI AND ML IN BUSINESS

COLLECTING AND STORING ESSENTIAL DATA	New systems for collecting, aggregating, and processing data may introduce **significant costs**. Having a clear understanding of **requirements** (what problem you are trying to solve) will help to contain costs.
CHANGING DATA LANDSCAPE	
MAKING INFORMED DECISIONS RELATED TO IMPLEMENTATION COSTS	
USING DATA LEGALLY AND ETHICALLY	
SELECTING AND IMPLEMENTING TECHNOLOGY	
IDENTIFYING STAFF WITH SPECIALIZED SKILLS	

IDENTIFY CHALLENGES OF AI AND ML IN BUSINESS

COLLECTING AND STORING ESSENTIAL DATA	With this increased focus on data (sensitive information) comes an increased **possibility for errors** in handling data in ways that are legal, ethical, and free of bias.
CHANGING DATA LANDSCAPE	
MAKING INFORMED DECISIONS RELATED TO IMPLEMENTATION COSTS	
USING DATA LEGALLY AND ETHICALLY	
SELECTING AND IMPLEMENTING TECHNOLOGY	
IDENTIFYING STAFF WITH SPECIALIZED SKILLS	

IDENTIFY CHALLENGES OF AI AND ML IN BUSINESS

COLLECTING AND STORING ESSENTIAL DATA	You may need to acquire numerous **new tools and technologies** to implement AI/ML solutions in your organization, including tools for data collection and processing, development and testing of solutions, presentation, and reporting.
CHANGING DATA LANDSCAPE	
MAKING INFORMED DECISIONS RELATED TO IMPLEMENTATION COSTS	
USING DATA LEGALLY AND ETHICALLY	
SELECTING AND IMPLEMENTING TECHNOLOGY	
IDENTIFYING STAFF WITH SPECIALIZED SKILLS	

IDENTIFY CHALLENGES OF AI AND ML IN BUSINESS

- COLLECTING AND STORING ESSENTIAL DATA
- CHANGING DATA LANDSCAPE
- MAKING INFORMED DECISIONS RELATED TO IMPLEMENTATION COSTS
- USING DATA LEGALLY AND ETHICALLY
- SELECTING AND IMPLEMENTING TECHNOLOGY
- **IDENTIFYING STAFF WITH SPECIALIZED SKILLS**

Organizations that successfully implement AI/ML solutions typically have at least **one data scientist** on staff. A more challenging path involves utilizing **technical professionals** who understand the basics of AI/ML, perhaps **consulting with data scientists** on a contract basis, and using common off-the-shelf tools and machine learning algorithms.

ACTIVITY TIME - LESSON 1 - TOPIC A

1. You have found a dataset to use for your machine learning project, but someone on the project team doesn't think it would work well as the basis for a machine learning project.

What might lead her to such a conclusion from looking at the data?

ACTIVITY TIME - LESSON 1 - TOPIC A

2. You find another source of data that you could use for the project. It includes a historic database of more than three million transactions. Some of the specific data you need are not present, but it might be possible to infer what you need from the data that exists.

Furthermore, some columns of data are missing as many as 5% of their data values. In this case, the statistician does not seem to think that the incomplete data is a problem, and she thinks that the dataset will actually work quite well for machine learning.

Why is this situation acceptable for a machine learning project?

ACTIVITY TIME - LESSON 1 - TOPIC A

3. You're working with a major online clothing retailer to enhance their online Search feature, which customers use to find articles of clothing they want to buy.

What types of improvements might be added to the Search feature through machine learning technology?

LESSON 1
Solving Business Problems Using AI and ML

Topic A	Identify AI and ML Solutions for Business Problems
Topic B	**Formulate a Machine Learning Problem**
Topic C	Select Approaches to Machine Learning

FORMULATE A MACHINE LEARNING PROBLEM

1	FRAME THE PROBLEM
2	IDENTIFY WHY THE PROBLEM MUST BE SOLVED
3	PROVIDE BACKGROUND INFORMATION THAT WILL HELP TO SOLVE THE PROBLEM
4	DETERMINE WHETHER THE PROBLEM IS APPROPRIATE FOR AI/ML

TASK (T): What the **algorithm should accomplish**—for example, "Predict the sale price for a house."

EXPERIENCE (E): What **dataset** the **algorithm uses** to learn—for example, "Records of real estate transactions for King County for the year 2019."

PERFORMANCE (P): How you will **evaluate the performance** of the model. How you measure this depends on the type of task the algorithm performs

If we take a **TASK**, and through **EXPERIENCE**, its **PERFORMANCE** becomes better, than we use AI / ML

FORMULATE A MACHINE LEARNING PROBLEM

1	FRAME THE PROBLEM
2	**IDENTIFY WHY THE PROBLEM MUST BE SOLVED**
3	PROVIDE BACKGROUND INFORMATION THAT WILL HELP TO SOLVE THE PROBLEM
4	DETERMINE WHETHER THE PROBLEM IS APPROPRIATE FOR AI/ML

RATIONALE: Identifying **why** you need to solve this problem will help you evaluate the result

BENEFITS: Identify **how** the organization will benefit from the solution.

LIFETIME AND UTILIZATION: Identify **who** will use the solution and how long it will be used

FORMULATE A MACHINE LEARNING PROBLEM

1	FRAME THE PROBLEM	**ASSUMPTIONS**: ● Sources of data that are acceptable or unacceptable to use. ● Special operating requirements, such as software environments you must use. ● Business context, such as project time frames, who will use the solution, and so forth. **REFERENCE PROBLEMS**: A list of similar problems you have solved through applied machine learning.
2	IDENTIFY WHY THE PROBLEM MUST BE SOLVED	
3	**PROVIDE BACKGROUND INFORMATION THAT WILL HELP TO SOLVE THE PROBLEM**	
4	DETERMINE WHETHER THE PROBLEM IS APPROPRIATE FOR AI/ML	

FORMULATE A MACHINE LEARNING PROBLEM

1	FRAME THE PROBLEM	Once you have clearly defined the problem, you can explore whether it is more appropriate to be **solved as an AI/ML problem or as a traditional programming problem.**
2	IDENTIFY WHY THE PROBLEM MUST BE SOLVED	
3	PROVIDE BACKGROUND INFORMATION THAT WILL HELP TO SOLVE THE PROBLEM	
4	**DETERMINE WHETHER THE PROBLEM IS APPROPRIATE FOR AI/ML**	

ACTIVITY TIME - LESSON 1 - TOPIC B

IOT Company is a global manufacturing company that produces eco-friendly heating and cooling systems for large buildings, such as hotels, apartment buildings, retail stores, office buildings, and factories. Many of the products they manufacture include cast metal parts. These parts are cast at IOT Company's foundry, where they use robotic inspection cameras to scan parts coming out of molds to identify cracks, voids, and other defects that render the parts unusable.

In this activity, you will consider which of the following machine learning outcomes might be used in this manufacturing scenario.

Regression, Classification, Clustering, Dimensionality Reduction

ACTIVITY TIME - LESSON 1 - TOPIC B

1. Based on scans of parts coming out of the mold, machine learning algorithms identify which parts are acceptable for use, and which parts are defective.
   What type of outcome would you look for in this example?

ACTIVITY TIME - LESSON 1 - TOPIC B

2. Over time, parts coming out of a mold tend to show increasingly more defects, to the point where eventually a threshold is passed. At that point, parts produced in the mold can't be used, and the mold must be cleaned. Casting faulty parts wastes production time, so the manufacturing team would like to predict when the threshold is about to be reached, so the mold can be cleaned before faulty parts are produced.
A machine learning algorithm can be used to examine the quality of parts produced and predict when the threshold is about to be reached.

What type of outcome are you looking for in this example?

LESSON 1
Solving Business Problems Using AI and ML

Topic A	Identify AI and ML Solutions for Business Problems
Topic B	Formulate a Machine Learning Problem
Topic C	**Select Approaches to Machine Learning**

OPEN SOURCE AI TOOLS

Programming Languages	Python, R, Java, C++, C#
Weka	Visual data mining and machine learning (no coding): Weka, Orange, KNIME, and SIMON
NumPy	Library for Python that supports advanced and more efficient mathematical functions (NumPy arrays - Multidimensional)
SciPy	It builds on NumPy by offering more powerful mathematical functions (scientific computing)

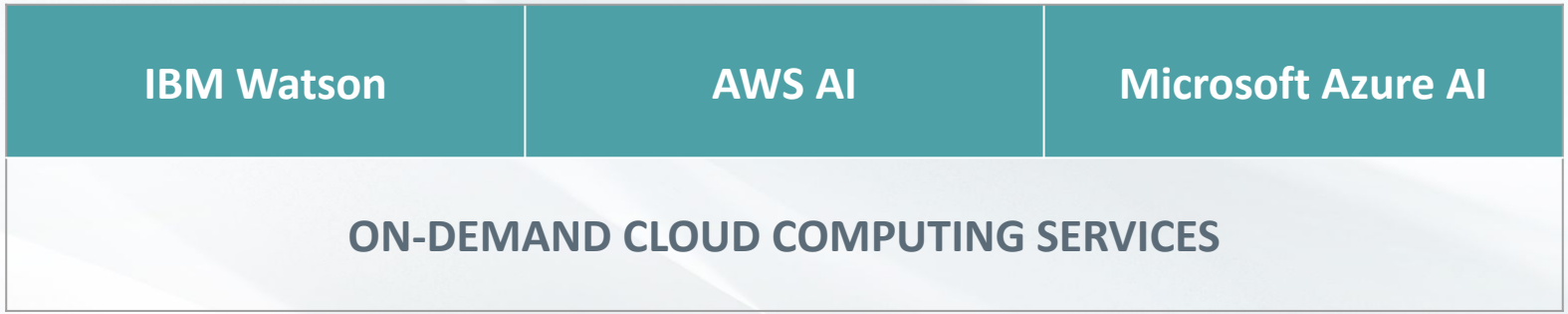
OPEN SOURCE AI TOOLS

Pandas	Part of SciPy. Similar to spreadsheet (rows and columns)
Matplotlib	Part of SciPy. Used for plotting data visually on graphs
Seaborn	Seaborn extends the functionality of Matplotlib by incorporating more types of plots and makes data visualization more attractive
scikit-learn	It provides support for fundamental supervised and unsupervised machine learning algorithms, including linear and logistic regression, decision trees, clustering, and so forth

OPEN SOURCE AI TOOLS

NLTK	Natural Language Toolkit (NLTK) is a suite of Python libraries that support natural language processing (NLP - Speech Recognition)
TensorFlow	Set up, train, and deploy artificial neural networks (deep learning)
Keras	ANN library in deep learning (useful for beginners)
Apache Spark MLlib	Framework for cluster computing

PROPRIETARY AI TOOLS



PRACTICAL EXERCISE



LESSON 1 SUMMARY ACTIVITY

What sorts of business problems do you expect to solve using AI and machine learning solutions?

Have you already selected software tools that you intend to use in your machine learning stack? If so, what tools are you using?

COURSE OUTLINES

LESSON 1	Solving Business Problems Using AI and ML
LESSON 2	**Preparing Data**
LESSON 3	Training, Evaluating, and Tuning a Machine Learning Model
LESSON 4	Building Linear Regression Models
LESSON 5	Building Forecasting Models
LESSON 6	Building Classification Models Using Regression and k-NN
LESSON 7	Building Clustering Models
LESSON 8	Building Decision Trees and Random Forests
LESSON 9	Building Support-Vector Machines
LESSON 10	Building Artificial Neural Networks
LESSON 11	Operationalizing Machine Learning Models
LESSON 12	Maintaining Machine Learning Operations

LESSON 2
Preparing Data

Topic A	**Collect Data**
Topic B	Transform Data
Topic C	Engineer Features
Topic D	Work with Unstructured Data

STRUCTURE OF DATA

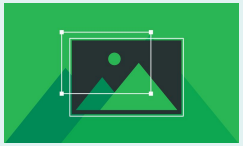
STRUCTURED DATA

format that facilitates searching, filtering, or extracting data (ex. spreadsheet or database)

Name	Age	DOB

UNSTRUCTURED DATA

Format that is not easy to search, filter, or extract data (ex. images, audio, video, social media posts)









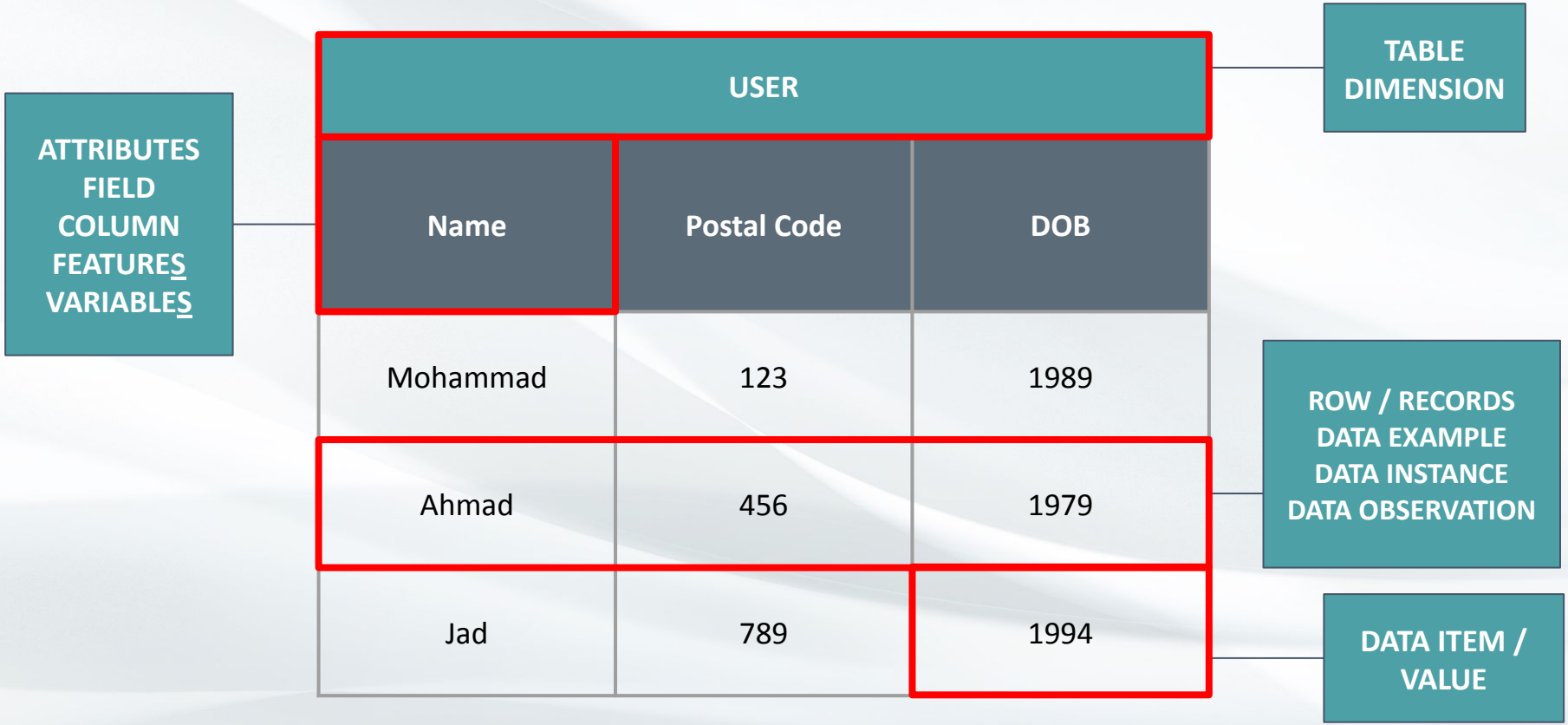
SEMI-STRUCTURED DATA

format that contain structured and unstructured data at the same time (ex. Email)

STRUCTURED
From: xyz@abc.com
To: mohammad@abc.com
Subject: Meeting

UNSTRUCTURED
Body, Attachments, Charts, Images

TERMS DESCRIBING PORTIONS OF DATA



DATA QUALITY ISSUES

IMBALANCED DATA	Suppose you have a dataset with **100,000 transactions**: **Fraudulent transactions: 500 (0.5%)** **Non-fraudulent transactions: 99,500 (99.5%)** Here, the dataset is highly imbalanced because one class (fraud = 500 cases) is extremely underrepresented compared to the other class (non-fraud = 99,500 cases).
IRRELEVANT FEATURES	
NON - REPRESENTATIVE DATA	
ERRORS, OUTLIERS, AND NOISE	

DATA QUALITY ISSUES

IMBALANCED DATA	While it is critical to have sufficient **quantity**, you should also strive for **sufficient quality (relevant)**. **BEFORE:** House ID | Bedrooms | Predicted Price **AFTER:** House ID | Bedrooms | **Location** | Predicted Price
IRRELEVANT FEATURES	
NON - REPRESENTATIVE DATA	
ERRORS, OUTLIERS, AND NOISE	

DATA QUALITY ISSUES

IMBALANCED DATA	At the same time, we need to make sure that we **don't have irrelevant features** **DATA REDUCTION (Practical Exercise)** EX: Estimating real estate prices should include the **number of bedrooms** and **location** (Longitude and Latitude are more important and precise **than Zip code**)
IRRELEVANT FEATURES	
NON - REPRESENTATIVE DATA	
ERRORS, OUTLIERS, AND NOISE	

DATA QUALITY ISSUES

IMBALANCED DATA	You sent a survey to **10,000 customers**. Only **200 customers** filled the survey (**2%**) **Result of Survey:** 150 Very Satisfied 🤩 | 50 unsatisfied 😡 **Analysis:** 75% of respondents are satisfied ⇒ Most of our 10,000 customers are happy **BUT** 2% only responded ⇒ **NON - REPRESENTATIVE DATA**
IRRELEVANT FEATURES	
NON - REPRESENTATIVE DATA	
ERRORS, OUTLIERS, AND NOISE	

DATA QUALITY ISSUES

IMBALANCED DATA	**ERRORS**: Wrong and Invalid data values Ex: GRADE: {84, 59, 92, **124**, 62} **Cause** Human entry mistakes, faulty equipment **Action** Data Validation + Filter **(Practical Exercise)**
IRRELEVANT FEATURES	
NON - REPRESENTATIVE DATA	
ERRORS, OUTLIERS, AND NOISE	

DATA QUALITY ISSUES

IMBALANCED DATA	**OUTLIERS**: Rare but valid (outside of normal distribution) Ex: Students' Ages At **School**: {9, 14, 12, **35**, 17}
IRRELEVANT FEATURES	
NON - REPRESENTATIVE DATA	**Cause** Rare events, extreme cases
ERRORS, OUTLIERS, AND NOISE	**(Practical Exercise)**

DATA QUALITY ISSUES

IMBALANCED DATA	**OUTLIERS**: **Action METHOD 1** 1- Calculate Z-Score (how many standard deviations a data point is away from the mean.) 2- Check if |Z|>3 then remove them About **68%** of values lie within **±1** standard deviation. About **95%** of values lie within **±2** standard deviations. About **99.7%** of values lie within **±3** standard deviations. Note: 3 is **NOT** a hard-rule **(Practical Exercise)**
IRRELEVANT FEATURES	
NON - REPRESENTATIVE DATA	
ERRORS, OUTLIERS, AND NOISE	

DATA QUALITY ISSUES

IMBALANCED DATA	**OUTLIERS**: **Action METHOD 2** 1- Calculate IQR (Q1 and Q3) 2- Calculate the Bounds 3- Identify outliers **(Practical Exercise)**
IRRELEVANT FEATURES	
NON - REPRESENTATIVE DATA	
ERRORS, OUTLIERS, AND NOISE	

DATA QUALITY ISSUES

IMBALANCED DATA	**OUTLIERS**: Z-Score is useful when Data is Normally Distributed, or I have Large Datasets with few outliers IQR is useful when I have skewed data, and it not affected by outliers Add a new value “200” on both methods and see the difference
IRRELEVANT FEATURES	
NON - REPRESENTATIVE DATA	
ERRORS, OUTLIERS, AND NOISE	

DATA QUALITY ISSUES

IMBALANCED DATA	**NOISE**: A heart rate monitor shows: 72, 73, 71, **200**, 72, 73 The sudden "200" might be electrical interference (noise), not the real heartbeat.
IRRELEVANT FEATURES	
NON - REPRESENTATIVE DATA	
ERRORS, OUTLIERS, AND NOISE	

OPEN DATASETS

University of California, Irvine Machine, Learning Repository	https://archive.ics.uci.edu/ml/index.php
Kaggle Datasets	https://www.kaggle.com/datasets
Amazon Datasets	https://registry.opendata.aws/
Microsoft Research Open Data	https://msropendata.com/
Open Media Library (OpenML)	https://www.openml.org/
Government Repositories	https://data.europa.eu/euodp/data/dataset Data.gov | Data.gov.uk | Data.gov.in | Open.canada.ca
Google Dataset Search	https://toolbox.google.com/datasetsearch

GUIDELINES FOR SELECTING A MACHINE LEARNING DATASET

Identify what you want to predict, so you can decide what data you need to collect.	Consider the **business problems** you want to **solve** and the **outcomes** you need to **produce** **EXAMPLE:** If it is a **supervised** machine learning problem, consider collecting data with **correct answers / labels**
Consider all data sources within your organization	
Consider using third-party data when you don't have your own data	
Establish repeatable mechanisms for collecting data	
Collect "right data," not necessarily "big data."	

GUIDELINES FOR SELECTING A MACHINE LEARNING DATASET

Identify what you want to predict, so you can decide what data you need to collect.	Use **existing data sources** that you already have at your company, such as paper records, records stored in document files, data collected from systems like CRM
Consider all data sources within your organization	
Consider using third-party data when you don't have your own data	
Establish repeatable mechanisms for collecting data	
Collect "right data," not necessarily "big data."	

GUIDELINES FOR SELECTING A MACHINE LEARNING DATASET

Identify what you want to predict, so you can decide what data you need to collect.	If you don't have your own data, consider using data accumulated by others. **EXAMPLE**: Google Dataset Search https://toolbox.google.com/datasetsearch
Consider all data sources within your organization	
Consider using third-party data when you don't have your own data	
Establish repeatable mechanisms for collecting data	
Collect "right data," not necessarily "big data."	

GUIDELINES FOR SELECTING A MACHINE LEARNING DATASET

Identify what you want to predict, so you can decide what data you need to collect.	Work with the **various data owners** to put systems into place to optimize the **standardization**, collection, and aggregation of data that will be used. Automation of the data pipeline can save time and reduce the possibility of human error.
Consider all data sources within your organization	
Consider using third-party data when you don't have your own data	
Establish repeatable mechanisms for collecting data	
Collect "right data," not necessarily "big data."	

GUIDELINES FOR SELECTING A MACHINE LEARNING DATASET

Identify what you want to predict, so you can decide what data you need to collect.	Getting the data right requires **not only technical know-how** (experience and good judgment in regard to the data needed to support the machine learning algorithms being used), **but also in regard to business knowledge** (knowing what data points adequately represent the business processes being modeled)
Consider all data sources within your organization	
Consider using third-party data when you don't have your own data	
Establish repeatable mechanisms for collecting data	
Collect "right data," not necessarily "big data."	

1. Operational Systems	The Farm 🌾	This is where the food is **first grown** (vegetables, fruits, milk).	Just like raw data is **first created.**
2. Operational Data Store (ODS)	The Market ⚙️	Farmers bring today's fresh food to the market. Everything is **sorted and kept ready** for people to see	This is like keeping today's **data fresh and ready**
3. Data Lake	The Big Storage Silo 🌊	Some food is stored in a giant warehouse **without cleaning** — potatoes, rice, even some mud still on them.	This is like a data lake: it holds **everything, clean or not (structured and unstructured)**.
4. Data Warehouse	The Food Factory 🏭	The food goes to a factory where it is **washed, cleaned,** and packaged. Now everything looks nice and is easy to use.	This is like the data warehouse, where data is **cleaned and organized**.
5. Data Mart	The Small Grocery Shop 🛒	Finally, food is sent to the **small neighborhood shop**, where only the things people need (bread, milk, apples) are available.	This is like the data mart: **only the needed data** for one team (like Sales or HR) is shown.

EXTRACT, TRANSFORM, AND LOAD (ETL)

Even when the original data sources are clean, problems may be introduced when data from different sources are combined.

(PRACTICAL EXERCISE)

- Different databases may use different schemas to organize data
- Columns may be named differently
- Data values may be represented by different data types or encoding schemes
- Records may overlap or contain duplicate data

You should resolve all of these problems in your data before attempting to apply machine learning algorithms.

EXTRACT, TRANSFORM, AND LOAD (ETL)

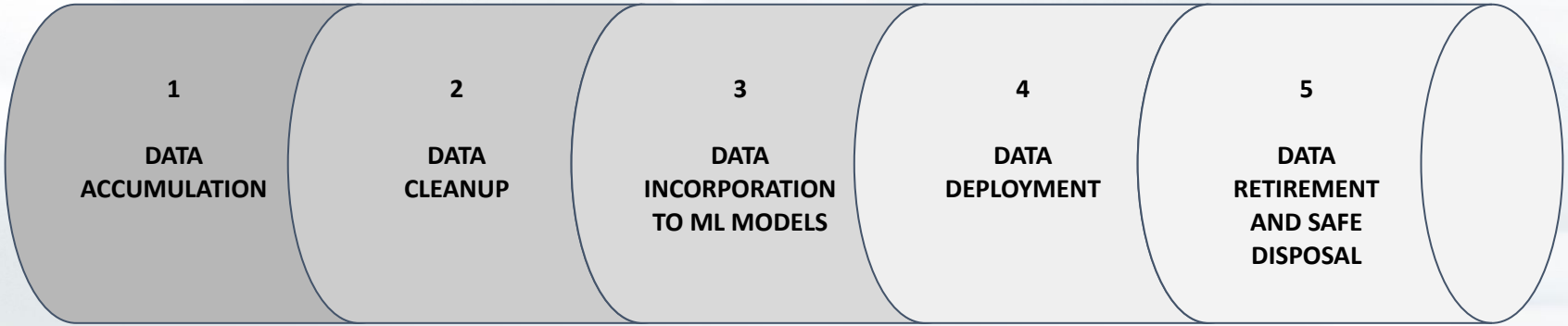
The process used to combine data from multiple sources is often called extract, transform, load (ETL).

1	**EXTRACT**	Pulls all or some of the data from various sources, which may have similar or dissimilar data structures
2	**TRANSFORM**	Converts data into a proper and uniform storage format or structure
3	**LOAD**	Inserts data into the destination where it will be stored for later modeling

MACHINE LEARNING PIPELINE

Instead of repeating the ETL manually every time to load new data, I apply the Machine Learning Pipeline, to repeat the ETL automatically.

Consider scripting your workflow from the start to make it easily repeatable and essentially self-documenting.



OBLIGATION TO PROTECT PII

PII: Personally Identifiable Information
Data that must be protected to ensure the privacy of the people described by that data, unless these data are anonymous

In order to protect the data, the company should restrict, as much as possible, the access to data for workers.

We can use the QUASI IDENTIFIERS (QI) to identify a person
EXAMPLE: Instead of searching by his first name and last name, we can use a combination of ZIP Code, Address, Age, … to identify this person

RELEVANT DATA PRIVACY LAWS

Companies should comply with regulations such as **EU GDPR** (EU General Data Protection Regulation - Protects Europe Citizens), **CCPA**, and **California's AB 1950** (protects California Citizens).

These standards can also be used internationally

Data privacy is as important as medical practice or else, they will be sued for negligence. For this reason, data (documents and records) should be kept as a proof

PRIVACY BY DESIGN

1. **PROACTIVE NOT REACTIVE; PREVENTATIVE NOT REMEDIAL**: Data privacy should be considered early and often (throughout the entire development lifecycle), not just after there is a problem.
2. **PRIVACY AS THE DEFAULT SETTING**: As initially installed, the application should be as private as possible. The user must opt in to decrease privacy to less private settings. By default, restrictions are placed on sharing, data collection, and data retention.
3. **PRIVACY EMBEDDED INTO DESIGN**: Privacy should be built into the design of the software. It should be explicitly included in processes like requirements identification, threat modeling, user interface design, testing, and so forth.

PRIVACY BY DESIGN

4. **FULL FUNCTIONALITY**: Positive-Sum, Not Zero-Sum: Customers value privacy. It is part of the value that customers pay for.
5. **END-TO-END SECURITY — FULL LIFECYCLE PROTECTION**: Privacy protections should follow the data wherever it goes—when it is first created, shared with others, archived, and deleted.
6. **VISIBILITY AND TRANSPARENCY — KEEP IT OPEN**: Privacy practices should be clear and overt, so users can have confidence in their privacy expectations. Policies and mechanisms should be in place to ensure users can address problems and have them resolved efficiently.
7. **RESPECT FOR USER PRIVACY — KEEP IT USER CENTRIC**: Users own their data. Data held in the software should be accurate, and the user must have the power to correct errors. The user can grant and revoke consent on the use of the data.

DATA PRIVACY PRINCIPLES AT ODDS WITH MACHINE LEARNING

GDPR Principle	Machine Learning Principle
Less data is better	More data is better
Decisions made about people must be transparent	It may be hard to completely know how decisions are made
Data usage must be revealed upfront	Data usage emerges through the training process
The user is king	The model is king

GUIDELINES FOR COMPLYING WITH DATA PRIVACY LAWS AND STANDARDS

- **IDENTIFY PERSONALLY IDENTIFIABLE INFORMATION (PII)**
- **IDENTIFY PERSONAL DATA AS DEFINED BY GDPR**: Any information that is used as an identification to any person is considered as a personal data
- **ENSURE PRIVACY POLICIES, TERMS, AND CONDITIONS ARE CLEAR**: It should include notes about the software update and how terms and conditions are changed, why and how the data is used
- **ENSURE SOFTWARE DOES NOT SHARE DATA WITHOUT CONSENT**: We should always get the user's approval before using any data. If the data should be used by other party, it should be anonymous

DATA ANONYMIZATION

Personally Identifiable Information (PII) may need to be anonymized before it can be processed and analyzed

- **REPLACEMENT**: Substitute any values that could be used to identify the user with different values
- **SUPPRESSION**: Omit (all or in part) any values that could be used to identify the user.
- **GENERALIZATION**: Substitute PII values with something less specific. (Ex. DOB with just decade)
- **PERTURBATION**: Make random changes to the data to corrupt values that could be used to identify the user.

ACTIVITY TIME

Preparing Data

ACTIVITY 2-1: Examining the Structure of a Machine Learning Dataset

ACTIVITY 2-2: Loading the Dataset

LESSON 2
Preparing Data

Topic A	Collect Data
Topic B	**Transform Data**
Topic C	Engineer Features
Topic D	Work with Unstructured Data

QUALITATIVE	ORDINAL	QUANTITATIVE	
1 NOMINAL (DISCRETE)	**2 ORDINAL (DISCRETE)**	**3 INTERVAL (DISCRETE OR CONTINUOUS)**	**4 RATIO (DISCRETE OR CONTINUOUS)**
Category DESCRIPTIVE | COUNTABLE | NON NUMERICAL | NON CONTINUOUS	Category + Ordered	Category + Ordered + Equal Intervals	Category + Ordered + Equal Intervals + Origin Zero (Zero means complete absence)
Gender (Male, Female) Hair Color (Black, Blonde) City (Dubai, Beirut, Riyadh)	Size (Small, Medium, Large) Grade (A, B, C, D, F) Happiness / Satisfaction Level (Unsatisfied, Neutral, Satisfied)	Temperature, Grade, Age (in Years)	Height, Weight, Duration, Age, Price
MODE	MEDIAN, MODE	MEAN, MEDIAN, MODE, RANGE, CORRELATION	MEAN, MEDIAN, MODE, RANGE

Corrupted or Unusable Data

Sometimes, you may receive a file containing corrupted or unusable data.

Such data can affect your AI model and your results.

For this case, you need to transform this data.

(PRACTICAL EXERCISE)

DEDUPLICATION

Deduplication is the process of identifying or removing the duplicates in a dataset.

(PRACTICAL EXERCISE)

IMPUTATION

The process of filling in missing data values that consists of using statistical calculations to determine what the missing values should be.

MEAN / MODE IMPUTATION	Calculate the **mean** or **mode** of all items that are **not missing** in that column, then use the result to **fill in** missing values. For example, to fill in a missing smartphone MSRP (Manufacturer's Suggested Retail Price), you would just take the **mean of all smartphone** MSRPs minus any missing values. This approach is **simple**, as it preserves the value of the mean/mode and sample size. **However**, it may **not be as good as** other methods listed here **(PRACTICAL EXERCISE)**
Substitution	
Hot-Deck Imputation	
Cold-Deck Imputation	
Regression Imputation	

IMPUTATION

The process of filling in missing data values that consists of using statistical calculations to determine what the missing values should be.

Mean / Mode Imputation	Use data from a **new record** that is **not in the sample**. For example, you might find **another source** of smartphone MSRPs that you can **substitute** the missing one with. In case there are **multiple matches**, you can choose the **most recent**, or decide **case by case**, or use the **average** of the values. In substitution, the choice is **manual** and **judgment-based** **(PRACTICAL EXERCISE)**
SUBSTITUTION	
Hot-Deck Imputation	
Cold-Deck Imputation	
Regression Imputation	

IMPUTATION

The process of filling in missing data values that consists of using statistical calculations to determine what the missing values should be.

Mean / Mode Imputation	Find records in the sample that have **similar values** on all other data items than the one that is missing, and **copy the missing value** from one of the **real** similar records. If there is more than one similar record, **randomly** select the one you copy from, **or** choose the **first one**. For example, you could find another smartphone in the dataset that has similar values for the other features, then **reuse** that smartphone's MSRP as the MSRP for the smartphone with the missing value. It **preserves realistic observed values** and the distribution of your dataset. **(PRACTICAL EXERCISE)**
Substitution	
HOT-DECK IMPUTATION	
Cold-Deck Imputation	
Regression Imputation	

IMPUTATION

The process of filling in missing data values that consists of using statistical calculations to determine what the missing values should be.

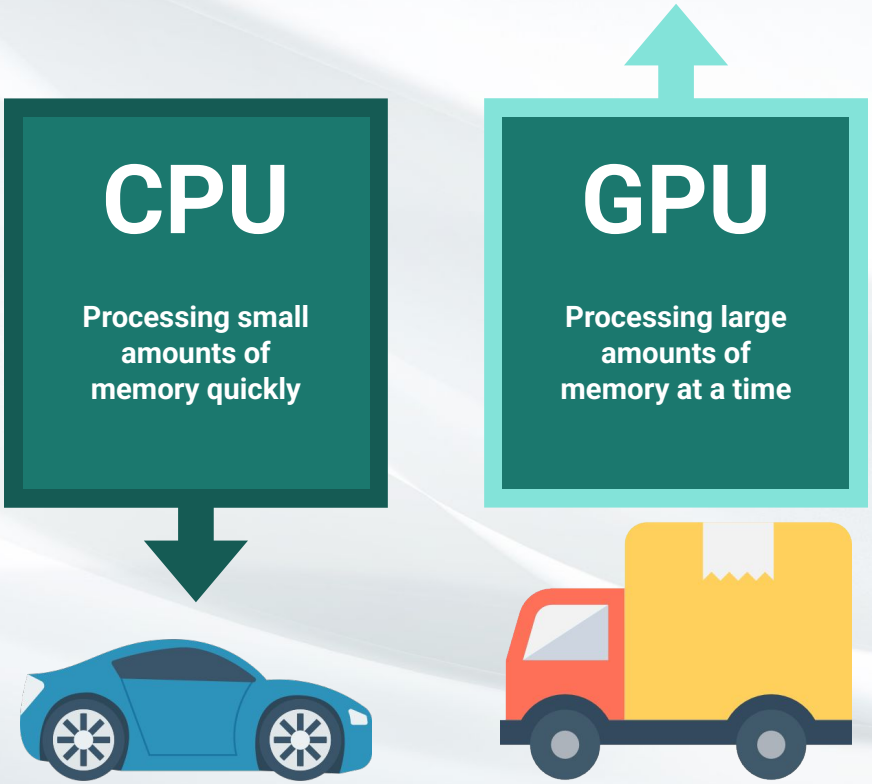
Mean / Mode Imputation	Similar to hot-deck imputation, but **instead** of pulling from the **same sample** that the missing value is in, you pull from an **external sample**. For example, you may have access to **another** electronics vendor's product info, so you'll try to find a **similar** smartphone in that sample. - **Substitution** = fill with **any** external info and choose the **"best"**. - **Hot Deck =** fill with **real** entry from **same** dataset (don't use mean) - **Cold Deck** = fill with a **specific**, predefined **external** dataset and Apply **systematic rule** (first, random, **mean** …) **(PRACTICAL EXERCISE)**
Substitution	
Hot-Deck Imputation	
COLD-DECK IMPUTATION	
Regression Imputation	

IMPUTATION

The process of filling in missing data values that consists of using statistical calculations to determine what the missing values should be.

Mean / Mode Imputation	Use a statistical model to identify what the missing value should be, based on data in the record. This method leverages **patterns** established among other records that are not missing the value to determine what the value should be. This is essentially a machine learning task that treats the record with the missing value as new data to estimate. Once the estimation is made, the record can be returned to the dataset to use on some other machine learning task. **(PRACTICAL EXERCISE)**
Substitution	
Hot-Deck Imputation	
Cold-Deck Imputation	
REGRESSION IMPUTATION	

HARDWARE REQUIREMENTS



GUIDELINES FOR EXPLORING THE STRUCTURE OF A DATASET

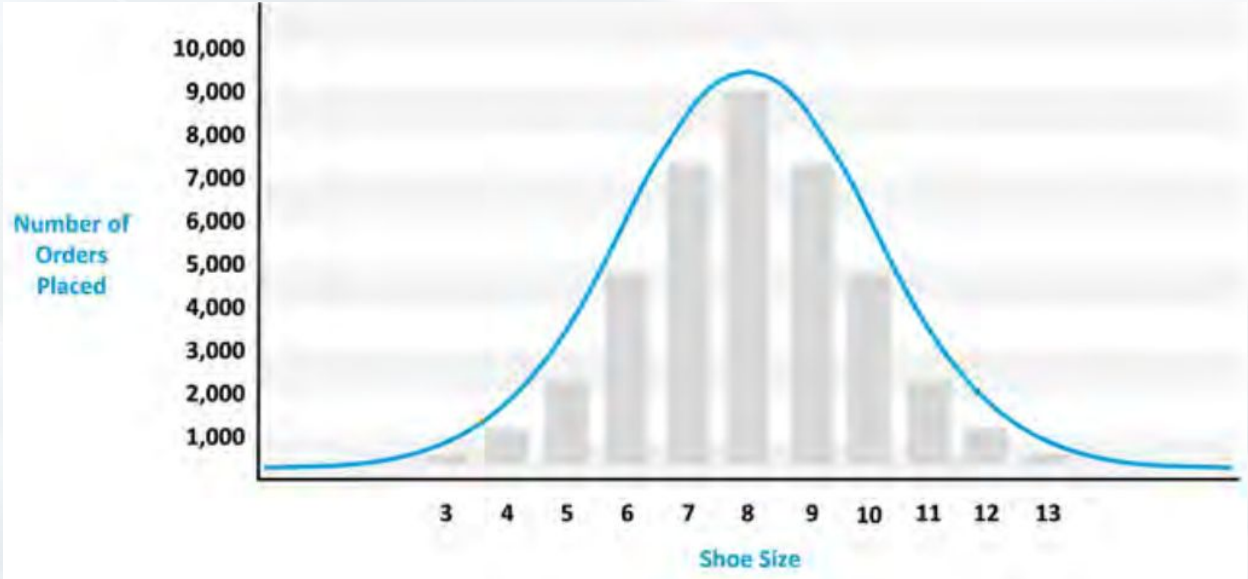
my_dataframe.**info()**	Returns the number of rows and columns in the dataframe
my_dataframe.**head()**	Returns a table displaying the first five rows of data in the dataframe
my_dataframe.**describe()**	Returns a table showing descriptive statistics for values in each column
pandas.**option_context()**	Change various settings used by pandas (Ex. Display output with 2 decimal places)

NORMAL DISTRIBUTION (GAUSSIAN CURVE)

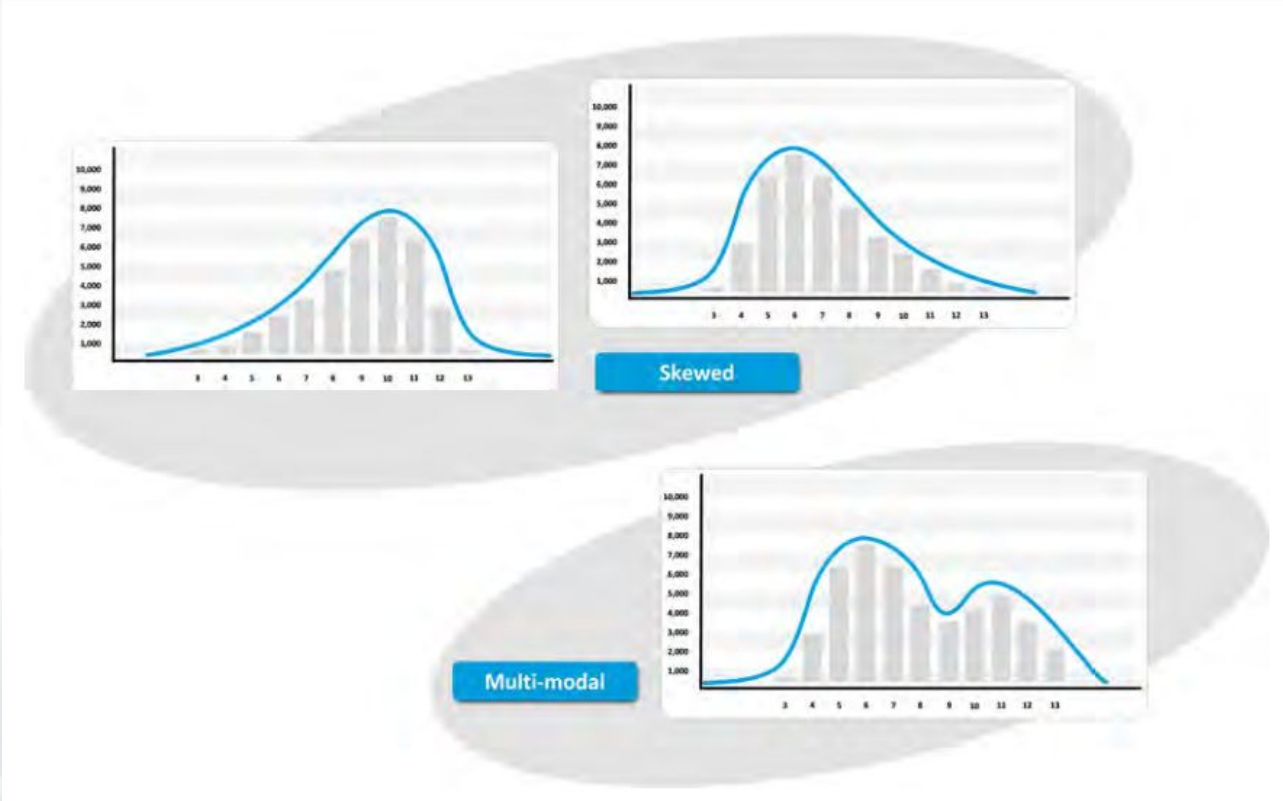
A function that represents the distribution of many random variables as a symmetrical bell-shaped graph.

- Bell-shaped
- Symmetrical
- Centered
-

(only one hump - one most common value)



NON - NORMAL DISTRIBUTIONS



DESCRIPTIVE STATISTICAL ANALYSIS (PRACTICAL EXERCISE)

Central Tendency (Measures of Center) (is the distribution centered?)	Mean
	Median
	Mode
Variability (Measures of dispersion, scatter, or spread)	Range
	Variance
	Standard Deviation
Skewness	
Kurtosis	

The mean is the **Average**
(**add** all numbers in the set, and then **divide** by the total count of numbers)

EXAMPLE: [12, 28, 41, 51, 68] ⇒ 40

BEST FOR: Continuous data in a symmetric distribution with no outliers

DESCRIPTIVE STATISTICAL ANALYSIS (PRACTICAL EXERCISE)

Central Tendency (Measures of Center) (is the distribution centered?)	Mean
	Median
	Mode
Variability (Measures of dispersion, scatter, or spread)	Range
	Variance
	Standard Deviation
Skewness	
Kurtosis	

The number value located in the **middle** of a **sorted set**

EXAMPLE: [12, 28, 41, 51, 68] ⇒ 41
EXAMPLE: [28, 41, 51, 68] ⇒ (41+51)/2 ⇒ (92)/2 ⇒ 46

BEST FOR: Distributions that are skewed or that have outliers

DESCRIPTIVE STATISTICAL ANALYSIS (PRACTICAL EXERCISE)

Central Tendency (Measures of Center) (is the distribution centered?)	Mean
	Median
	Mode
Variability (Measures of dispersion, scatter, or spread)	Range
	Variance
	Standard Deviation
Skewness	
Kurtosis	

The number that occurs **most often** within a set of numbers

EXAMPLE: [12, 14, 16, 16, 16, 17] ⇒ 16

BEST FOR: Qualitative and **categorical data** [“Public”, “Private”]

DESCRIPTIVE STATISTICAL ANALYSIS (PRACTICAL EXERCISE)

Central Tendency (Measures of Center) (is the distribution centered?)	Mean
	Median
	Mode
Variability (Measures of dispersion, scatter, or spread)	**Range**
	Variance
	Standard Deviation
Skewness	
Kurtosis	

The **difference** between the **smallest** and **largest** values in the data

Identifying the minimum and maximum values gives you a sense of where the **data is spread.**

EXAMPLE: [12, 28, 41, 51, 68] ⇒ 68 - 12 ⇒ 56

DESCRIPTIVE STATISTICAL ANALYSIS (PRACTICAL EXERCISE)

Central Tendency (Measures of Center) (is the distribution centered?)	Mean
	Median
	Mode
Variability (Measures of dispersion, scatter, or spread)	**Range**
	Variance
	Standard Deviation
Skewness	
Kurtosis	

Interquartile Range (**IQR**): divides the range into **four subsets**, where each subset contains the **same number of values**

IQR = Last value of Q3 - Last value of Q1

It helps to show where the majority of values lie

If IQR is HIGH, your data is spread out
if IQR is LOW, your data is more clustered

DESCRIPTIVE STATISTICAL ANALYSIS (PRACTICAL EXERCISE)

Central Tendency (Measures of Center) (is the distribution centered?)	Mean
	Median
	Mode
Variability (Measures of dispersion, scatter, or spread)	Range
	Variance
	Standard Deviation
Skewness	
Kurtosis	

Variance measures how spread out your data is from the mean or how far each value in the dataset is from the mean

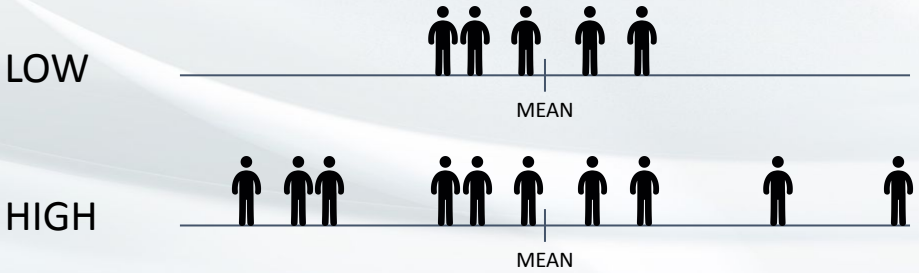
DESCRIPTIVE STATISTICAL ANALYSIS (PRACTICAL EXERCISE)

Central Tendency (Measures of Center) (is the distribution centered?)	Mean
	Median
	Mode
Variability (Measures of dispersion, scatter, or spread)	Range
	Variance
	Standard Deviation
Skewness	
Kurtosis	

Deviation just means how **far** from the normal. The Standard Deviation is a measure of how **spread out** our numbers are.

VISUALIZE STANDARD DEVIATION: EX1 | EX2

So, using the Standard Deviation we have a "**standard**" way of knowing what is normal, and what is extra large or extra small.

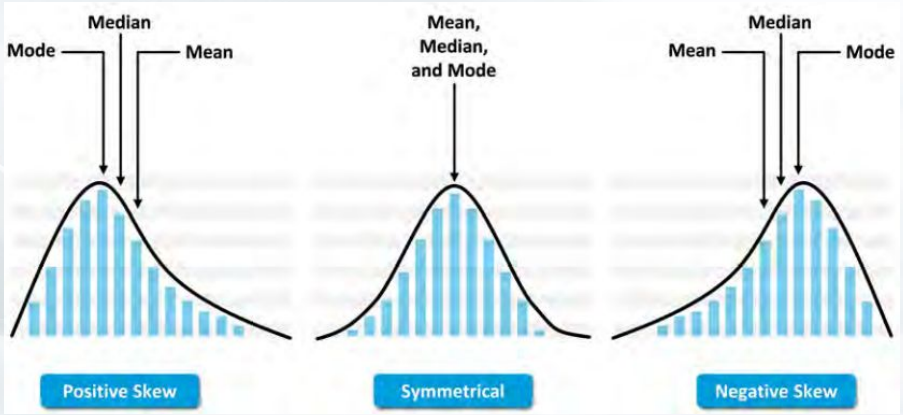


neither a higher nor a lower value is inherently better.

DESCRIPTIVE STATISTICAL ANALYSIS

Central Tendency (Measures of Center) (is the distribution centered?)	Mean
	Median
	Mode
Variability (Measures of dispersion, scatter, or spread)	Range
	Variance
	Standard Deviation
Skewness	
Kurtosis	

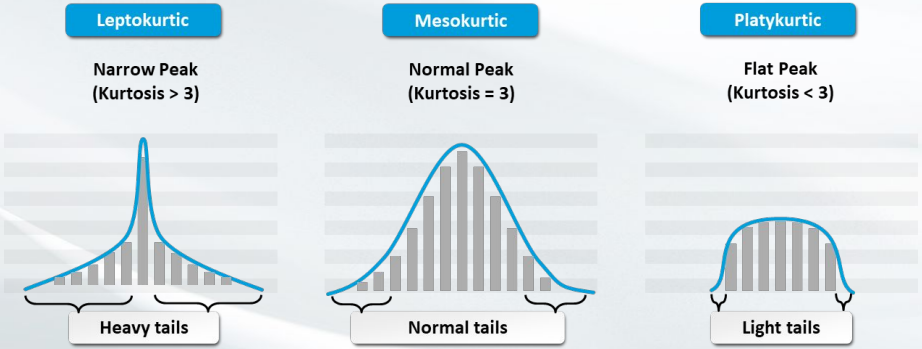
Skewness describes the extent to which a **distribution's symmetry differs from a normal distribution** (zero skew, or symmetrical), either to the right (positive skew) or to the left (negative skew).



DESCRIPTIVE STATISTICAL ANALYSIS

Central Tendency (Measures of Center) (is the distribution centered?)	Mean
	Median
	Mode
Variability (Measures of dispersion, scatter, or spread)	Range
	Variance
	Standard Deviation
Skewness	
Kurtosis	

In a normal bell-shaped distribution, there are tails on the left and right sides. The kurtosis coefficient is a measure of the shape of these tails.

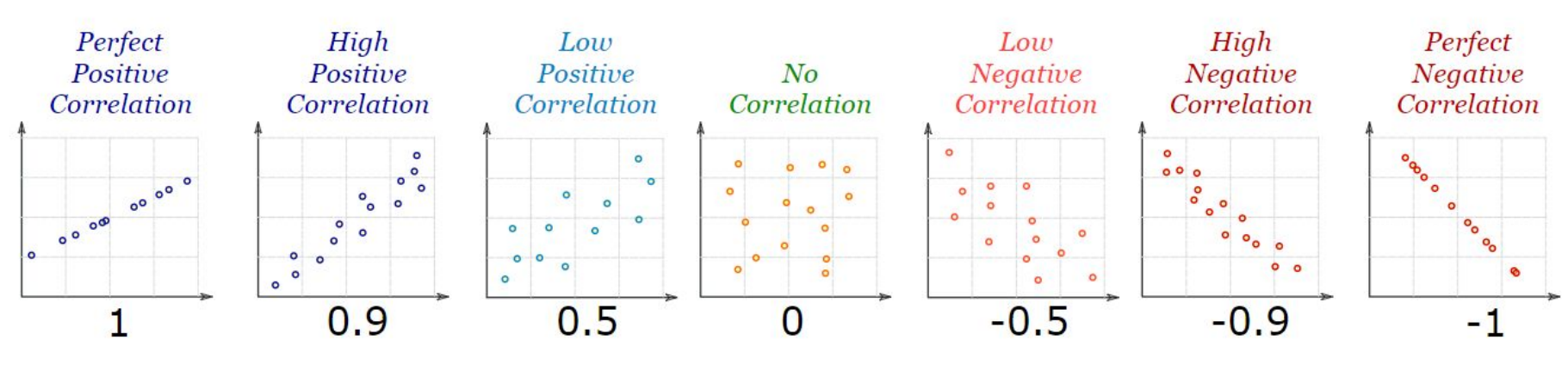


CORRELATION

Correlation is used to assess the strength of a linear relationship between two continuous numeric variables: r (Ex. Weight & Height).

it shows whether 2 variables are related to each other or not

GAME



GUIDELINES FOR ANALYZING A DATASET

Identify Noise within a Dataset	Specific data items
	Columns
	Rows
Reveal Outliers in a Dataset	Use statistics measures to reveal extreme values
	Use visualization tools to reveal extreme values
	Use machine learning algorithms to reveal outliers
Use Python to Analyze Correlations	my_dataframe.corr()
	my_dataframe.sort_values()

- A particular **data value** may be **missing**
- It may contain an extreme or atypical value (**outlier**) that does not follow the general pattern established by other data items in the same column

WHAT TO DO: Anomalous data items may need to be **modified** to adhere to the general pattern, or the **record** that contains them may need to be **dropped** from the training dataset

GUIDELINES FOR ANALYZING A DATASET

Identify Noise within a Dataset	Specific data items
	Columns
	Rows
Reveal Outliers in a Dataset	Use statistics measures to reveal extreme values
	Use visualization tools to reveal extreme values
	Use machine learning algorithms to reveal outliers
Use Python to Analyze Correlations	my_dataframe.corr()
	my_dataframe.sort_values()

- Some of the **columns** you have initially selected for use in a model may have little or **no correlation** to the target
- Some of the **columns** may be **redundant** since they are **dependent on other columns** that are already included in the model.

EXAMPLE: Cars and Weather

WHAT TO DO: Unnecessary columns may need to be dropped from the training dataset.

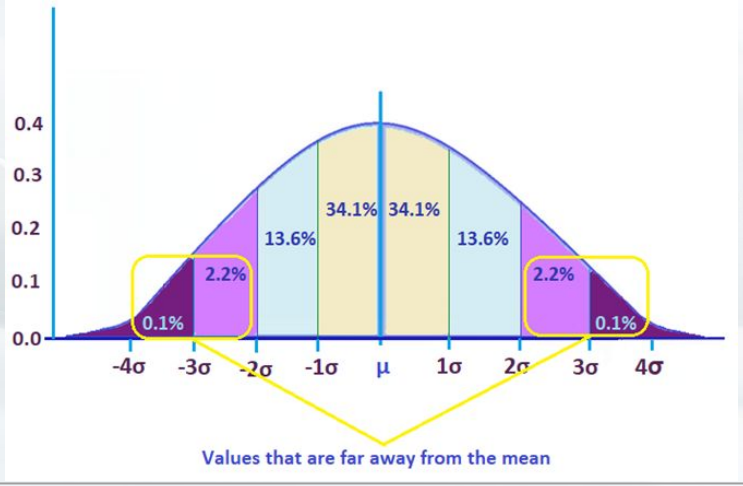
GUIDELINES FOR ANALYZING A DATASET

Identify Noise within a Dataset	Specific data items	In some cases, the individual data items within a particular row/record **may each be fine, but together they may represent an anomalous combination**, one that defies the overall pattern established by data in other records. Unusual or anomalous data combinations may **mislead the algorithm** during training. EXAMPLE: Weight 75 KG | Category Airplane WHAT TO DO: Anomalous rows may need to be dropped from the training dataset.
	Columns	
	Rows	
Reveal Outliers in a Dataset	Use statistics measures to reveal extreme values	
	Use visualization tools to reveal extreme values	
	Use machine learning algorithms to reveal outliers	
Use Python to Analyze Correlations	my_dataframe.corr()	
	my_dataframe.sort_values()	

GUIDELINES FOR ANALYZING A DATASET

Identify Noise within a Dataset	Specific data items
	Columns
	Rows
Reveal Outliers in a Dataset	**Use statistics measures to reveal extreme values**
	Use visualization tools to reveal extreme values
	Use machine learning algorithms to reveal outliers
Use Python to Analyze Correlations	my_dataframe.corr()
	my_dataframe.sort_values()

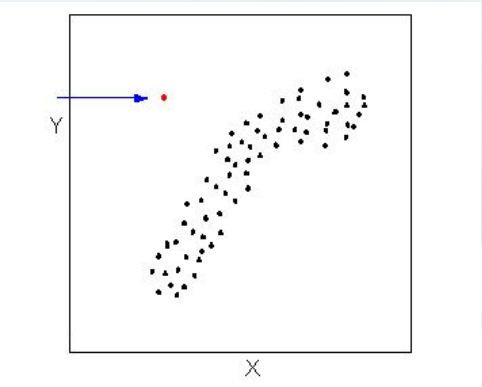
For example, calculate the **standard deviation** for every value in the dataset, and **filter** them to reveal values with a standard deviation from the mean of more than **2 or 3**, which are the **outliers**



GUIDELINES FOR ANALYZING A DATASET

Identify Noise within a Dataset	Specific data items
	Columns
	Rows
Reveal Outliers in a Dataset	Use statistics measures to reveal extreme values
	Use visualization tools to reveal extreme values
	Use machine learning algorithms to reveal outliers
Use Python to Analyze Correlations	my_dataframe.corr()
	my_dataframe.sort_values()

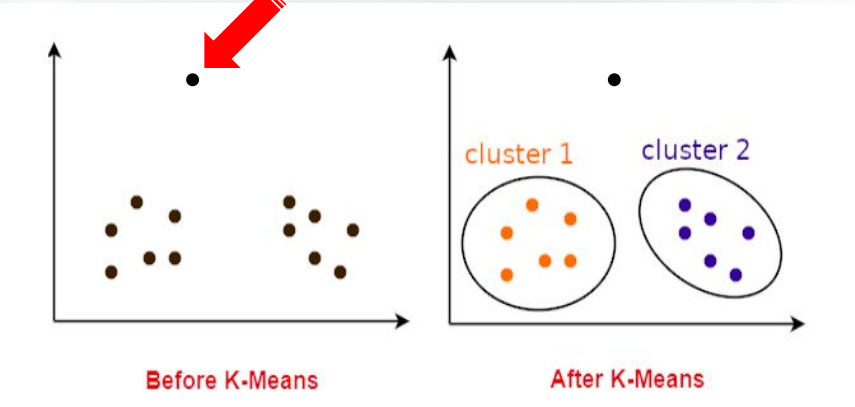
For example, use **visualizations** such as a histogram or box-and-whisker plot for **distributions** extending out in long, skinny (leptokurtic) tails, which may **represent outliers** in the data. In a scatterplot, look for values far away from any clusters.



GUIDELINES FOR ANALYZING A DATASET

Identify Noise within a Dataset	Specific data items
	Columns
	Rows
Reveal Outliers in a Dataset	Use statistics measures to reveal extreme values
	Use visualization tools to reveal extreme values
	Use machine learning algorithms to reveal outliers
Use Python to Analyze Correlations	my_dataframe.corr()
	my_dataframe.sort_values()

For example, the **k-means** clustering algorithm can identify natural clusters of data based on multiple values within a dataset. Identify those values with the **greatest distance** from any cluster as outliers.



GUIDELINES FOR ANALYZING A DATASET

Identify Noise within a Dataset	Specific data items	Returns a table showing the **correlation coefficient** (a floating point number) for each pair of columns in the dataframe. You can specify the type of correlation to be performed, with Pearson standard correlation being the default type.
	Columns	
	Rows	
Reveal Outliers in a Dataset	Use statistics measures to reveal extreme values	
	Use visualization tools to reveal extreme values	
	Use machine learning algorithms to reveal outliers	
Use Python to Analyze Correlations	**my_dataframe.corr()**	
	my_dataframe.sort_values()	

GUIDELINES FOR ANALYZING A DATASET

Identify Noise within a Dataset	Specific data items	Sorts the values along either axis. You can specify a list of names to sort by and specify whether the sort should be in **ascending** or **descending** order. The function returns a **dataframe** with the **values sorted**. You can assign the return value back to the original dataframe if you want the original to contain the sorted results.
	Columns	
	Rows	
Reveal Outliers in a Dataset	Use statistics measures to reveal extreme values	
	Use visualization tools to reveal extreme values	
	Use machine learning algorithms to reveal outliers	
Use Python to Analyze Correlations	my_dataframe.corr()	
	my_dataframe.sort_values()	

ACTIVITY TIME

Preparing Data

ACTIVITY 2-3: Exploring the General Structure of the Dataset

ACTIVITY 2-4: Analyzing a Dataset Using Statistical Measures

LESSON 2
Preparing Data

Topic A	Collect Data
Topic B	Transform Data
Topic C	**Engineer Features**
Topic D	Work with Unstructured Data

DATA PREPROCESSING

DATA PREPROCESSING

The task of applying various transformation and encoding techniques to data so that it can be interpreted and analyzed by a machine learning algorithm.

FEATURE ENGINEERING

The technique of **generating** and extracting features from data in order to **improve** the ability for a machine learning model to make estimations.

FEATURE SCALING

The task of **transforming** the values of multiple **features** so that those values are all on a **similar scale**.

FEATURE ENGINEERING

(PRACTICAL EXERCISE)

BEFORE
ADDRESS
305, AL MUROOJ, RIYADH, RIYADH, 123456

AFTER				
HOUSE NUMBER	**STREET NAME**	**CITY**	**STATE**	**POSTAL CODE**
305	AL MUROOJ	RIYADH	RIYADH	123456

FEATURE SCALING

PROBLEM: If you are comparing Age (0 to 100 years) to Income (0 to 1,000,000 SAR), then the income values may indicate better results, but that's not the case. (measuring all ingredients with same spoon)

NORMALIZATION: Convert the dataset to a range **between 0 and 1**
EXAMPLE: Youngest age is 0 and Highest age is 1 (apply the same for Income). This way, we have normalized dataset **(PRACTICAL EXERCISE)**

STANDARDIZATION: Converting the dataset to statistical measures by plotting them in a frequency distribution and calculate the Z-SCORE (The number of standard deviations from the mean) **(PRACTICAL EXERCISE)**

DATA ENCODING

Let's suppose I have a dataset containing Qualitative Data (Categorical Variables). For example, **Eye Color** (Blue, Brown, Hazel) or **City** (Beirut, Dubai, Riyadh).

PROBLEM: I **cannot** do descriptive statistics for **text**

SOLUTION: I need to **convert** them to **integers**.
That's why, I use **DATA ENCODING**

DEFINITION: Change Qualitative Data (**text**) To Quantitative (**numbers**)

DATA ENCODING

LABEL ENCODING
ONE-HOT ENCODING
BINARY ENCODING
FREQUENCY-BASED ENCODING
TARGET MEAN ENCODING
HASH ENCODING

Applied to **ORDINAL DATA** **(PRACTICAL EXERCISE)**
[small, medium, large] ⇒ [0, 1, 2]

This should **NOT** be applied to Nominal Data like countries [UK, France, Spain] ⇒ [0, 1, 2]. It is wrong because it doesn't mean that Spain is the highest

BEFORE		AFTER
SIZE		SIZE
SMALL		0
MEDIUM		1
LARGE		2

DATA ENCODING

- LABEL ENCODING
- **ONE-HOT ENCODING**
- BINARY ENCODING
- FREQUENCY-BASED ENCODING
- TARGET MEAN ENCODING
- HASH ENCODING

Applied to **NOMINAL DATA**
(Disadvantage: It increases the number of columns which decreases efficiency / accuracy with some algorithms

(PRACTICAL EXERCISE)

BEFORE		AFTER		
COUNTRY		isUK	isFrance	isSpain
UK		1	0	0
FRANCE		0	1	0
SPAIN		0	0	1

DATA ENCODING

- LABEL ENCODING
- ONE-HOT ENCODING
- **BINARY ENCODING**
- FREQUENCY-BASED ENCODING
- TARGET MEAN ENCODING
- HASH ENCODING

Applied to **NOMINAL DATA**
(for 127 categories, One Hot Encoding will have 127 features, while for Binary encoding, we will need just 7 features)

BEFORE		AFTER	
COUNTRY		COUNTRY1	COUNTRY2
UK		0	1
FRANCE		1	0
SPAIN		1	1

DATA ENCODING

- LABEL ENCODING
- ONE-HOT ENCODING
- BINARY ENCODING
- **FREQUENCY-BASED ENCODING**
- TARGET MEAN ENCODING
- HASH ENCODING

Calculate how frequently each class occurs within the training set, and you use this number as the code for that class

BEFORE
COUNTRY
UK
FRANCE
SPAIN
UK
SPAIN
UK

AFTER	
COUNTRY	FBE
UK	3
FRANCE	1
SPAIN	2

(PRACTICAL EXERCISE)

DATA ENCODING

- LABEL ENCODING
- ONE-HOT ENCODING
- BINARY ENCODING
- FREQUENCY-BASED ENCODING
- **TARGET MEAN ENCODING**
- HASH ENCODING

We count the number of activated times for each country and divide them by the number of occurrences

BEFORE	
COUNTRY	**TARGET**
UK	0
FRANCE	1
SPAIN	1
UK	0
SPAIN	0
UK	1

AFTER	
COUNTRY	**TME**
UK	0.33
FRANCE	1
SPAIN	0.5

DATA ENCODING

- LABEL ENCODING
- ONE-HOT ENCODING
- BINARY ENCODING
- FREQUENCY-BASED ENCODING
- TARGET MEAN ENCODING
- **HASH ENCODING**

Map a particular text string **to** a number value. It is based on the characters in the text string, and will produce the same number value each time a particular text string is provided to it. So, **DO NOT USE IT** when you have text having the same number of characters (ex: Pizza and Fries)

BEFORE		AFTER
COUNTRY		COUNTRY
UK		c9e8fc8d
FRANCE		a7c9c55e
SPAIN		23d8352c

CONTINUOUS VS DISCRETE

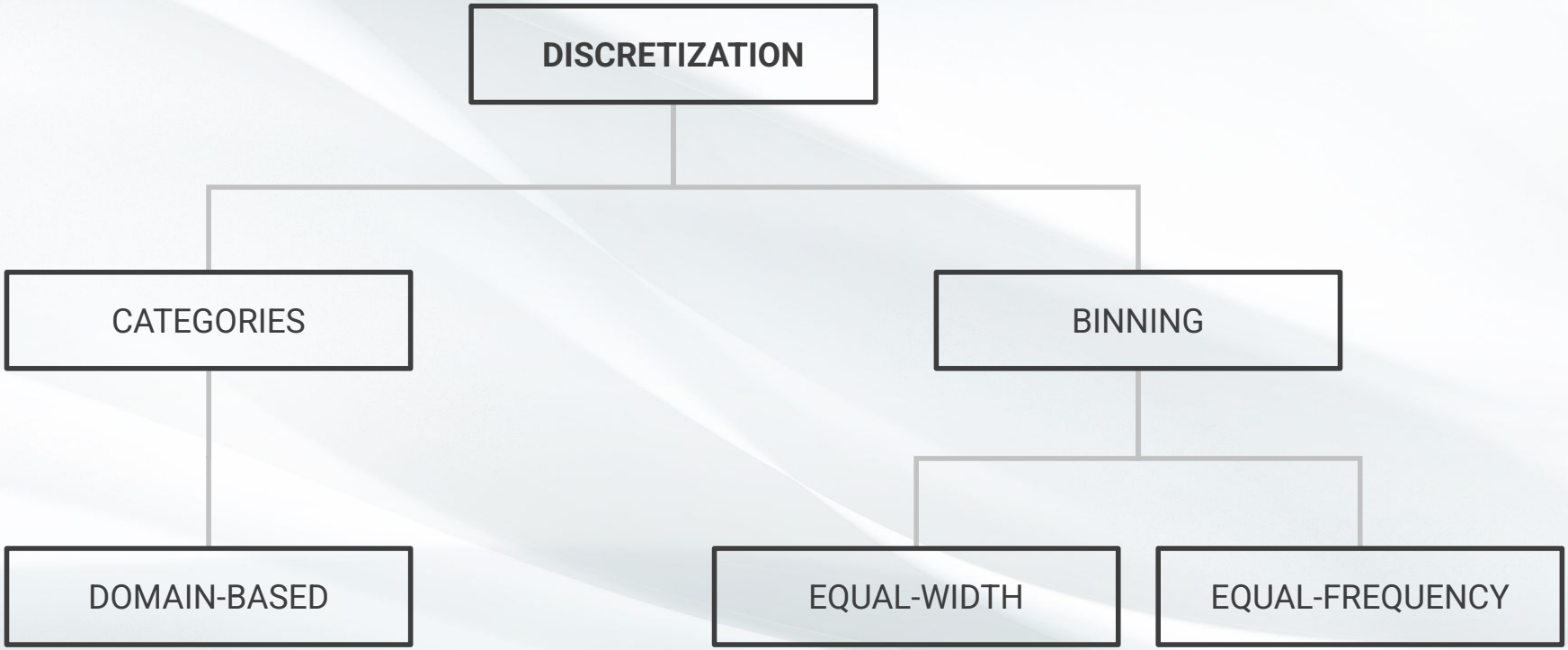
CONTINUOUS	DISCRETE
quantitative variable whose values are uncountable and can extend infinitely within a certain range.	values are countable and limited because there is a definite gap between each value in a range of values.
Example: Age (between 20 and 29) 20.5, 20.523, 24.745	Example: Age (between 20 and 29) 20, 21, 22, 23, 24, 25, 26, 27, 28, 29

(PRACTICAL EXERCISE)

CONTINUOUS VARIABLE DISCRETIZATION

PROBLEM: Some algorithms are very sensitive to continuous numbers. Sometimes, it is **not logical** to split for **each possible value** (ex. 15.5, 15.8, 15.9, …)

SOLUTION: That's why we do **DISCRETIZATION**: The process of converting a **continuous variable** into a **discrete variable**



(PRACTICAL EXERCISE)

FEATURE SPLITTING

In some cases, a feature in your dataset might benefit from being **partitioned into two or more features (columns)**.

For example, let's say a **Product Name** feature contains both a **family name** of the product (such as Ansible) and a specific **model identifier** (such as R73).

So, one of the values might be 'Ansible R73' ⇒ Ansible | R73

(PRACTICAL EXERCISE)

DIMENSIONALITY REDUCTION

PROBLEM

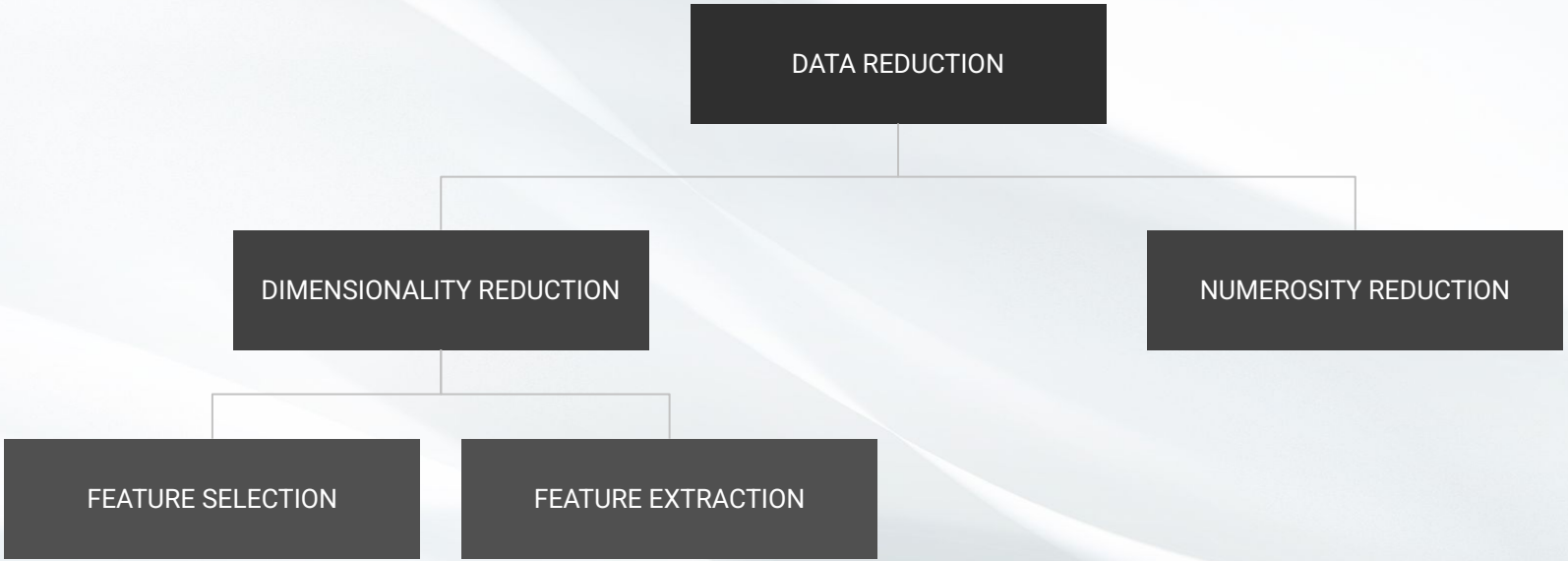
Data fed to the machine learning model may be **very large** or has many features (columns) so it becomes **harder to visualize**, takes more computation time and requires more storage

GOAL

Reducing the number of random variables

SOLUTION

Dimensionality Reduction: Process of **simplifying** a dataset by **eliminating** redundant or irrelevant features


DATA REDUCTION
DIMENSIONALITY REDUCTION
NUMEROSITY REDUCTION
FEATURE SELECTION
FEATURE EXTRACTION

DIMENSIONALITY REDUCTION

FEATURE SELECTION	You select a **subset** of the **original features**. The model trains on this subset **rather than** the whole dataset. This subset includes **relevant** and/or unique features, and excludes features deemed redundant or irrelevant to the problem. Useful in datasets having large number of features compared to actual data examples
FEATURE EXTRACTION	**Algorithms**: Chi-Square, ANOVA, Correlation, Mutual Information, Variance Threshold, Forward/Backward Selection, RFE, LASSO, Tree-based Feature Importance **(PRACTICAL EXERCISE)**

LOCATION	NUMBER OF ROOMS	NUMBER OF BATHROOMS	NAME OF THE BUILDING	AGE OF THE BUILDING

DIMENSIONALITY REDUCTION

FEATURE SELECTION	You build **new features** from the **original features**. We **combine** multiple correlated features into one. For example, if you're trying to predict Hard Disk Storage drive failure, the feature "**age**" probably correlates highly with the features "**reads**" and "**writes**," so the three can be **merged into one** derivative feature that represents the drive's logical usage.
FEATURE EXTRACTION	Feature extraction is particularly useful in computer vision applications like **image processing**. **(PRACTICAL EXERCISE)** **Algorithms**: PCA, LDA, ICA, Kernel PCA, t-SNE, UMAP, Autoencoders, Word Embeddings, Fourier/Wavelet, HOG, MFCC

ACTIVITY TIME

Preparing Data

ACTIVITY 2-5: Analyzing a Dataset Using Visualizations

LESSON 2
Preparing Data

Topic A	Collect Data
Topic B	Transform Data
Topic C	Engineer Features
Topic D	**Work with Unstructured Data**

UNSTRUCTURED DATA

- Textual Data
- Image Data
- Video Data
- Audio Data

UNSTRUCTURED DATA

- **Textual Data**
- Image Data
- Video Data
- Audio Data

TEXTUAL DATA

PROBLEM: Imagine you have 40,000 different words in your language.
If you use one-hot encoding (the old way):

The word "cat" would look like [1, 0, 0, 0, … 0] (with 39,999 zeros!)
The word "dog" would look like [0, 1, 0, 0, … 0]

This is very big and wasteful — almost all zeros.

SOLUTION: Word Embedding

WORD EMBEDDING

BEFORE: cat, dog, lion, pizza, cake, happy, sad
AFTER: (Simple example)

Word	Small Code (Vector)	Meaning
cat	[1, 0, 0]	an animal
dog	[1, 0, 0]	also an animal
lion	[1, 0, 0]	also an animal
pizza	[0, 1, 0]	food
cake	[0, 1, 0]	food
happy	[0, 0, 1]	feeling
sad	[0, 0, 1]	feeling

WORD EMBEDDING

Embedding: The process of condensing a language vocabulary into small-dimensional vectors. **(Used for speech recognition)**

- These vectors are **more efficient than traditional encoding schemes**.
  - Example: One-hot encoding a word into 40,000 possible words.
    - Would have 39,999 values of 0—**very inefficient**.
- Embedding represents a word as its own vector.
  - Usually no more than a few hundred dimensions.
  - Vectors constitute overall embedded space.
  - Vector occupies a point in this space.
- Network trains vectors on some language input to **find similarities** between words.
  - Similar words placed close together in embedded space.
  - Network can group together words in meaningful ways.
- Consider leveraging pre-trained embeddings.
- If training your own embeddings, consider the following tools:
  - Word2vec
  - Doc2vec
  - fastText

WORD EMBEDDING

BEFORE: cat, dog, lion, pizza, cake, happy, sad
AFTER: (Real example)

Word	Small Code (Vector)	Meaning
cat	[0.21, 0.85, 0.10]	an animal
dog	[0.22, 0.83, 0.12]	also an animal
lion	[0.28, 0.80, 0.05]	also an animal
pizza	[0.90, 0.15, 0.02]	food
cake	[0.88, 0.18, 0.01]	food
happy	[0.05, 0.20, 0.95]	feeling
sad	[0.02, 0.25, 0.92]	feeling

BAG OF WORDS (PRACTICAL EXERCISE)

A list of each individual word in a text without respect to grammar, punctuation, etc. **(used for text analysis) (We don't care about order)**

- Used as a way to represent text for training, before being embedded as vectors.
- Example: "I really enjoyed this movie more than I enjoyed the first one"
  - List: ['I', 'really', 'enjoyed', 'this', 'movie', 'more', 'than', 'I', 'enjoyed', 'the', 'first', 'one']
  - Dictionary: {'I': 2, 'really': 1, 'enjoyed': 2, 'this': 1, 'movie': 1, 'more': 1, 'than': 1, 'the': 1, 'first': 1, 'one': 1}
- Similar approach: bag-of-n-grams.
  - Each element is an n-gram rather than a word.
  - Commonly used to represent features for language classification.

TOKENIZATION

Tokenization partitions a document into smaller units. In fact, **n-grams** are a form of tokenization. There is also **word tokenization**, where a document is simply broken up into words, and character tokenization, where each character is partitioned.

An example of character tokenization applied to the word "**gleams**" and put into a list: ['g', 'l', 'e', 'a', 'm', 's']

(PRACTICAL EXERCISE)

N-GRAMS

Sometimes, you want to keep small pieces **stuck together**, not just single blocks. An n-gram is when you take n words at a time as a group.

Example (sentence = "I love popcorn"):

1-gram (unigram): "I", "love", "popcorn"
2-gram (bigram): "I love", "love popcorn"
3-gram (trigram): "I love popcorn"

This helps the computer know that "love popcorn" **comes together**, not just random "love" and random "cats."

STOP WORDS (PRACTICAL EXERCISE)

A stop word is any word that is **very common** in text and whose **inclusion** typically provides **little to no value**. Words like "**the**," "**a**," "**an**," and so on, are usually excised (**removed**) from a document and put into a list of stop words. When an algorithm operates on text, it can refer to this list and ignore any such words.

Other than universally common words, a stop word can also be **any word that appears very frequently** in the document in question.

So, a list of stop words might be created by taking the frequency of all words in a document and extracting the top results.

STEMMING (PRACTICAL EXERCISE)

Stemming = cutting off common suffixes/prefixes to get a “root-like” word.

running → run
played → play
quickly → quick
shadows → shadow
does → do

stemming can be crude:
leaves → leav (not “leaf”)

LEMMATIZATION

Lemmatization addresses the **shortcomings of stemming** through morphology; that is, the analysis of word parts and structures and how they change. Lemmatization attempts to **derive the lemma**, or the **canonical dictionary form** of a word. The lemma can be derived through cutting off a suffix, like in stemming, but also through more intelligent methods, like analyzing word tense and grammatical mood.

When "leaves" is lemmatized, the outcome will be: 'leaf'. This is the base dictionary form of the word. **However**, lemmatization isn't perfect, as "leaves" in a different context **might be a verb** whose lemma is actually "**leave**"—**to depart**.

UNSTRUCTURED DATA

- Textual Data
- **Image Data**
- Video Data
- Audio Data

LATENT REPRESENTATION

Turning a picture into a set of numbers that capture its essence.

In other words, **compress** this kind of data to eliminate "noise" in the image and **only retain the features** of the image that are most useful for determining patterns.

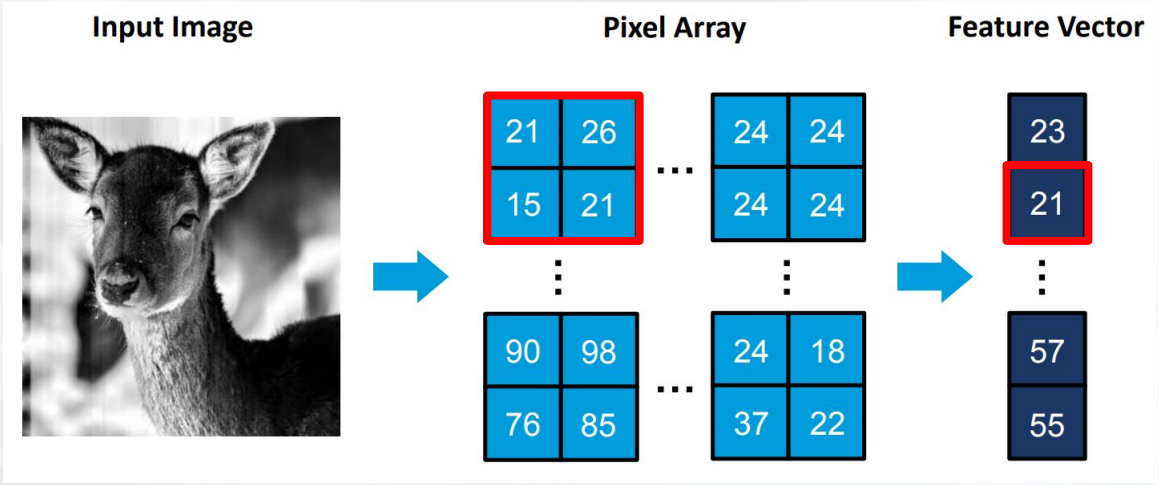


IMAGE GRAYSCALE CONVERSION

Converting an image from **Color to Grayscale** (Data size is **reduced** by ⅔ (faster to process). Although color information can be useful, most algorithms perform **just fine** with grayscale images

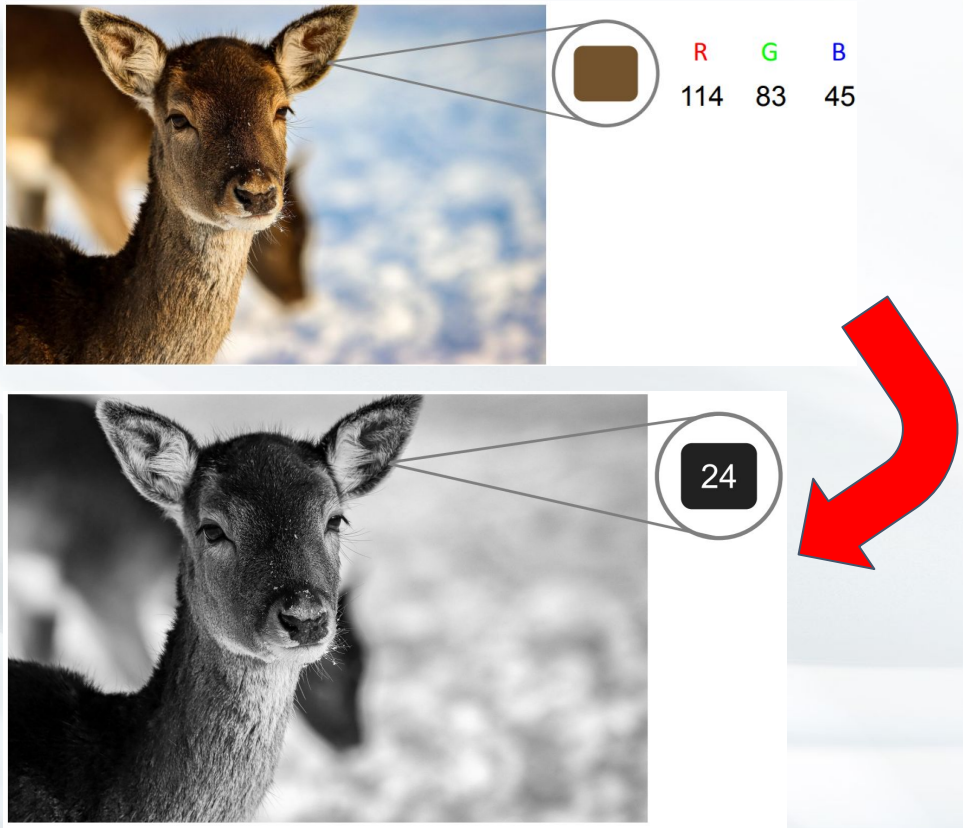


IMAGE SCALING





IMAGE RESHAPING

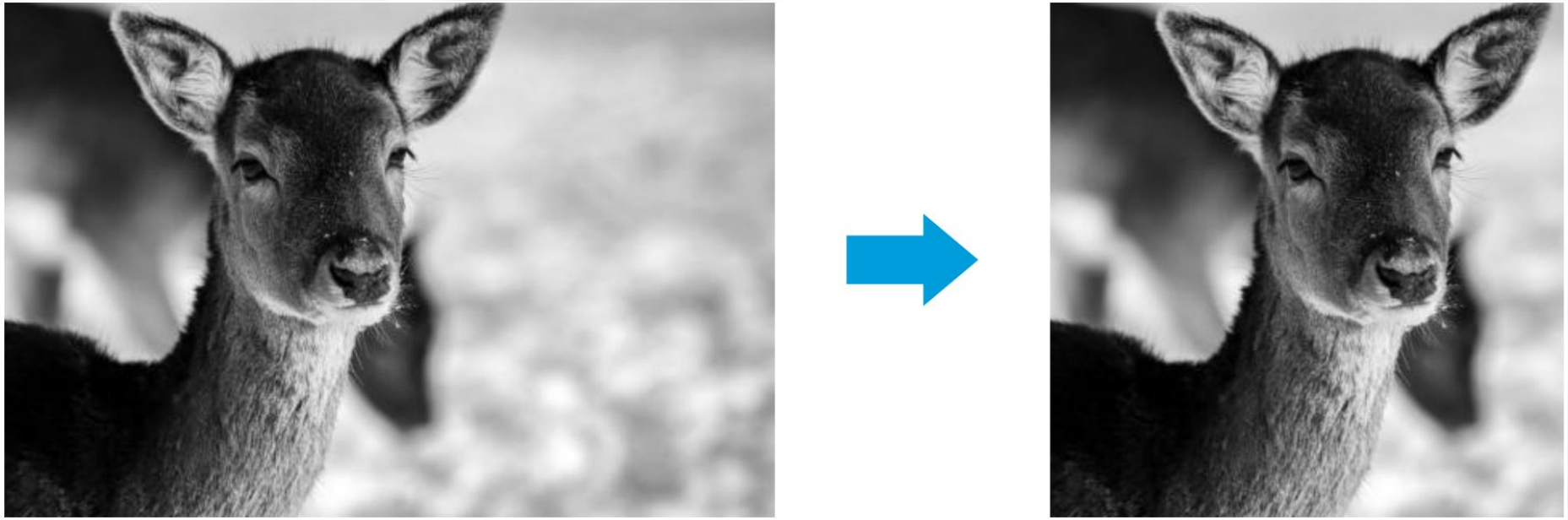


IMAGE PERTURBATION

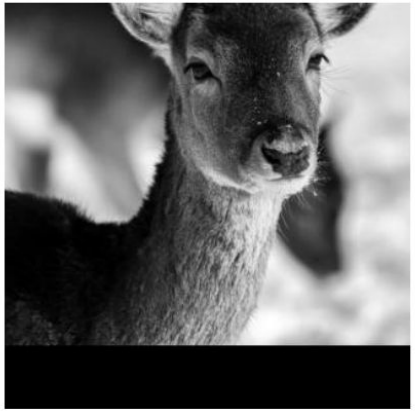
Flip (Horizontal)



Rotate (CCW 30°)



Offset (*y* 50px)



Add Noise (Random)



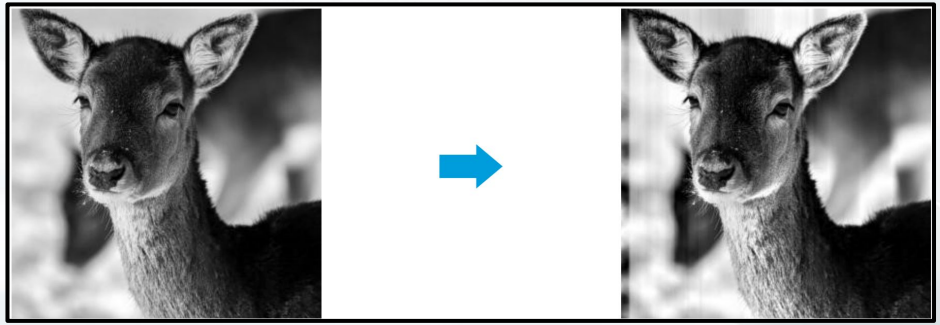
IMAGE NORMALIZATION AND STANDARDIZATION

Just like we did Normalization and Standardization for datasets before.
In this example, the outcome of normalization is **not dramatically different**, at least from a human point of view.

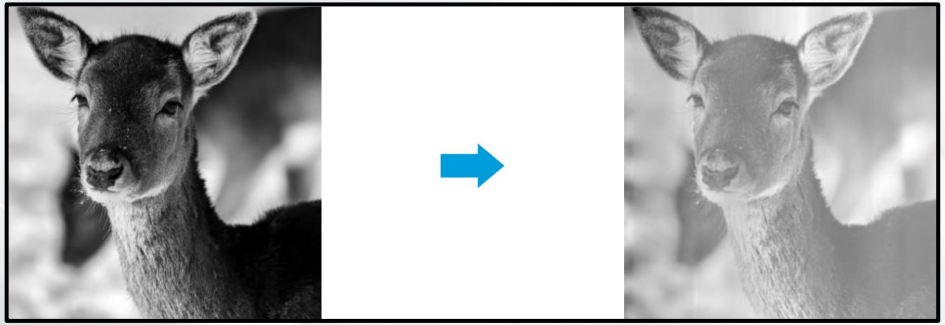


IMAGE NORMALIZATION AND STANDARDIZATION

Just like we did Normalization and Standardization for datasets before.
In this example, the outcome of normalization is not dramatically different, at least from a human point of view.



Normalized Image



Standardized Image

UNSTRUCTURED DATA

- Textual Data
- Image Data
- **Video Data**
- Audio Data

VIDEO DATA PREPROCESSING

Frame Rate	If your video is 10 minutes long and shot at 60 fps, it will contain 36,000 frames (10 min * 60 sec * 60 fps)
Data Storage	
Resolution Scaling	Solution
Consistent Transformation	Reduce frame rate from 60 fps to 30 fps.
Frame Sequence	

VIDEO DATA PREPROCESSING

Frame Rate	The amount of storage space it takes to hold all of the frames from a video as individual image files will almost always be greater than the storage space taken up by the video file itself
Data Storage	
Resolution Scaling	
Consistent Transformation	
Frame Sequence	

VIDEO DATA PREPROCESSING

Frame Rate	You should almost always **scale down** video frames to a much smaller resolution. Even **low-resolution** video can slow a network down due to the sheer number of frames it must look at. Reduced resolution from 1920×1080 to 180×180.
Data Storage	
Resolution Scaling	
Consistent Transformation	
Frame Sequence	

VIDEO DATA PREPROCESSING

Frame Rate	• Cropped to a 1:1 square shape • Converted to grayscale • Rotated 10 degrees clockwise • Flipped vertically • Applied blur effect
Data Storage	
Resolution Scaling	
Consistent Transformation	
Frame Sequence	

VIDEO DATA PREPROCESSING

Frame Rate	Neural networks will **need to consider** video frames within an **overall sequence**. Make sure you are **naming and categorizing** frames in a sequence, such as 000001.jpg, 000002.jpg, and so on. An **iterative naming** scheme like this will keep the frames in the **correct order**.
Data Storage	
Resolution Scaling	
Consistent Transformation	
Frame Sequence	

UNSTRUCTURED DATA

- Textual Data
- Image Data
- Video Data
- **Audio Data**

AUDIO DATA PRIMER

Like image data, audio data **must be transformed before** it can be processed by a neural network or other machine learning model.

First, you need to understand the three aspects of sound waves that are most relevant to audio processing:

- **Amplitude**: The magnitude of the signal, usually measured in **decibels** (dB)
- **Frequency**: The number of times a sound wave is repeated per second, usually measured in hertz (Hz)
- **Time**: The duration of the signal, usually measured in seconds

FOURIER TRANSFORMATION

Problem: If you record a chord from a piano (C + E + G): In the time domain, it just looks like a complex wave — lots of ups and downs.
Your eyes can't tell there are three notes inside, let alone which ones.
Solution: Fourier Transformation, so we move from **time-domain** (raw waveform) → **frequency-domain** (spectrum).

Imagine mixing different colors of paint (red + blue + green).
If I just look at the paint, I see one color.
Fourier Transform is like a magic tool that says:
"This paint has 50% red, 30% blue, 20% green."
SIMULATOR

AUDIO DATA PREPROCESSING

Reduce Sampling Rate	Although 44.1 kHz is a good sampling rate for audio playback, neural networks **can work** with **lower sampling** sizes. 22.05 kHz is good enough for most neural networks. Reducing sampling rate can **save storage** space and processing time, as **the smaller** the sampling rate, the **less information** there is.
Reduce bit depth/bit rate	
Convert stereo to mono	
Normalize/standardize	
Add or remove silence	

AUDIO DATA PREPROCESSING

Reduce Sampling Rate	Bit depth is another dimension by which the amount of information in an audio file is measured. It is essentially the number of binary digits that represent one sample. **So, you may wish to reduce either, or both**
Reduce bit depth/bit rate	
Convert stereo to mono	
Normalize/standardize	
Add or remove silence	

AUDIO DATA PREPROCESSING

Reduce Sampling Rate	The source audio may have sound recorded in **two separate channels** (stereo sound). It's usually beneficial to convert stereo tracks into mono to **reduce complexity**. A neural network may not need separate channels to perform its assigned task, assuming no important spatial information will be lost.
Reduce bit depth/bit rate	
Convert stereo to mono	
Normalize/standardize	
Add or remove silence	

AUDIO DATA PREPROCESSING

Reduce Sampling Rate	You can apply these **familiar feature scaling** techniques to audio just like you can with other types of data. The purpose remains the same: to **emphasize the distribution of values rather than absolute values.**
Reduce bit depth/bit rate	
Resolution Scaling	
Normalize/standardize	
Add or remove silence	

AUDIO DATA PREPROCESSING

Reduce Sampling Rate	Adding and removing silence serve a **similar purpose** as reshaping or perturbing an image; Most neural networks expect audio input to be the same length, so you might want to add or **remove silence** to **short or long** files as necessary
Reduce bit depth/bit rate	
Resolution Scaling	
Normalize/standardize	
Add or remove silence	

ACTIVITY TIME

Preparing Data

ACTIVITY 2-6

NOTE: In the lab, Activity 2-6 is not available, you just have to continue executing the cell entitled
Split the Data Into Training and Testing Sets and Labels

LESSON 2 SUMMARY ACTIVITY

In your machine learning projects, what data sources will you use, and what sorts of tasks will be particularly challenging to collect, aggregate, and prepare that data for machine learning?

Will you primarily use existing data for your machine learning projects, or will you have to put new systems in place to collect the data you will use?

COURSE OUTLINES

LESSON 1	Solving Business Problems Using AI and ML
LESSON 2	Preparing Data
LESSON 3	**Training, Evaluating, and Tuning a Machine Learning Model**
LESSON 4	Building Linear Regression Models
LESSON 5	Building Forecasting Models
LESSON 6	Building Classification Models Using Regression and k-NN
LESSON 7	Building Clustering Models
LESSON 8	Building Decision Trees and Random Forests
LESSON 9	Building Support-Vector Machines
LESSON 10	Building Artificial Neural Networks
LESSON 11	Operationalizing Machine Learning Models
LESSON 12	Maintaining Machine Learning Operations

LESSON 3
Training, Evaluating, and Tuning a Machine Learning Model

Topic A **Train a Machine Learning Model**
Topic B Evaluate and Tune a Machine Learning Model

TRADITIONAL PROGRAMMING VS MACHINE LEARNING

TRADITIONAL PROGRAMMING		MACHINE LEARNING
Analyze the problem	1	Analyze the problem
Produce / revise the sample data set	2	Select algorithm and prepare datasets **(T,E)**
Write code	3	Algorithm uses training dataset to learn **(T)**
Evaluate the output	4	Apply model to evaluation dataset **(E)**
Revision if needed	5	Revision if needed
Deploy	6	Deploy
MORTGAGE / LOAN TABLE AT A BANK	**EX**	**PREDICT THE PRICE OF A HOUSE**

STEPS FOR MACHINE LEARNING MODELS



STEP 3	
1	The practitioner selects an algorithm to use
2	The practitioner feeds data into the algorithm. This is called the training data, or training sample
3	The algorithm outputs a model based on the training data.
4	The practitioner feeds the model new data that it hasn't seen before.
5	The model makes an estimation (i.e., a prediction or some other decision) about this new data.

WHAT IS AN ALGORITHM

A set of rules that defines how a problem solving operation is performed

EXAMPLE: Sort the below steps to get an algorithm

- ENTER THE ROOM
- OPEN THE DOOR
- CLOSE THE DOOR
- UNLOCK THE DOOR

DIFFERENCES AMONG MACHINE LEARNING TYPES

SUPERVISED	SEMI-SUPERVISED	UNSUPERVISED
		

MACHINE LEARNING OUTCOMES

<table>
<tr><th></th><th>LEARNING</th><th>OUTCOME</th><th>EXAMPLES</th></tr>
<tr><td rowspan="4">MACHINE LEARNING</td><td rowspan="2">SUPERVISED</td><td>REGRESSION</td><td rowspan="4">Given one or more input ⇒
predict resulting outcome variable

EXAMPLE: Given the number of years of experience an employee has in a particular field ⇒ return a value that predicts the employee's salary

Ex: Weather Forecasting, Market Forecasting, Predicting Life Expectancy

ALGORITHMS
Linear Regression - Regularized Linear Regression - Iterative Linear Regression - CART - C4.5 - SVM</td></tr>
<tr><td>CLASSIFICATION</td></tr>
<tr><td rowspan="2">UNSUPERVISED</td><td>CLUSTERING</td></tr>
<tr><td>DIMENSIONALITY REDUCTION</td></tr>
<tr><td>REINFORCEMENT</td><td></td></tr>
</table>

MACHINE LEARNING OUTCOMES

<table>
<tr><th></th><th>LEARNING</th><th>OUTCOME</th><th>EXAMPLES</th></tr>
<tr><td rowspan="4">MACHINE LEARNING</td><td rowspan="2">SUPERVISED</td><td>REGRESSION</td><td rowspan="4">Given a new instance of data ⇒ Identify the class it belongs to

EXAMPLE:
Given various images of vehicules (with labels) ⇒ Identify which one is a car and which one is a motorcycle

Ex: Fraud Detection, Image Classification, Diagnostics

ALGORITHMS
Logistic Regression - Multinomial Logistic Regression - KNN - CART - C4.5 - SVM</td></tr>
<tr><td>CLASSIFICATION</td></tr>
<tr><td rowspan="2">UNSUPERVISED</td><td>CLUSTERING</td></tr>
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MACHINE LEARNING OUTCOMES

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<tr><td rowspan="4">MACHINE LEARNING</td><td rowspan="2">SUPERVISED</td><td>REGRESSION</td><td rowspan="4">Without any prior knowledge of a dataset's structure, identify components that belong grouped together
EXAMPLE:
Given various images of vehicules (without labels) ⇒ Identify which one is a car and which one is a motorcycle

Ex. Recommender Systems (Netflix), Targeted Marketing, Customer Segmentation

ALGORITHMS
k-Means Clustering, Hierarchical Clustering (HAC and HDC)</td></tr>
<tr><td>CLASSIFICATION</td></tr>
<tr><td rowspan="2">UNSUPERVISED</td><td>CLUSTERING</td></tr>
<tr><td>DIMENSIONALITY REDUCTION</td></tr>
<tr><td>REINFORCEMENT</td><td></td></tr>
</table>

MACHINE LEARNING OUTCOMES

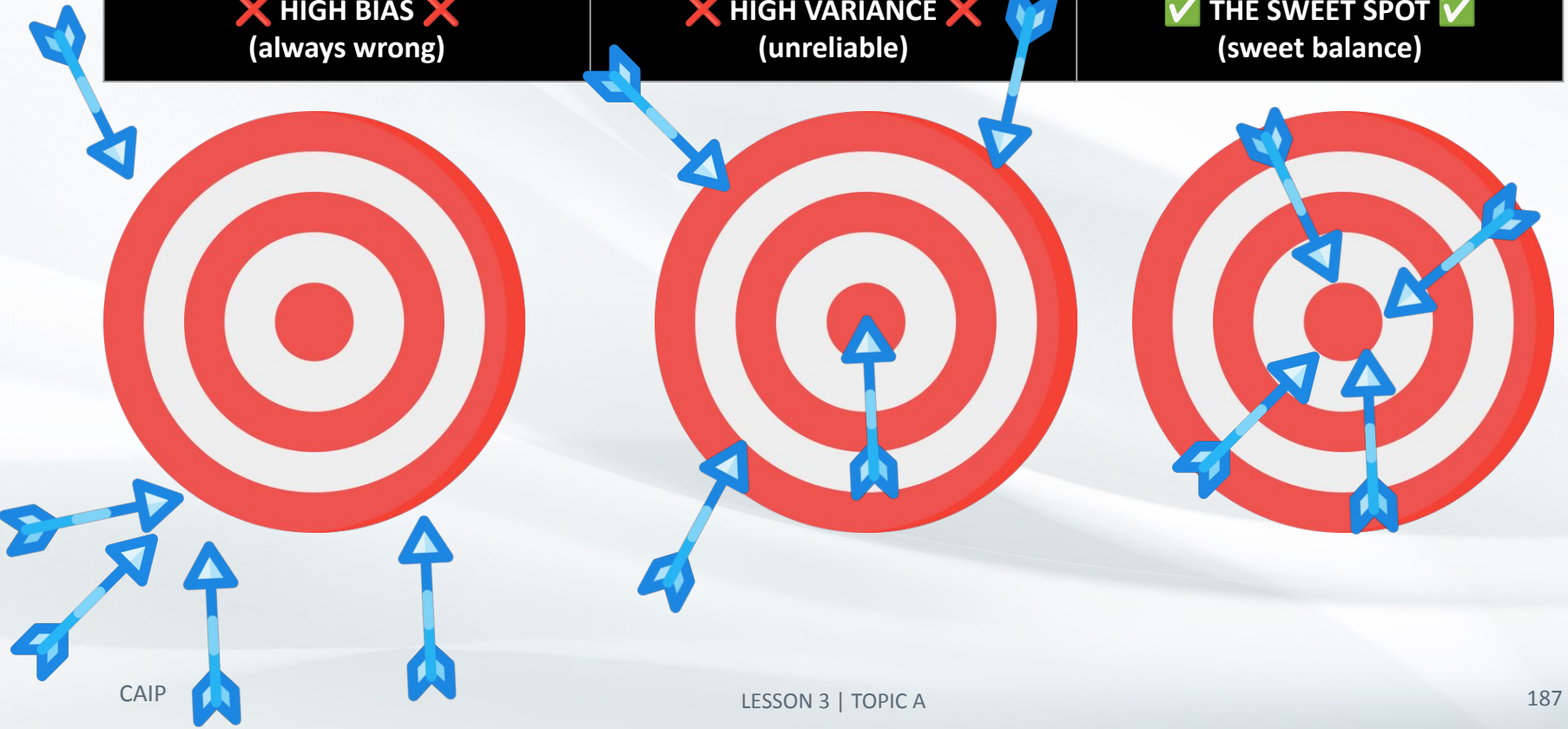
	LEARNING	OUTCOME	EXAMPLES
MACHINE LEARNING	SUPERVISED	REGRESSION	Given a machine learning model containing **multiple input variables**, **eliminate** the variables that are **not required** to produce the intended outcome EXAMPLE: Big Data Visualization
		CLASSIFICATION	
	UNSUPERVISED	CLUSTERING	
		DIMENSIONALITY REDUCTION	
	REINFORCEMENT		

MACHINE LEARNING OUTCOMES

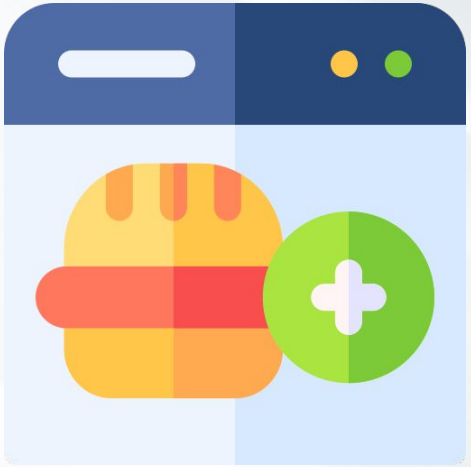
	LEARNING	OUTCOME	EXAMPLES
MACHINE LEARNING	SUPERVISED	REGRESSION	Real Time Decisions Teaching a robot to walk Learning Tasks Self Driving Car Chess Playing
		CLASSIFICATION	
	UNSUPERVISED	CLUSTERING	
		DIMENSIONALITY REDUCTION	
	REINFORCEMENT		

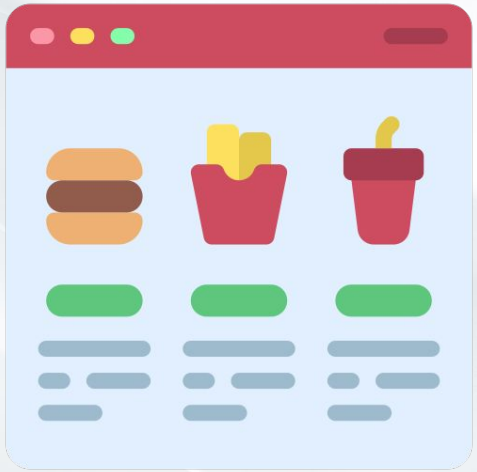
THE BIAS-VARIANCE TRADEOFF

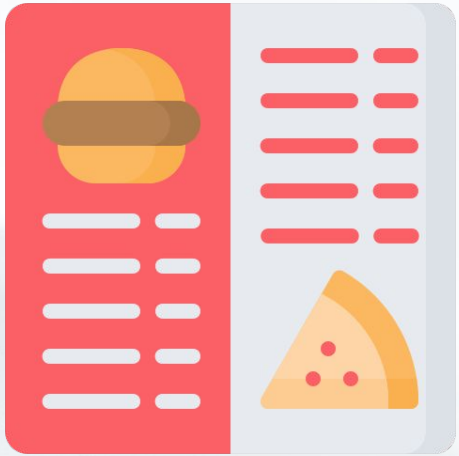
❌ HIGH BIAS ❌ (always wrong)	❌ HIGH VARIANCE ❌ (unreliable)	✅ THE SWEET SPOT ✅ (sweet balance)



THE BIAS-VARIANCE TRADEOFF

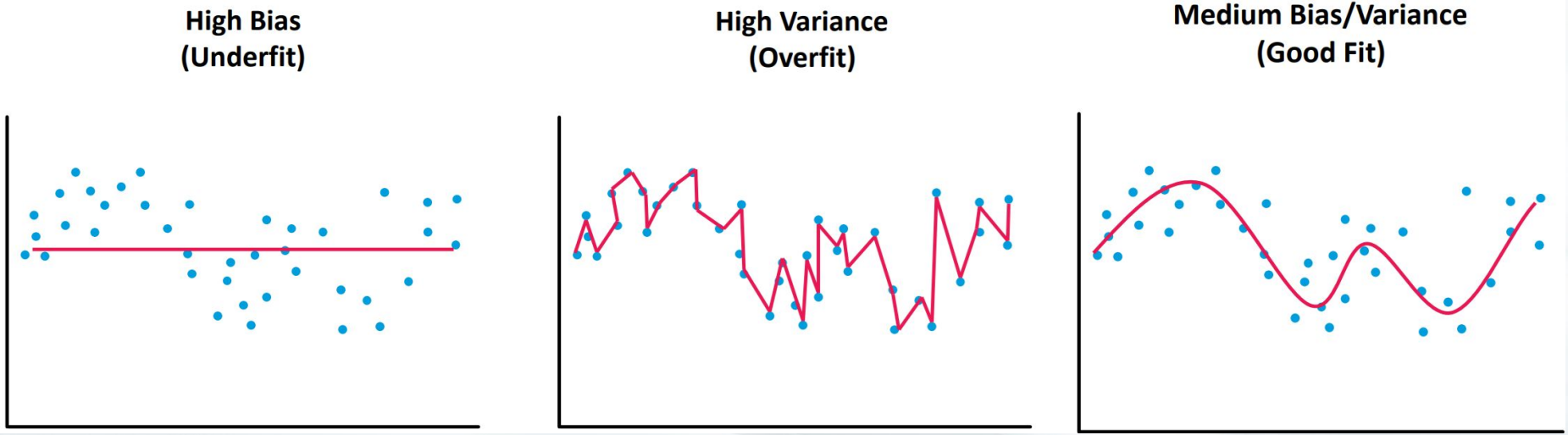






HIGH BIAS	HIGH VARIANCE	THE SWEET SPOT
Always go to the **same restaurant** **Simple** Menu = **Simple** Order **Not enough** knowledge (**underfit**) Lower predictive performance Low Variance & High Bias	Visit **a lot** of restaurants (variation) Adaptable to different datasets **Complex** Menu = **Complex** Order Start of **confusion** (**overfitting**)	Try **random** restaurants **Balanced** Menu = **Balanced** Order **Know** what to order (skillful)

THE BIAS-VARIANCE TRADEOFF



THE BIAS-VARIANCE TRADEOFF

HIGH BIAS

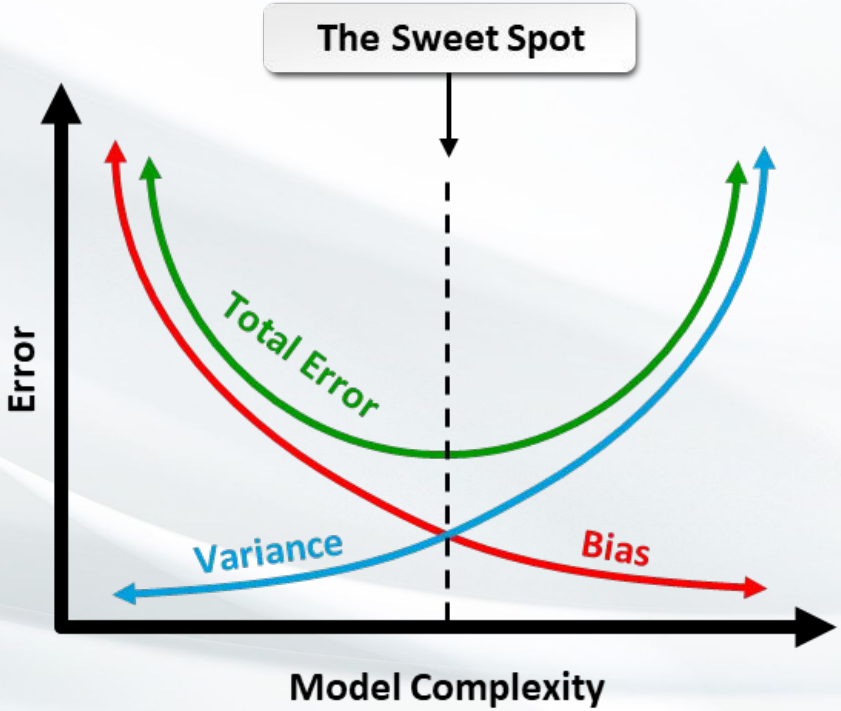
- May **underfit** the training set
- More **simplistic = Lower predictive performance**
- Less likely to be influenced by true relationships between features and target outputs

HIGH VARIANCE

- May **overfit** the training set
- More **complex**
- More likely to be influenced by false relationships between features and target outputs (“noise”)

THE SWEET SPOT

- **Good enough** fit on training datasets
- Just **complex enough**
- Skillful in finding true relationships between features and target outputs while not overly influenced by noise



BIAS

Bias is a systematic error in data analysis that results in a faulty model.

SELECTION BIAS	I want to predict the success rate of a particular treatment over time. I **SELECTED ONLY** 2 patients out of the total
REPORTING BIAS	I want to predict the success rate of a particular treatment over time. I **REPORTED ONLY** the male patients
ATTRITION BIAS	I want to predict the success rate of a particular treatment over time. I **REMOVED** the data of patients who **DROPPED** out of treatment

OVERFITTING

You may tune the model to very closely **match the unique data** in the **training** dataset, but find that the model **does not generalize well** to new datasets (**testing dataset**).

EXAMPLE: You practiced more than 100 hours on your home circuit until you mastered it and gained new skills, but these skills are not applied to other circuits



GENERALIZATION

PROBLEM	I used to be a person who only talks to people who support the same football club!
SOLUTION	A friend came to me and he told me that I need to be more **GENERAL** and accept other people supporting other football clubs



GENERALIZATION

When you want your machine learning prediction model to **work well on GENERAL data** (new data sets) and **not the one used to train on**

EXAMPLE: You want your students to **win a race** not only on your home circuit but also on **any other circuit**



HOLDOUT METHOD (One Way To Split Our Data)

PROBLEM:
If you give a child **10 math questions** to try to solve, and then you test him using one of the 10 questions, he might get 10/10 right because he probably memorized them — but he may still fail a **new test** with **different number**

SOLUTION:
You give a child **8 math questions** to try to solve, and you secretly keep **2 math questions hidden,** and then you test him using one of the **2 new questions**. If he solved them correctly, it means he really learned math, not just memorized the practice questions (8 math questions)

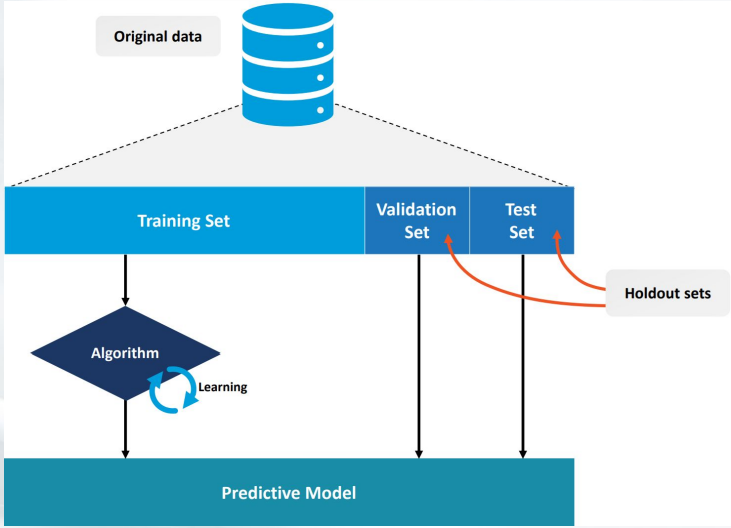
HOLDOUT METHOD (One Way To Split Our Data)



(PRACTICAL EXERCISE)

HOLDOUT METHOD (One Way To Split Our Data)

Training Set (classes)	Data you use to **train** (or fit) the model. In supervised machine learning, the learning algorithm **operates on the training set**
Validation Set (test bank)	The model **doesn't learn** from this dataset. You use it to evaluate how well the model can perform on a new dataset that it wasn't trained on (**used to tune performance**)
Test Set (real test)	This is another testing set. You should **not use it** to tune performance (like you would use a validation set), but rather only for testing the final fit of the model



MODEL PARAMETER VS HYPERPARAMETER

MODEL PARAMETER	HYPERPARAMETER
Imagine You Are Painting a Picture You add blue, red, yellow here and there. The **picture** “remembers” where you put each color, and **absorbs** the colors. They are inside the picture (**internal**). They change while you paint. The final picture is shaped by them.	Before you start painting, you decide: Which brushes to use 🖌️ (big or small brush) How many colors you will have 🎨 How long you will spend painting ⏱️ These are **external** factors
Parameter that is **derived** from the machine learning model as it undergoes the training process. Model parameters are actually "**learned**" and **Configured** by the model, and not by us (AI Practitioners)	Parameter that is set on the **algorithm itself (before I start)** and **not the learning model (not after I start)**. Hyper parameters are provided **before training**, typically by us

MODEL PARAMETER VS HYPERPARAMETER

House	Size (m^2)	Bedrooms	Location Score	Price (Label)
1	120	3	8	$250,000
2	80	2	6	$180,000
3	150	4	9	$320,000

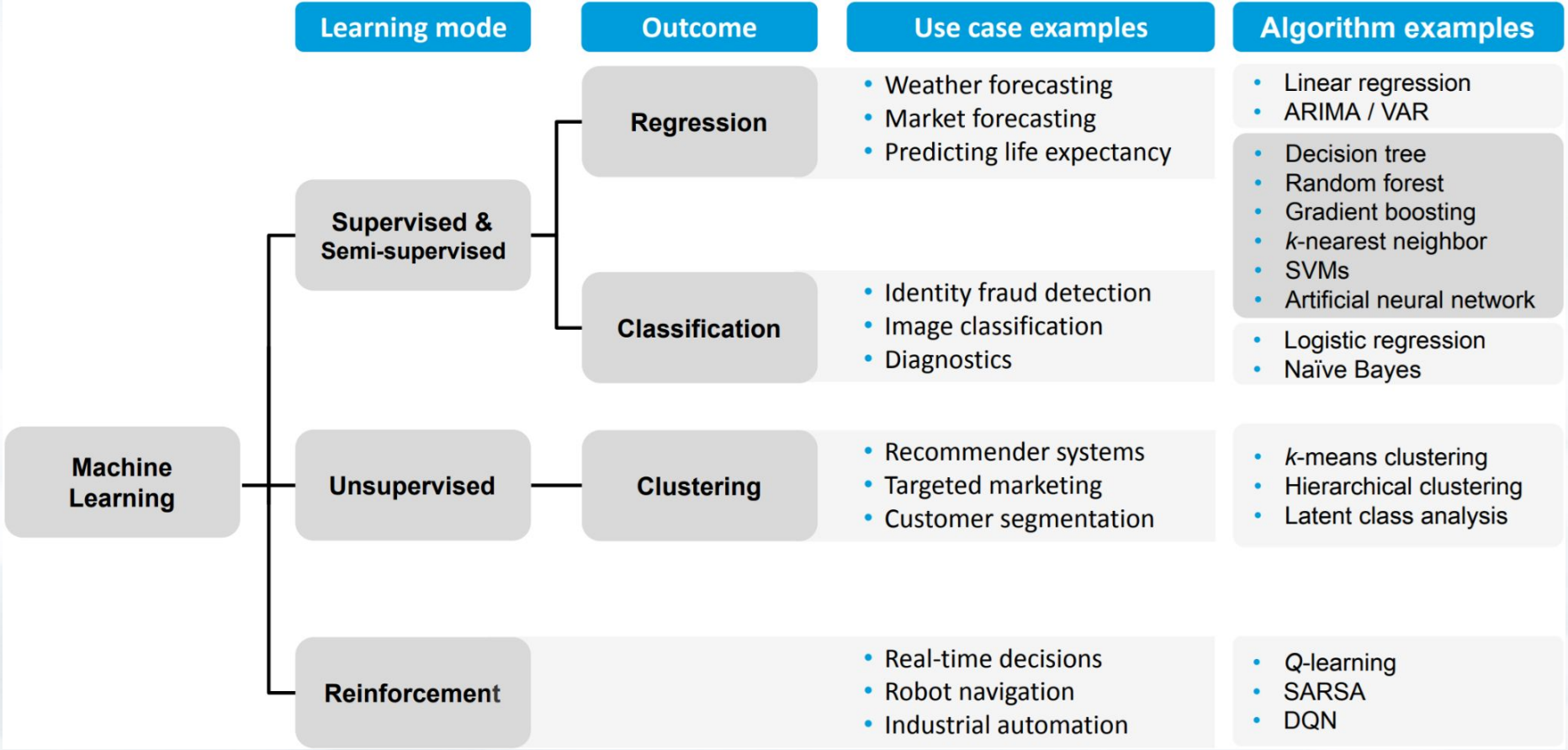
MODEL PARAMETER	HYPERPARAMETER
Weight for Size: 1200 (means each m^2 adds $1200 to price)	Learning Rate: 0.1
Weight for Bedrooms: 15000 (each bedroom adds $15,000)	Number of Iterations: 1000
Weight for Location Score: 20000 (each point adds $20,000)	Train/Test Split: 80% training, 20% holdout

DESIGN OF EXPERIMENTS

You begin with a **hypothesis**, and then you **systematically change** the variables that you **can control** to see their impact on the variables you can't directly control.

EXAMPLE: I change the number of bedrooms to see how it affects the price

INDEPENDENT VARIABLES	The variables you can directly change (Ex. Number of bedrooms)
DEPENDENT VARIABLES	The variables that you don't control, which might change indirectly as a result (Ex. Price)


Learning mode
Outcome
Use case examples
Algorithm examples
Machine Learning
Supervised & Semi-supervised
Regression
Weather forecasting
Market forecasting
Predicting life expectancy
Linear regression
ARIMA / VAR
Classification
Identity fraud detection
Image classification
Diagnostics
Decision tree
Random forest
Gradient boosting
k-nearest neighbor
SVMs
Artificial neural network
Logistic regression
Naïve Bayes
Unsupervised
Clustering
Recommender systems
Targeted marketing
Customer segmentation
k-means clustering
Hierarchical clustering
Latent class analysis
Reinforcement
Real-time decisions
Robot navigation
Industrial automation
Q-learning
SARSA
DQN

ACTIVITY TIME

Training, Evaluating, and Tuning a Machine Learning Model

ACTIVITY 3-1: Train a Machine Learning Model

NOTE: In the lab, you just have to continue executing the cell entitled **Build and test a linear regression model - Round 1**

LESSON 3
Training, Evaluating, and Tuning a Machine Learning Model

Topic A Train a Machine Learning Model

Topic B Evaluate and Tune a Machine Learning Model

ITERATIVE TUNING

A typical approach to **improving** the effectiveness of a model is to use **different samples** for training and testing, to reflect the types of challenges the model will encounter when used with real data.

As you **test the model over multiple iterations**, you'll make **refinements**, and test it again



EVALUATION METRICS

Let's imagine you are a **jury member** in a **food competition**, on which basis are you going to **evaluate the food** in front of you?

Taste, Presentation, Completeness, Consistency, …

The same thing is applied to Machine Learning Algorithms

A method of assessing the skill, performance, and characteristics of a model based on a specific measurement.

GOODHART'S LAW (Charles Goodhart)

“When a measure becomes a target, it ceases (stops) to be a good measure.”

You might optimize a model so that it performs **extremely well** on **one metric**, but it might also **perform poorly on other**, equally important metrics. The model's overall skill could suffer as a result.

Keep in mind that there is **no one objectively "perfect" metric** for every circumstance, and that you may need to consider multiple metrics in the evaluation process.

RANDOMNESS IN MACHINE LEARNING

- Two different algorithms may produce similar outcomes
- Two different algorithms may have different performance characteristics
  - RUN TIME: Algorithm A - 2 seconds | Algorithm B - 7 minutes
- One algorithm may perform best on small datasets, while another tends to work best on large datasets.

UNCERTAINTY AND HOW TO OVERCOME IT

- Running the **same** algorithm **many times** on several different samples
- Running several **different** algorithms, testing the resulting models, and picking the one that produces the best results
- Running several different algorithms, allowing each one to cast its "vote" for the answer, and taking the "wisdom of the crowd" as the answer
- Selecting the **right data** (data that is relevant to the problem)
- Reducing **noise** (data values that are incorrect or misleading, making it harder for machine learning algorithms to find the important patterns in the data)

CROSS VALIDATION

PROBLEM

Let's suppose I have **overfitting**, or a database containing 1,000 records. I cannot use the 1,000 records just to **train** my model. I won't have any data left to **test** the model. A model may **perform well on training data** but **not do so well** when it's faced with **a new dataset** it hasn't seen before (**test data**)

SOLUTION

Cross Validation: provides a way to **compare different** machine learning methods, and measure how well the model is able to generalize to new test data by **splitting** our data into training and testing

CROSS VALIDATION - FOOTBALL EXAMPLE

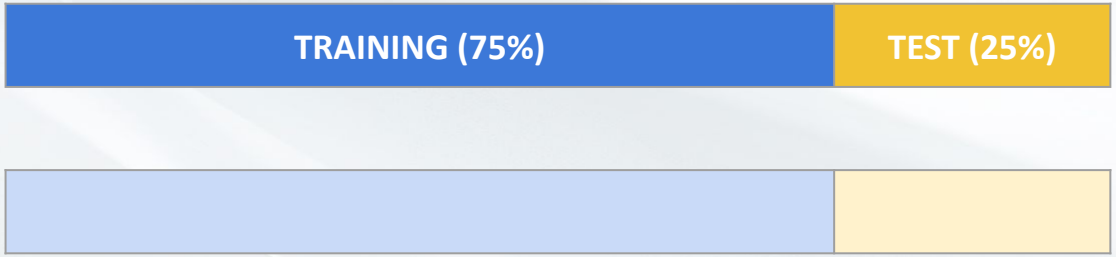
In Football, the coach tries various formations and writes down the score for each formation and then compares them and chooses which one is the best for his team



SPLITTING THE DATASET

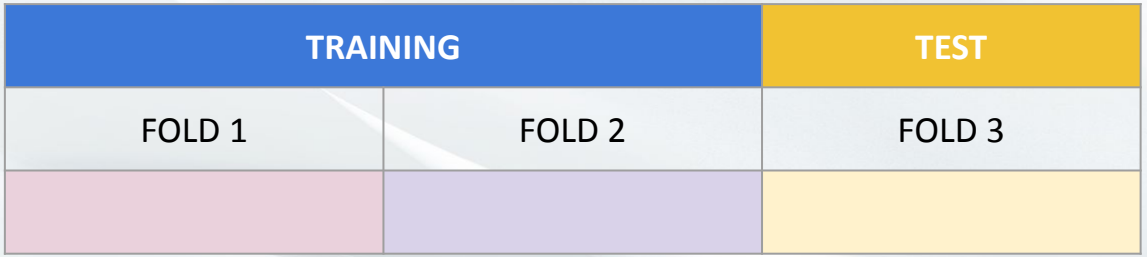
Holdout Method
k-fold Cross Validation
Stratified *k*-fold Cross Validation
Leave-*p*-out Cross Validation

Dataset is split into **2**: **training** dataset and **test** dataset



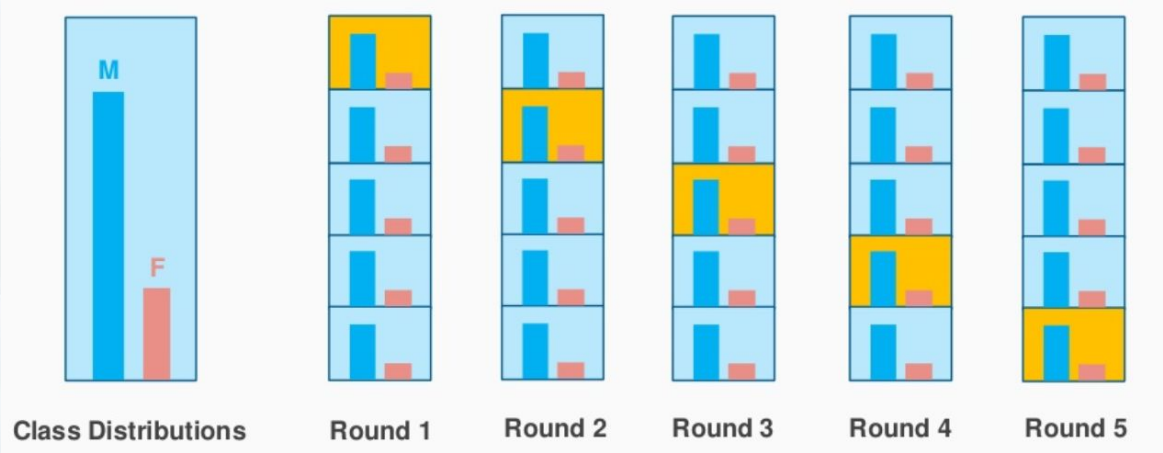
SPLITTING THE DATASET

Holdout Method	Dataset is split into k **"random"** groups (folds): **test** dataset and the remaining groups (k-1) from the **training** set Repeat k times, rotating the test fold each time. Example: 3-fold Cross Validation
***k*-fold Cross Validation**	
Stratified *k*-fold Cross Validation	
Leave-*p*-out Cross Validation	**(PRACTICAL EXERCISE)**



SPLITTING THE DATASET

Holdout Method	Splits the data into k folds, making sure each fold is an **appropriate representative** of the original data. (same proportionality, class distribution, mean, variance, etc)
k-fold Cross Validation	
Stratified *k*-fold Cross Validation	
Leave-*p*-out Cross Validation	



SPLITTING THE DATASET

Holdout Method	A k-fold cross validation method, but with k equal to all data points in the set (n). So, n − p data points are used in training, and p data points are used to test Example: p=1  Advantage: minimize bias in small datasets **(PRACTICAL EXERCISE)**
k-fold Cross Validation	
Stratified *k*-fold Cross Validation	
Leave-*p*-out Cross Validation	

FEATURE TRANSFORMATION FUNCTIONS

NORMALIZATION	Scales data items to fit within a range such as 0 to 1 NOTE: May **leave** the distribution **skewed**.
STANDARDIZATION	Converting the dataset to statistical measures by plotting them in a frequency distribution and calculate the **Z-SCORE** (The number of standard deviations from the mean)
Log	Obtained by calculating the log base 10 of x, log base e of x (ln(x)), or log base 2 of x. NOTE: It **reduces skewness** and can be used only on positive numbers
CUBE ROOT	Obtained by raising each data item to a power of 1 / 3. NOTE: Reduces skewness, but **not as strongly as** a log function.

ACTIVITY TIME

Training, Evaluating, and Tuning a Machine Learning Model

ACTIVITY 3-2: Dealing with Outliers

ACTIVITY 3-3: Scaling and Normalizing Features

ACTIVITY 3-4: Refitting and Testing the Model

NOTE: In the lab, you just have to continue executing the cell entitled **Identify outliers**

LESSON 3 SUMMARY ACTIVITY

What strategies might you employ when determining which algorithm you'll use initially?

What concerns do you have about bias in your own datasets?

COURSE OUTLINES

LESSON 1	Solving Business Problems Using AI and ML
LESSON 2	Preparing Data
LESSON 3	Training, Evaluating, and Tuning a Machine Learning Model
LESSON 4	**Building Linear Regression Models**
LESSON 5	Building Forecasting Models
LESSON 6	Building Classification Models Using Regression and k-NN
LESSON 7	Building Clustering Models
LESSON 8	Building Decision Trees and Random Forests
LESSON 9	Building Support-Vector Machines
LESSON 10	Building Artificial Neural Networks
LESSON 11	Operationalizing Machine Learning Models
LESSON 12	Maintaining Machine Learning Operations

LESSON 4
Building Linear Regression Models

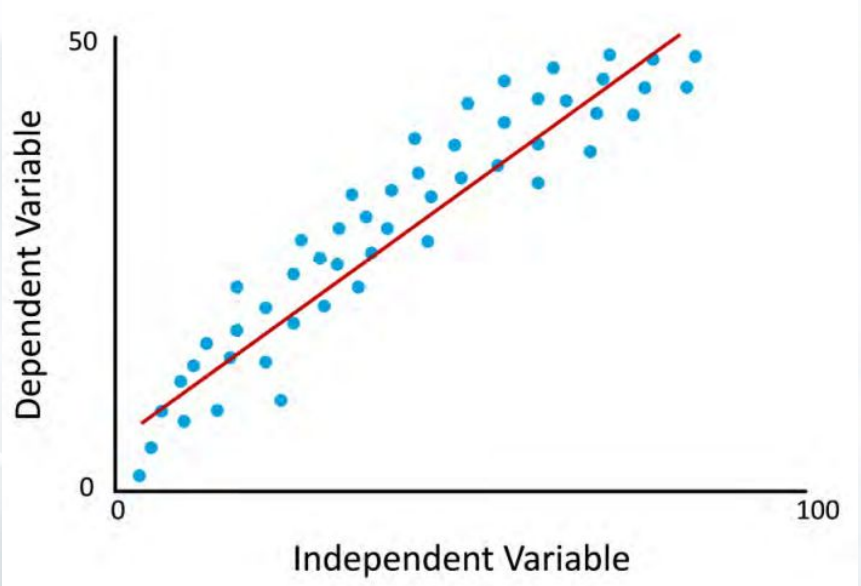
Topic A	**Build Regression Models Using Linear Algebra**
Topic B	Build Regularized Linear Regression Models
Topic C	Build Iterative Linear Regression Models

LINEAR REGRESSION

Regression analysis is the technique of identifying the **relationships between variables**.

- **Dependent Variable**: one whose variation you are interested in studying (Ex. House Price)
- **Independent Variable**: has a potential effect on the variation of dependent variables (Ex. Number of Bedrooms)

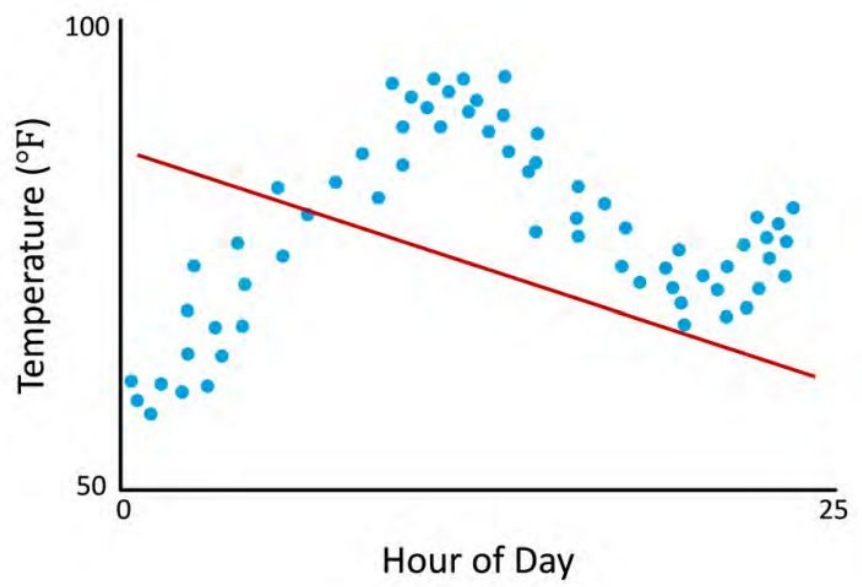
(PRACTICAL EXERCISE)

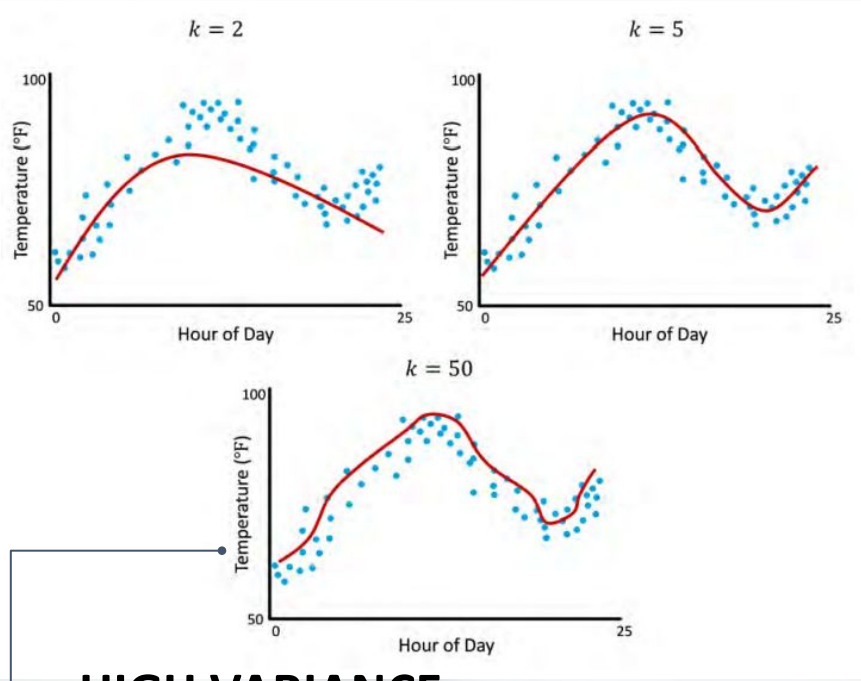


SIMULATOR	GAME

LINEAR MODEL WITH HIGHER ORDER FITS

(PRACTICAL EXERCISE)





HIGH BIAS

Underfitting - Simple Model

HIGH VARIANCE

Overfitting - Complex Model

COST FUNCTION - THE LESS, THE BETTER

Imagine you are training **alone** for your next 5K Mini Marathon

PROBLEM:

- You will **not eat well**, you will **not train well**
- you are **not measuring your timings** ⇒ you will **not know** whether you are **improving or not**

SOLUTION:

I hire a personal trainer. He **measures my timings**, and he tells me whether I am improving or not, and whether I need to push more or not.

The expected time to finish the 5K: 30 min
My actual time that it took me to finish the 5k: 45 min
Difference: late arrival by 15 min (I need to push more)

OUR TRAINER = COST FUNCTION

PROBLEM	Our model is not predicting the results perfectly
SOLUTION	**Minimize** the difference (amount of error) between the predicted value and the actual value

ERROR	=	Actual Value	-	Predicted Value
ERROR	=	y	-	ŷ (Pronounced: y hat)
ERROR	=	32°C	-	35°C

COST FUNCTION (PRACTICAL EXERCISE)

MEAN SQUARED ERROR (MSE)

The value of MSE is **always positive or greater than zero**. A value **close to zero** will represent **better quality** of the estimator / predictor (regression model).

Once the error (difference) between the predicted value and the actual value is calculated, that error is squared. Then, the average of these squared errors is taken.

$$J(\boldsymbol{\theta}) = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

MEAN ABSOLUTE ERROR (MAE)

Calculates the average difference in without considering the sign (i.e., positive or negative) of those values

$$J(\boldsymbol{\theta}) = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

COST FUNCTION - THE LESS, THE BETTER

ACTUAL VALUE y	PREDICTED VALUE ŷ	ERROR DIFFERENCE	MSE (MEAN SQUARE ERROR)	MAE (MEAN ABSOLUTE ERROR)
100	130	-30	900	30
150	170	-20	400	20
200	220	-20	400	20
250	260	-10	100	10
300	325	-25	625	25
		RESULT (MEAN)	**485**	**21**

COST FUNCTION - WHICH ONE IS BETTER

MSE minimizes errors that are less than 1	$0.5^2 = 0.25$	$0.25 < 0.5$
MSE exaggerates errors that are greater than 1	$5^2 = 25$	$25 > 5$

If you want big mistakes to count a lot, use MSE.
If you want all mistakes to count the same, use MAE.

R vs R^2

R describes the relationship, and R^2 grades the model.
Use R to choose features and R^2 to judge the model.

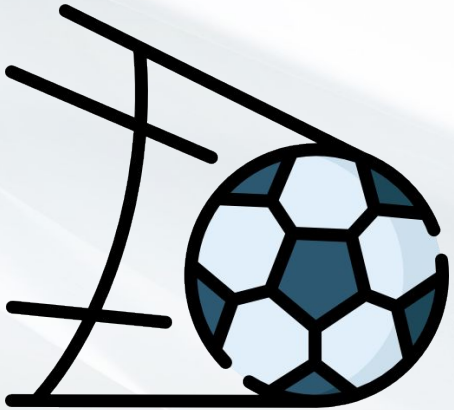
COEFFICIENT OF DETERMINATION - R^2 (SCORE)

PROBLEM: How to know whether my model is good or not?
SOLUTION: Use **R^2 (SIMULATOR 1 | SIMULATOR2)**
Scoring method for evaluating the performance of a linear regression model. **It compares the actual values to the mean**

EXAMPLE: **R^2 is always between 0 and 1**. If the regression line were able to perfectly pass through every data point, the R^2 would be 1; on the other hand, as the line misses more and more points, its R^2 decreases, as it becomes less and less able to explain the variance.

NOTE: It's usually preferable to **minimize a cost function** like MSE rather than attempt to **optimize R^2**

COMPARISON: R^2 AND MSE





MACHINE LEARNING	R^2	MSE
FOOTBALL	GOALS	YELLOW CARDS
OBJECTIVE	↗️	↘️

COMPARISON: R^2 AND MSE

$$\mathrm{MSE}(y, \hat{y}) = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$$\mathrm{R}^2(y, \hat{y}) = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

EXAMPLE: If we are comparing 2 models (algorithms). One predicts the **INCOME** (Ex. USD 20,000) and the other one predicts the **AGE** (Ex. 40)

Since the Income and the Age are on a different scale (20,000 VS 40), then R^2 will make it easier to state which model is performing better (because it is **normalized**)

DISADVANTAGES OF LINEAR REGRESSION

- Bad with large datasets and it will lead to memory issues (computational cost of inverse matrix calculations)
- it is prone to **overfitting** (my model is giving me **excellent** results for my **training dataset**, but giving me **very bad** results for my **test dataset**)

SOLUTION: Apply **regularization** (next topic)

ACTIVITY TIME

Building Linear Regression Models

LINEAR REGRESSION USING GOOGLE SHEETS

ACTIVITY 4-1: Building a Regression Model Using Linear Algebra (DATASET)

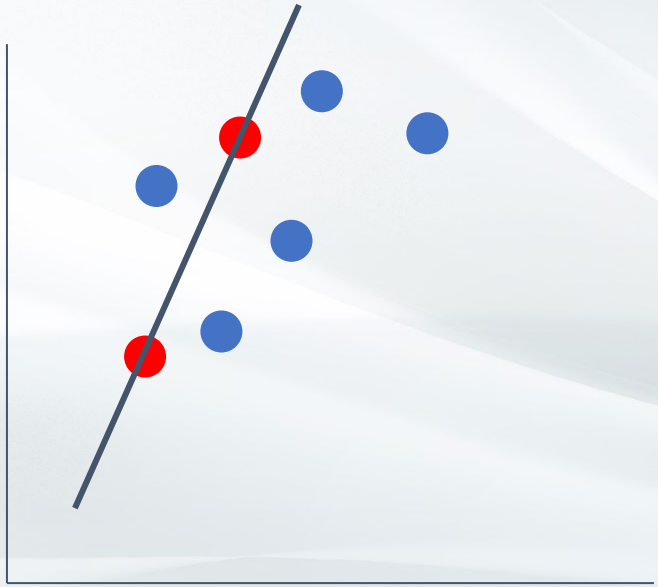
ACTIVITY 4-2: (Optional) Building a Linear Regression Model to Predict Diabetes Progression

LESSON 4
Building Linear Regression Models

Topic A	Build Regression Models Using Linear Algebra
Topic B	**Build Regularized Linear Regression Models**
Topic C	Build Iterative Linear Regression Models

REGULARIZATION TECHNIQUES

PROBLEM: Normal Linear regression can be prone to **overfitting** if the **error value is zero** (line fits the training data perfectly), or if **MSE and MAE values are very high** (indicating poor generalization).



TRAINING DATA

TESTING DATA

NOTE: REGULARIZATION *is one way to achieve our goal which is* ***GENERALIZATION***

REGULARIZATION TECHNIQUES



GOAL

*Increase **BIAS** to improve **VARIANCE** (GENERALIZATION capability)*

- TRAINING DATA
- TESTING DATA

REGULARIZATION TECHNIQUES

PROBLEM: Normal Linear regression is prone to overfitting.
SOLUTION: We do Regularization (improve the skill of your machine learning model - **Regularization Improves Generalization**)

It is the process of simplifying a model by applying **constraints** to the model parameters. It does this by applying the hyperparameter.

"For every extra mistake you make, you have to pay a small fine."

You only perform regularization **during training**, and **NOT** after training and while evaluating the model's performance

REGULARIZATION TECHNIQUES

Ridge Regression	Ridge regression uses a mathematical function called an **ℓ_2 norm** to implement its regularization term (minimize the effect of features). The ℓ_2 norm is the **sum of square** coefficients, and the **objective is to minimize it (doesn't reach 0)**.
Lasso Regression	In other words, the main idea behind Ridge Regression is to find a **new line** that **doesn't fit** the **Training Data**. This helps to keep the model parameter weights small, **reducing overfitting**.
Elastic Net Regression	$$J(\boldsymbol{\theta}) = \mathrm{MSE}(\boldsymbol{\theta}) + \lambda \sum_{i=1}^{d} \theta_i^2$$

REGULARIZATION TECHNIQUES

Ridge Regression	**PROBLEM**: I have a lot of features (variables) and I need to enhance my model **SOLUTION**: LASSO: I need to set the least important of them to 0
Lasso Regression	Lasso regression uses the ℓ_1 norm to implement its regularization term (removes features). The ℓ_1 norm **forces the coefficients of the least relevant features to 0**—in other words, **removing them** from the model
Elastic Net Regression	

$$J(\boldsymbol{\theta}) = \text{MSE}(\boldsymbol{\theta}) + \lambda \sum_{i=1}^{d} |\theta_i|$$

REGULARIZATION TECHNIQUES

Ridge Regression	If you are **unsure** whether to use Ridge or Lasso, how about **using both of them**?
Lasso Regression	Elastic net regression uses a **weighted average of both ridge and lasso regression** as part of its regularization term. It is therefore an attempt to leverage the best of both ℓ1 and ℓ2 norms
Elastic Net Regression	$$J(\boldsymbol{\theta}) = \text{MSE}(\boldsymbol{\theta}) + \lambda\left(\frac{1-\alpha}{2}\sum_{i=1}^{d}\theta_i^2 + \alpha\sum_{i=1}^{d}\lvert\theta_i\rvert\right)$$

REGULARIZATION TECHNIQUES

Ridge Regression	An α of 0 uses the Ridge ℓ2 regularization term $$J(\boldsymbol{\theta}) = \text{MSE}(\boldsymbol{\theta}) + \lambda\left(\frac{1-0}{2}\sum_{i=1}^{d}\theta_i^2 + 0\sum_{i=1}^{d}\lvert\theta_i\rvert\right)$$
Lasso Regression	An α of 1 uses the Lasso ℓ1 regularization term $$J(\boldsymbol{\theta}) = \text{MSE}(\boldsymbol{\theta}) + \lambda\left(\frac{1-1}{2}\sum_{i=1}^{d}\theta_i^2 + 1\sum_{i=1}^{d}\lvert\theta_i\rvert\right)$$
Elastic Net Regression	The objective is to **find a value between 0 and 1** that optimizes the regularization of the model.

COMPARISON: RIDGE VS LASSO VS ELASTIC NET

RIDGE REGRESSION	LASSO REGRESSION	ELASTIC NET
Suitable in datasets featuring a **large number of features**	Suitable in datasets that have only a **small or moderate number of features**	Suitable in datasets that have only a **small or moderate number of features**
The value minimizes **but will not reach** 0 (reduce overfitting)	The value minimizes **until it reaches** 0	**value between 0 and 1** that optimizes the regularization of the model
All features will **stay the same**	Features will be **reduced / removed** (feature selection)	better in situations where **multiple features exhibit high correlation**

ACTIVITY TIME

Building Linear Regression Models

ACTIVITY 4-3: Building a Regularized Linear Regression Model

LESSON 4
Building Linear Regression Models

Topic A	Build Regression Models Using Linear Algebra
Topic B	Build Regularized Linear Regression Models
Topic C	**Build Iterative Linear Regression Models**

WHY WE NEED THE ITERATIVE APPROACH

The closed-form approach to linear regression, whether regularized or not, is **good at solving problems** in certain types of datasets.

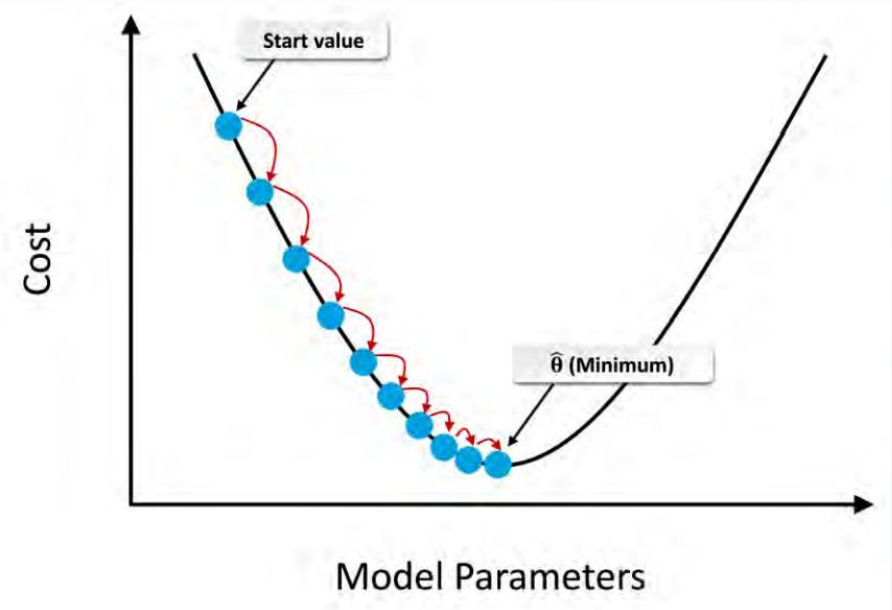
PROBLEM: However, you may encounter datasets on which the **normal equation struggles** (Large datasets, large features and large number of records).

SOLUTION: We use **iterative approaches**.
(PRACTICAL EXERCISE)

GRADIENT DESCENT (COST OPTIMIZATION)

Used to **tune the model parameters** over **several iterations** by taking gradual "**steps**" **down** a slope, toward a minimum error value.

1. **Select a value** for θ by starting at a random location
2. Calculate the **partial derivative** of each model parameter with respect to the cost function
3. **Update each value of θ**

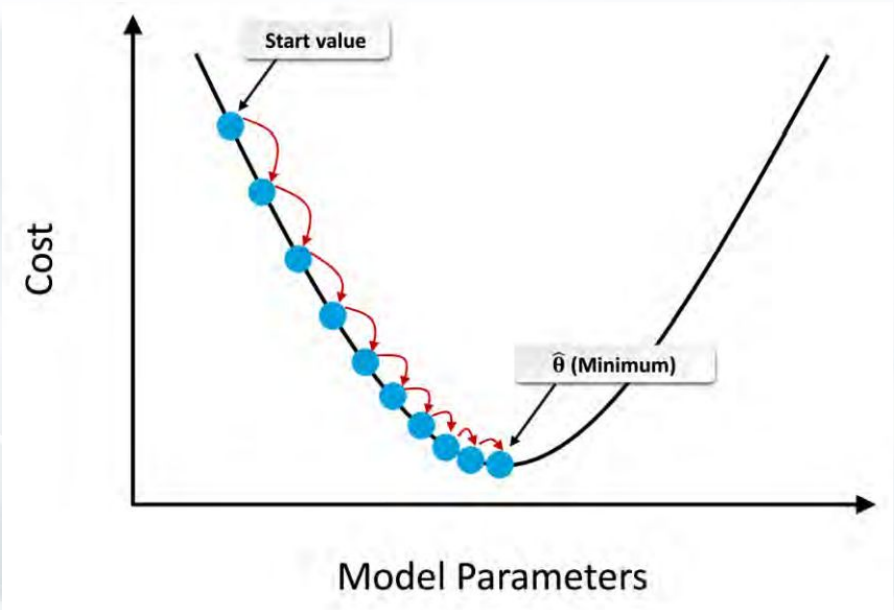


SIMULATOR 1	SIMULATOR 2	SIMULATOR 3

GRADIENT DESCENT

Gradient descent **performs better**, than the normal equation, in datasets that have a **large number of features** or examples, because calculating the inverse of a very large matrix of values **takes a great deal of memory**

GOAL: We use GRADIENT DESCENT to minimize MSE

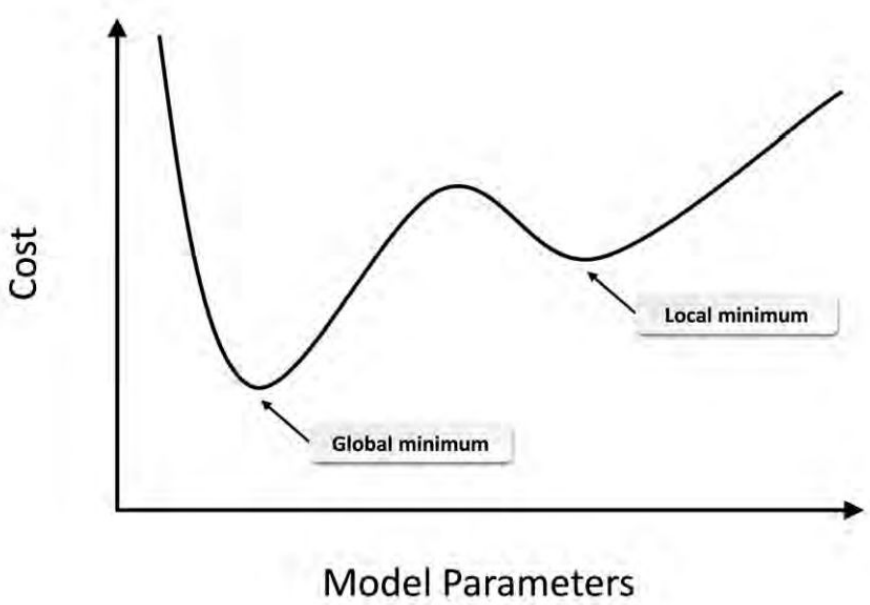


GLOBAL MINIMUM VS. LOCAL MINIMA

LOCAL MINIMA (MINIMUM)
Lowest point for any one valley (may be several)

GLOBAL MINIMUM
Lowest point overall
(only one)

EXAMPLE: MSE Cost Function is convex (Valley with one minimum). Others are non-convex

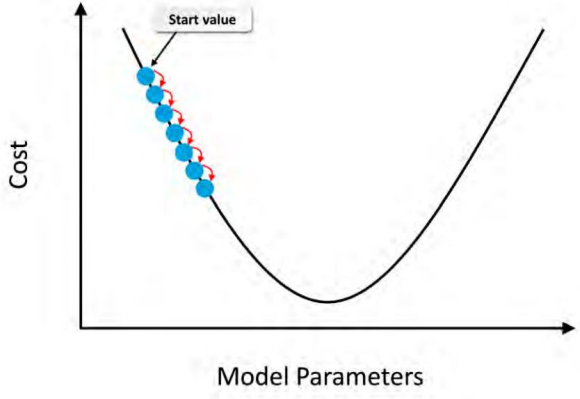


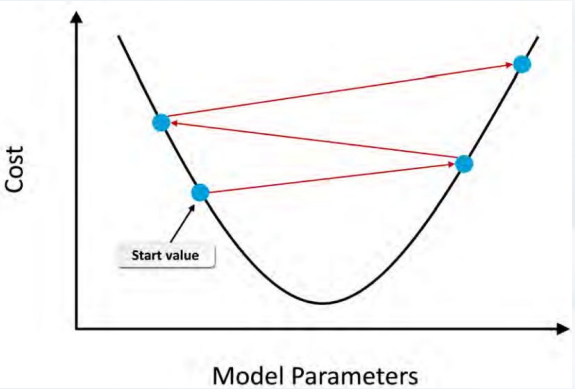
LEARNING RATE

The size of each "step" down the valley

If the learning rate is **small**, **more "steps"** will be required to converge at the minimum.

If the learning rate is **too large**, each "step" may jump to the opposite curve and actually end up higher than it was before.



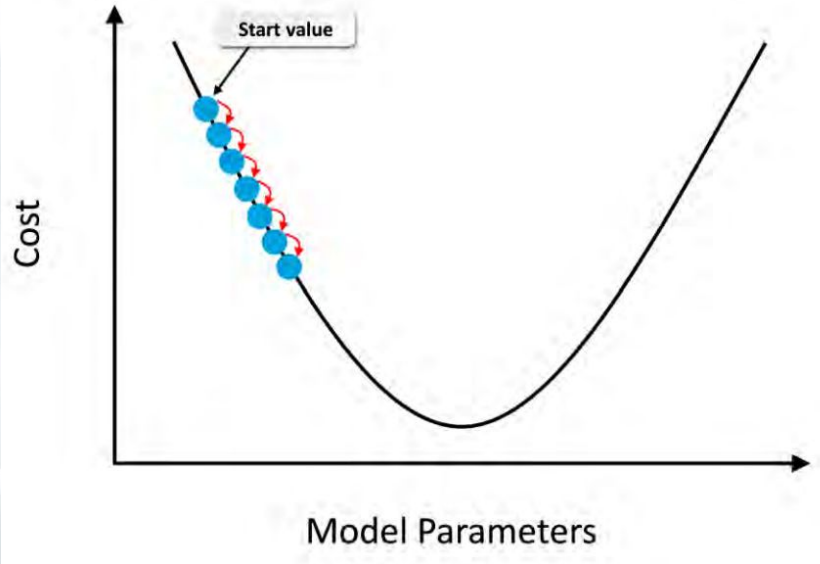


LEARNING RATE

The size of each "step" down the valley

The learning rate is a **hyperparameter (α)** of the model that you must tune to get the best results.

If the learning rate is **small**, **more "steps"** will be required to converge at the minimum. Too small, and the descent could end up taking a very long time. SIMULATOR



GRADIENT DESCENT TECHNIQUES

BATCH GRADIENT DESCENT (SIMULATOR) (ANIMATION)	• Uses **entire training** set to calculate stepwise gradients. • Each example has its own gradient. • Model params updated based on average gradient of all examples. • Not commonly used; takes a **very long time** to compute gradients for big data
STOCHASTIC GRADIENT DESCENT (ANIMATION)	• Selects **example at random** and computes gradient at every step. • **Much quicker** than BGD. • Example update may cancel out updates from previous example. • Progress is made slowly in correct direction.
STOCHASTIC AVERAGE GRADIENT	• Similar to SGD, but has a **memory** of previously computed gradients. • Converges faster than SGD. • **Only supports ℓ_2 regularization.** • May have performance issues in large datasets.
MINI-BATCH GRADIENT DESCENT	• Middle ground between BGD and SGD. • Selects a **group of examples at random**. • Steps in direction of average gradient from all examples in group. • May perform better than SGD, especially when backed by GPUs.

ACTIVITY TIME

Building Linear Regression Models

ACTIVITY 4-4: Building an Iterative Linear Regression Model

DEFINITION eta0: if our improvement in the error over a certain number of steps is not changed by more than 0.05 then we will stop, because we are close enough

LESSON 4 SUMMARY ACTIVITY

What type of data in your organization do you think might be conducive to a linear regression analysis?

Do you believe your data is best approached by a closed-form solution or an iterative one? Why?

COURSE OUTLINES

LESSON 1	Solving Business Problems Using AI and ML
LESSON 2	Preparing Data
LESSON 3	Training, Evaluating, and Tuning a Machine Learning Model
LESSON 4	Building Linear Regression Models
LESSON 5	**Building Forecasting Models**
LESSON 6	Building Classification Models Using Regression and k-NN
LESSON 7	Building Clustering Models
LESSON 8	Building Decision Trees and Random Forests
LESSON 9	Building Support-Vector Machines
LESSON 10	Building Artificial Neural Networks
LESSON 11	Operationalizing Machine Learning Models
LESSON 12	Maintaining Machine Learning Operations

LESSON 5
Building Forecasting Models

Topic A **Build Univariate Time Series Models**
Topic B Build Multivariate Time Series Models

DEFINITIONS

Forecasting: A task that involves **making predictions** about future events based on the analysis of **relevant past events**

Univariate Time Series: A univariate time series includes a **single variable** whose change is recorded in **consistent increments of time**.

EXAMPLE:

Month	Number of Passengers
2025-01	30
2025-02	45
2025-03	48

Autoregressive Integrated Moving Average (ARIMA)

AUTO**R**EGRESSIVE (p) **I**NTEGRATED (d) **M**OVING **A**VERAGE (q)

Imagine you have a toy car that moves forward every day.
Sometimes it moves a little, sometimes a lot, and you want to guess where the car will be tomorrow.

AR → "Remember the past" (hyperparameter: p)

I → "Look at changes, not just where you are" (hyperparameter: d)

MA → "Listen to the mistakes you made before" (hyperparameter: q)

Autoregressive Integrated Moving Average (ARIMA)

AUTOREGRESSIVE (p) INTEGRATED (d) MOVING AVERAGE (q)

Think of AR as saying: "If my car moved **far yesterday**, it will **probably** move **far** today **too**."
It uses **past values** to guess the next one.

Example:
Before yesterday, you sold 9 ice creams, yesterday you sold 10 ice creams

So you guess today you'll sell **about 10 again**, because it usually follows the past pattern.

Autoregressive Integrated Moving Average (ARIMA)

AUTOREGRESSIVE (p) INTEGRATED (d) MOVING AVERAGE (q)

p → AR order (Auto-Regressive part)

Meaning: How many past values we look at to predict the future.
Think: "How far back should I remember?"

If p = 1 → use yesterday's value only.
If p = 2 → use yesterday and the day before.

Autoregressive Integrated Moving Average (ARIMA)

AUTOREGRESSIVE (p) **INTEGRATED (d)** MOVING AVERAGE (q)

"I" is about looking at the change (**difference**), not the actual number.
It says: "Let's see how much the car moved, not where the car is standing."

Day 1: you sold 8
Day 2: you sold 10 (change = +2)
Day 3: you sold 12 (change = +2 again!)
So you guess tomorrow you will sell **12 + 2 = 14**.
You're watching the increase, not the absolute number.

Autoregressive Integrated Moving Average (ARIMA)

AUTOREGRESSIVE (p) **INTEGRATED (d)** MOVING AVERAGE (q)

d ⟶ I order (Integrated / Differencing)

Meaning: How many times we take the difference to remove trends and make data stationary.

Think: “How many times do I look at changes instead of absolute values?”

d=0 ⟶ use raw data.
d=1 ⟶ use first differences (today – yesterday).
d=2 ⟶ use second differences (difference of differences).

Autoregressive Integrated Moving Average (ARIMA)

AUTOREGRESSIVE (p) INTEGRATED (d) **MOVING AVERAGE (q)**

"If I made a mistake yesterday, I will fix it today."

It uses **past errors** (how wrong your guesses were) to **improve future guesses**.

You guessed 10 sales yesterday but sold 12 (**error = +2**).

Today you add that extra 2 to your guess to get closer.

Autoregressive Integrated Moving Average (ARIMA)

AUTOREGRESSIVE (p) INTEGRATED (d) **MOVING AVERAGE (q)**

q → MA order (Moving Average part)

Meaning: How many past forecast errors we use to adjust today's prediction.

Think: "How many past mistakes should I learn from?"

q=1 → adjust using yesterday's error.
q=2 → adjust using yesterday's and the day before's error.

LESSON 5
Building Forecasting Models

Topic A Build Univariate Time Series Models

Topic B Build Multivariate Time Series Models

MULTIVARIATE TIME SERIES

A multivariate time series includes **multiple variables** whose changes are recorded in **consistent increments of time**. Just like in other applications of machine learning, most problems have **several factors** that influence the outcome of a prediction or other model estimation.

In a multivariate time series, you're interested in analyzing how **multiple variables** relate to **one another** within the **context of time**.

In other words, the variables not only depend on their own values in the past, but on the values of other variables as well.

VECTOR AUTOREGRESSION (VAR)

Vector autoregression (VAR) is an alternative to ARIMA that can forecast multivariate time series data. It is able to map the relationship between **multiple variables** by making each variable in time a function of its own past values and the past values of all other variables.

Just like in ARIMA, the "autoregressive" part of VAR refers to the **gap** between each observation in time as a **lag**.

LESSON 5 SUMMARY ACTIVITY

In your machine learning projects, to whom will you typically have to present your findings, if anyone?

Where do you see your machine learning models being deployed, and how will this affect your management and maintenance of the model?

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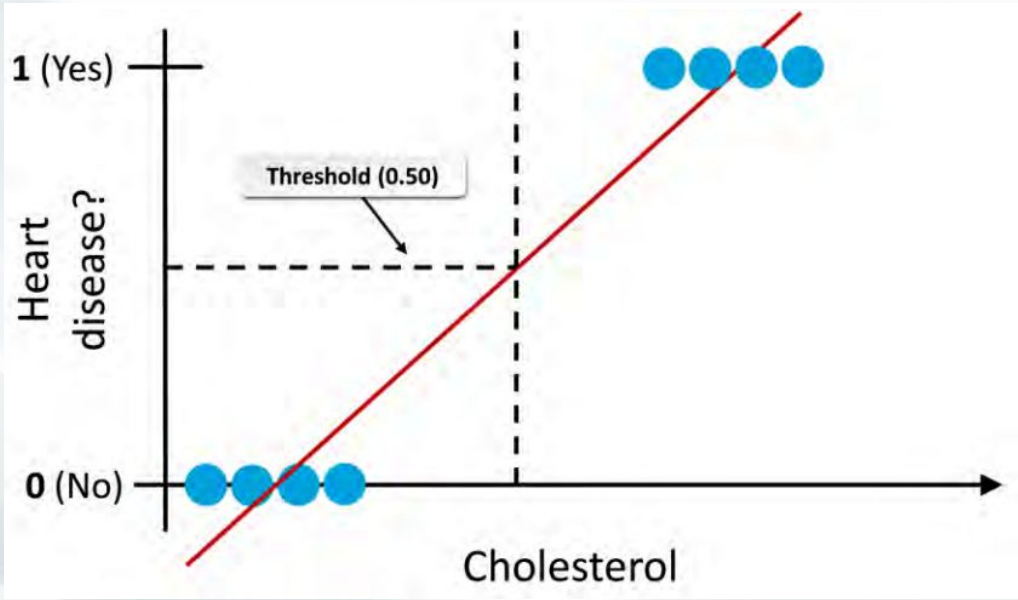
LESSON 6
Building Classification Models Using Regression and k-NN

Topic A	**Train Binary Classification Models Using Logistic Regression**
Topic B	Train Binary Classification Models Using k-Nearest Neighbor
Topic C	Train Multi-Class Classification Models
Topic D	Evaluate Classification Models
Topic E	Tune Classification Models

LINEAR REGRESSION SHORTCOMINGS

Consider a machine learning model in which you're trying to **predict heart disease** in patients (1 for yes, 0 for no).

One of the features in this dataset is a patient's **cholesterol**. If you graphed this function using standard **linear regression**, it might look something like this

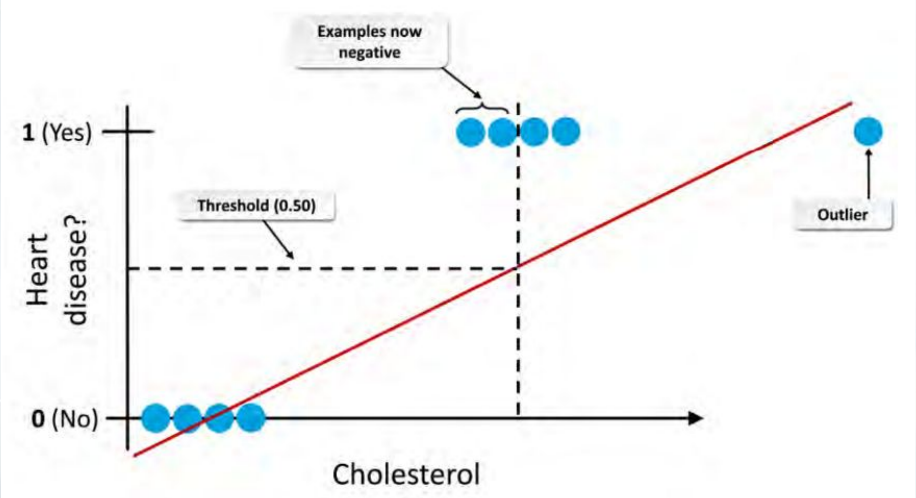


(PRACTICAL EXERCISE)

LINEAR REGRESSION SHORTCOMINGS

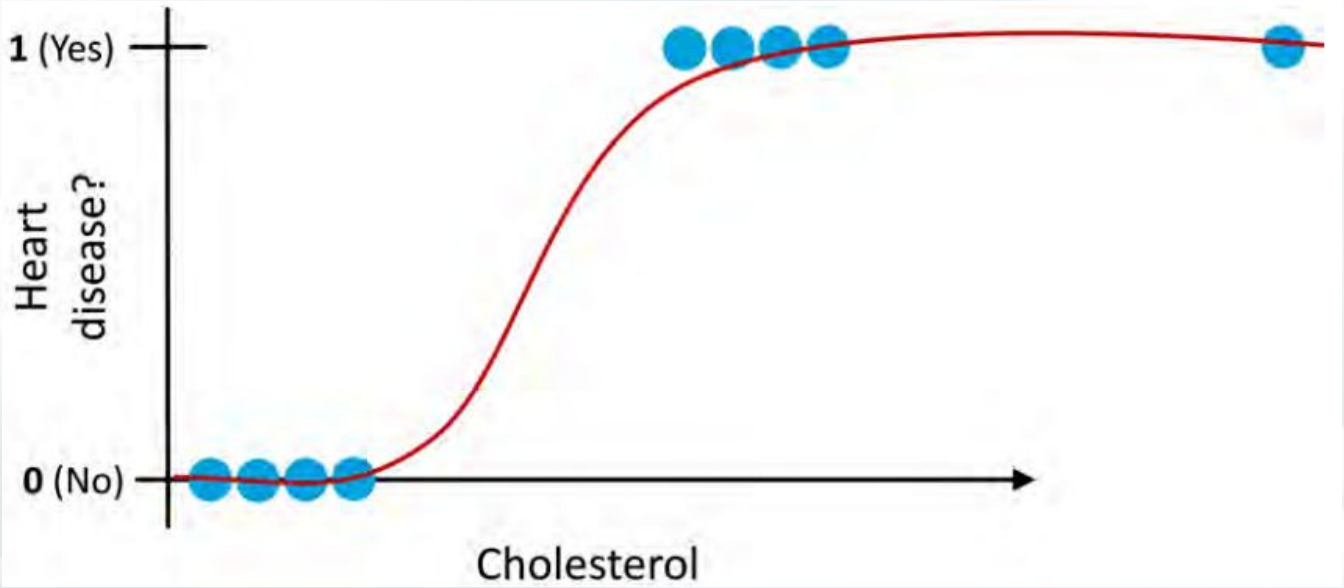
What if there is an **outlier**?
As you can see, this skews the line of best fit. In order to maintain a 0.50 threshold, the threshold **must move** to a new position.

Now, some examples that **were classified as positive** (has heart disease) are **now being classified as negative** (no heart disease).



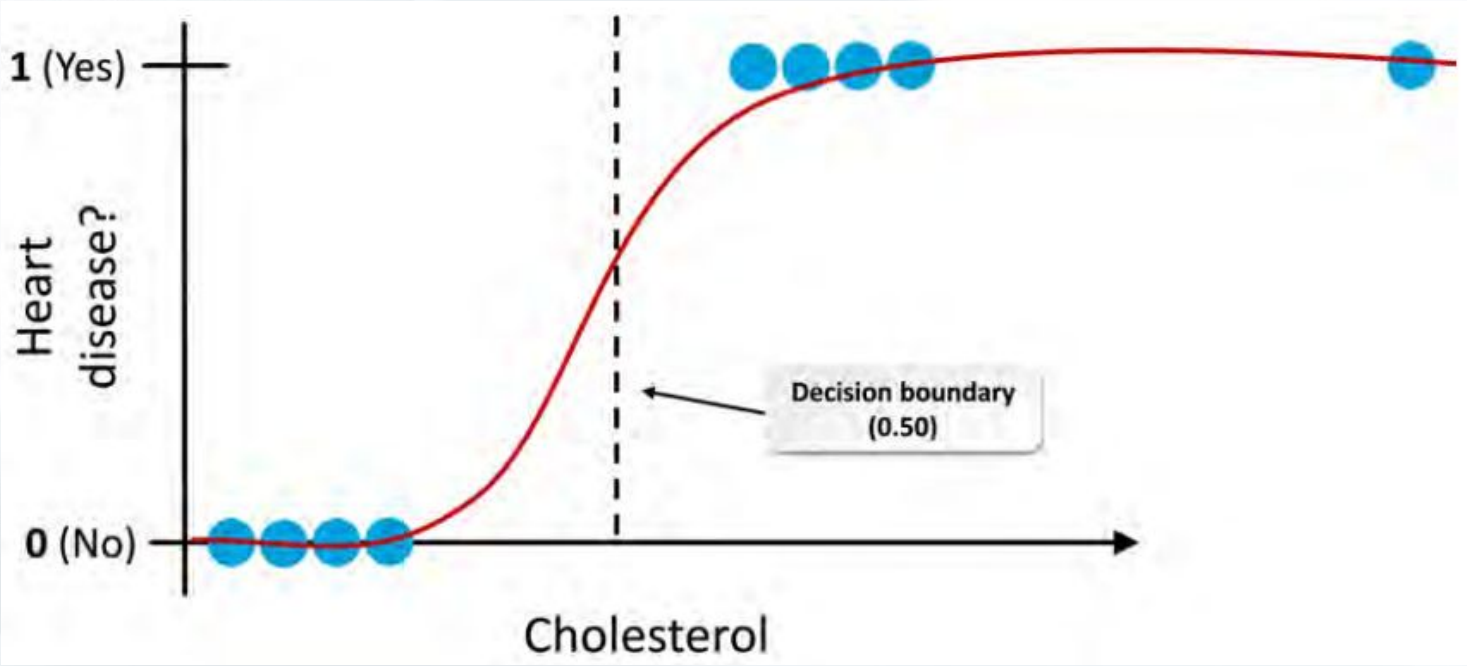
LOGISTIC REGRESSION

Type of linear regression in which the output is a **prediction between 0 and 1**. Instead of a straight line, the logistic function takes an **S shape (sigmoid function)**



DECISION BOUNDARY

The division line that separates negative classes and positive classes in a classification problem



COST FUNCTION FOR LOGISTIC REGRESSION

Unlike with linear regression, you cannot effectively use the mean squared error (**MSE**) cost function with logistic regression. This would result in a **non-convex** function with **multiple local minima**, making it **difficult to minimize** the cost function.

Instead, you can use the following cost function for logistic regression called **LOG LOSS**:

$$\text{if } y = 1, c(\theta) = -\log(\hat{p})$$
$$\text{if } y = 0, c(\theta) = -\log(1 - \hat{p})$$

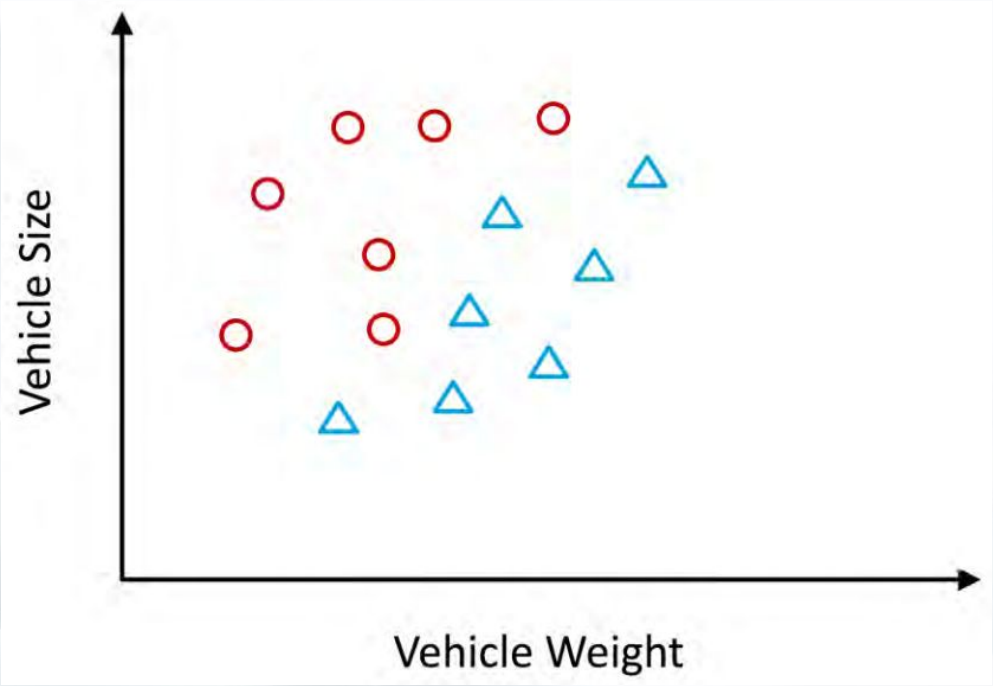
LESSON 6
Building Classification Models Using Regression and k-NN

Topic A	Train Binary Classification Models Using Logistic Regression
Topic B	**Train Binary Classification Models Using k-Nearest Neighbor**
Topic C	Train Multi-Class Classification Models
Topic D	Evaluate Classification Models
Topic E	Tune Classification Models

LOGISTIC REGRESSION SHORTCOMINGS

Consider the following graph, in which the features of vehicle **weight** and vehicle **size** are plotted against each other.

The purpose is to **classify** the vehicle as either a **car** or **plane** (red circles vs. blue triangles).

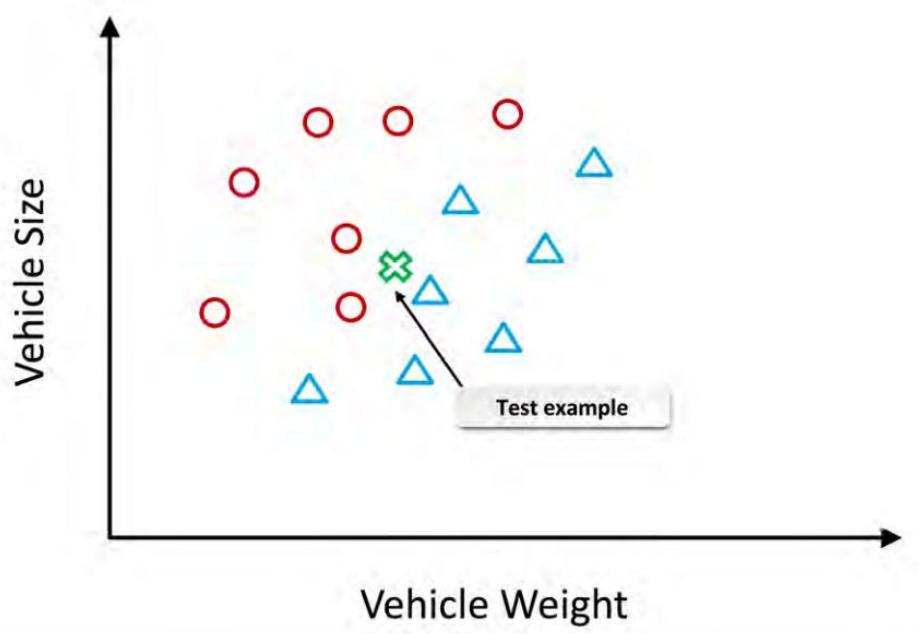


LOGISTIC REGRESSION SHORTCOMINGS

You could use a logistic regression model to predict these classifications.

However, consider how you might determine the class for a new test example, depicted in the following figure as a green X (test example)

Rather than using logistic regression, you can use **KNN**

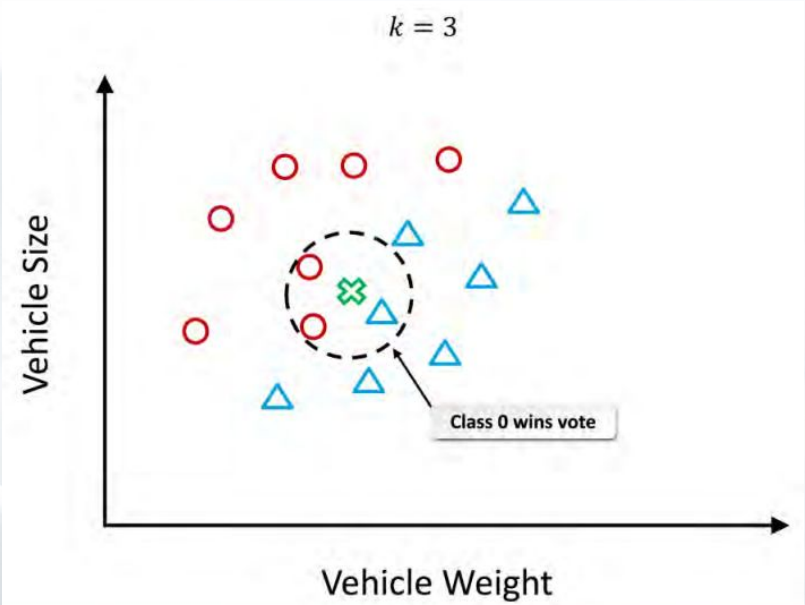


k-Nearest Neighbor (k-NN)

Classify data examples based on their **similarities** to other data examples within the feature space. The **k** in k-NN defines the number of the data examples that are the **closest neighbors (nearest in distance)** of the example in question.

Using k, k-NN takes a **plurality vote** of these neighboring points to determine how to classify the example in question.

SIMULATOR + **PRACTICAL EXERCISE**



k DETERMINATION - 1st OPTION: MANUAL

	HIGHER *K*	LOWER *K*
ADVANTAGES	Less “noisy” (effect of anomalies reduced)	keeping these boundaries distinct
DISADVANTAGES	Less distinct boundaries and increased computational overhead	less effective at minimizing the effect of noise on the data

k DETERMINATION - 2nd OPTION: BOOTSTRAPPING

- Bootstrapping: Taking the **square root** of the total number of data examples in the training set
  - Example: If I have 16 records, *k* is preferably to be 4

NOTE: Always make k odd when you have only two classifiers (to avoid a tie in the vote)

LOGISTIC REGRESSION VS KNN

- k-NN does not really train a model in the same way as a typical machine learning algorithm like logistic regression.
- The k-NN model doesn't really improve its classification abilities through learning.
- k-NN can take a very long time to actually make a prediction when there are many data examples and features
- **Logistic regression**, because it has been trained to determine optimal model parameters, is **much faster** in making predictions on new data

LOGISTIC REGRESSION VS KNN - SUMMARY

- **k-NN** is ideal when datasets are **small** and implementation must be simple
- **Logistic Regression** is ideal when datasets are **large** and predictions must be generated relatively quickly

ACTIVITY TIME

Building Classification Models Using Regression and k-NN

KNN Using Google Sheets

ACTIVITY 6-1: Training Binary Classification Models

ACTIVITY 6-2: (Optional) Creating a Logistic Regression Model

LESSON 6
Building Classification Models Using Regression and k-NN

Topic A	Train Binary Classification Models Using Logistic Regression
Topic B	Train Binary Classification Models Using k-Nearest Neighbor
Topic C	**Train Multi-Class Classification Models**
Topic D	Evaluate Classification Models
Topic E	Tune Classification Models

TRAIN MULTI-CLASS CLASSIFICATION MODELS

Not all classification problems are binary in nature—some require more than just a 0 or a 1. In this topic, you'll train a model to handle cases in which there are multiple ways to classify a data example.

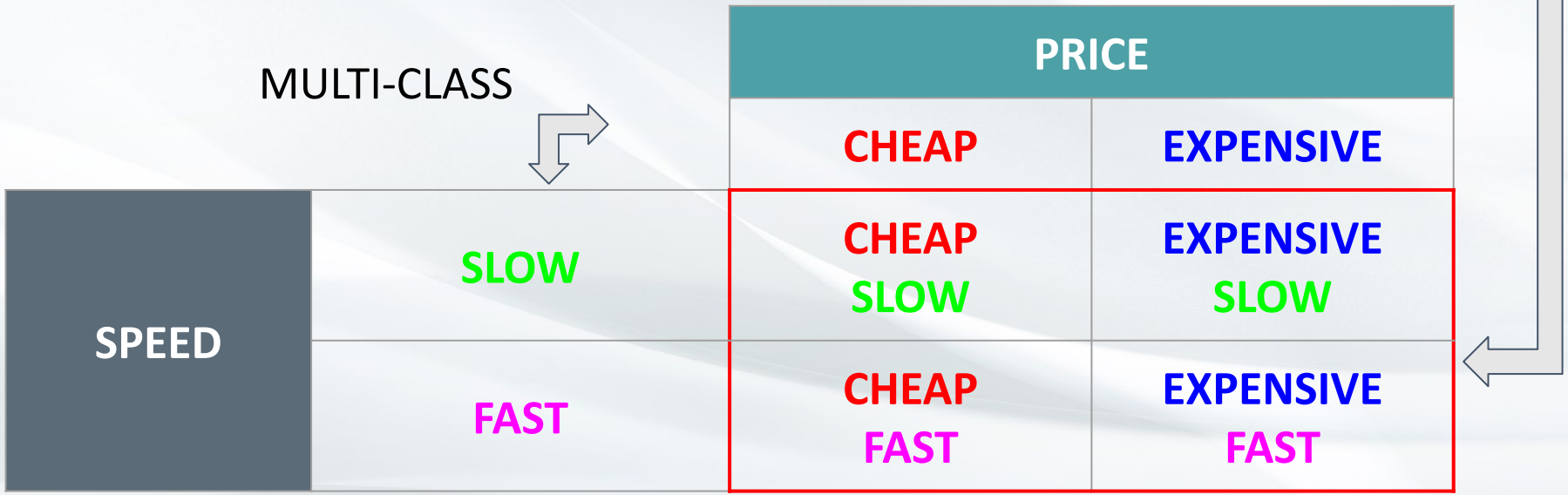


MULTI LABEL CLASSIFICATION

Data example can be given multiple labels
Ex. You need to analyze cars

MULTI-LABEL

MULTI-CLASS



MULTI CLASS CLASSIFICATION

Data example can be placed into one of three or more classes
An example cannot belong to class A and B and C; it must belong to A **or** B **or** C.

Ex. You need to analyze text in a sentence
A text can be a **Noun**, **Verb**, **Adjective**, **Adverb**, …

NOTE:
For **multi-label** problems, consider training **multiply binary classifiers**.
For **multi-class** problems, consider training a **multinomial logistic regression model**.

MULTINOMIAL LOGISTIC REGRESSION

Sometimes called **softmax regression**, is a method for solving multi-class problems. (Based on **PROBABILITIES** and NOT BINARY CLASSIFICATION)

The end result is that, for each data example, the softmax function determines the class with the highest probability, where all probabilities add up to 1.

Example: "I saw that movie yesterday" ⇒ Probabilities for "saw":
84% verb | 13% noun | 3% adjective
Thus, the algorithm predicts that the word "saw" in this sentence is a verb.

COST FUNCTION: CROSS-ENTROPY

Cost function used to evaluate the performance of a softmax function. It penalizes low probability scores for a particular class. The lower the probability, the higher the cost.

Value between 0 and 1 (the less the better)
When there are only two classes (binary classification), the cross entropy function is the same as the log loss function mentioned earlier

Remember: **MSE** is used in **Linear Regression**
Log Loss is used in **Logistic Regression**
Cross-Entropy is used in **Multinomial Logistic Regression**

ACTIVITY TIME

Building Classification Models Using Regression and k-NN

ACTIVITY 6-3: Training a Multi-Class Classification Model

LESSON 6
Building Classification Models Using Regression and k-NN

Topic A	Train Binary Classification Models Using Logistic Regression
Topic B	Train Binary Classification Models Using k-Nearest Neighbor
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Topic D	**Evaluate Classification Models**
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EVALUATE CLASSIFICATION MODELS

PROBLEM: It's not **enough** to just **train a model** you think is best, and then **call it a day**. Unless you're using a **very simple dataset** or you get **lucky**, the default parameters aren't going to give you the best possible model for solving the problem.

SOLUTION: CLASSIFIER PERFORMANCE MEASUREMENT

CLASSIFIER PERFORMANCE MEASUREMENT

- There are **various** ways to measure the **performance of the model**
- **Each way** will help us optimize the model for a **particular situation**

EX. I have a jury. each person in the jury focuses on a certain skill (acting, voice, look).

1. ACCURACY
2. PRECISION
3. RECALL
4. F1 SCORE

SIMULATOR SIMULATOR2

CONFUSION MATRIX

Which one is more dangerous?
Example: evaluating the strength of the wing of an airplane.
How many false negative I'm allowed to have

		ACTUAL	
		YES	NO
PREDICTION	YES	True Positive	False Positive
	NO	False Negative	True Negative

CONFUSION MATRIX

		ACTUAL	
		HEART DISEASE	NO HEART DISEASE
PREDICTION	HEART DISEASE	43 TRUE POSITIVE	6 FALSE POSITIVE
	NO HEART DISEASE	3 FALSE NEGATIVE	212 TRUE NEGATIVE

CONFUSION MATRIX

For example, in some situations, it might be better to have a model that ensures there will be **no false negatives** at the expense of **allowing perhaps 30% of positive predictions to be false**.

For example, you'd probably want your machine learning model to identify **every bridge that has a structural problem** before you have a **failure** that harms someone. Saving lives would outweigh the cost of having someone investigate bridges that don't actually require any repairs.

In other situations, it might be better to ensure there are **no false negatives** while **allowing a small number of false positives**.

ACCURACY

Measure of how **frequently** each prediction is correctly deemed positive or negative (Number between 0 and 1)

FORMULA:

Correct Predictions (TP + TN)

All Predictions (TP + TN + FP + FN)

		ACTUAL	
		YES	NO
PREDICTION	YES	True Positive	False Positive
	NO	False Negative	True Negative

USEFUL: when class label values are **balanced**

QUESTION: Can you give an example in which accuracy is useless?

EXAMPLE WHERE ACCURACY IS USELESS

		ACTUAL	
		COLLAPSE	NO COLLAPSE
PREDICTION	COLLAPSE	5 TRUE POSITIVE	2 FALSE POSITIVE
	NO COLLAPSE	1 FALSE NEGATIVE	512 TRUE NEGATIVE

Note: Bridge collapses are exceedingly rare.
Can you please calculate the accuracy?

EXAMPLE WHERE ACCURACY IS USELESS

The accuracy is **extremely high**—close to 99%.
This makes accuracy nearly **useless**, as the machine learning model will become **no better** at predicting the rare cases in which a bridge shows signs of imminent collapse.

Conclusion: Accuracy is therefore only useful in datasets where the label data is **balanced**. Accuracy is **not useful in imbalanced** classes.

PRECISION (aka Positive Predictive Value)

How many **PREDICTED** values as **YES** are **ACTUAL YES**

Out of the **total predicted** positive results, **how many** results were **actually** positive?

Measure of how often the positives identified by the learning model are true positives (Number between 0 and 1)

		ACTUAL	
		YES	NO
PREDICTION	YES	True Positive	False Positive
	NO	False Negative	True Negative

FORMULA:

Correct Positive Predictions (TP)

All Positive Instances (TP + FP)

USEFUL: when class label values are **imbalanced**

PRECISION (aka Positive Predictive Value)

EXERCISE: What is the precision for the previous dataset?

However, the problem with precision in this case is that it doesn't address cases in which a **bridge will collapse** but the model **doesn't think it will** (i.e., false negatives). Even just a single bridge collapsing is intolerable because of the damage it causes

RECALL (aka SENSITIVITY or TPR)

How many **ACTUAL** values as **YES** are **PREDICTED YES**

FORMULA:
Correct Positive Predictions (TP)

All Relevant Instances (TP + FN)

		ACTUAL	
		YES	NO
PREDICTION	YES	True Positive	False Positive
	NO	False Negative	True Negative

USEFUL: when class label values are **imbalanced**

Can you calculate the Recall for the bridge example? What do you conclude

RECALL (aka SENSITIVITY or TPR)

The recall would be 5 (TP) / 5 (TP) + 1 (FN), or around 83%.

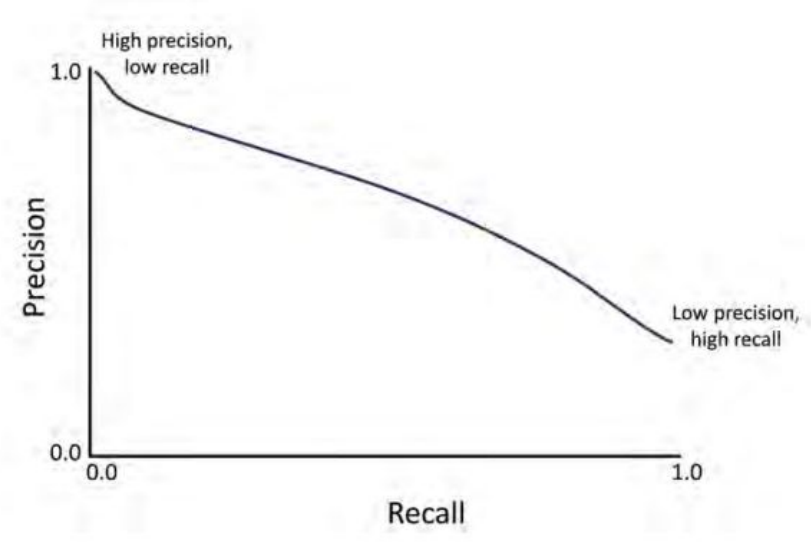
Now, you have a better idea of how well your model performs with respect to its ultimate purpose—**minimizing bridge collapse and the damage it can cause**.

Technically, the model could improve its recall by predicting that **all bridges are about to collapse**, leading to a recall of 100%, but this would make the model **useless for prioritizing repair efforts**.

PRECISION–RECALL TRADEOFF

If the **FALSE POSITIVE RATE** is **more important** to you to minimize it, we use **PRECISION**

If the **FALSE NEGATIVE RATE** is **more important** to you to minimize it, we use **RECALL**



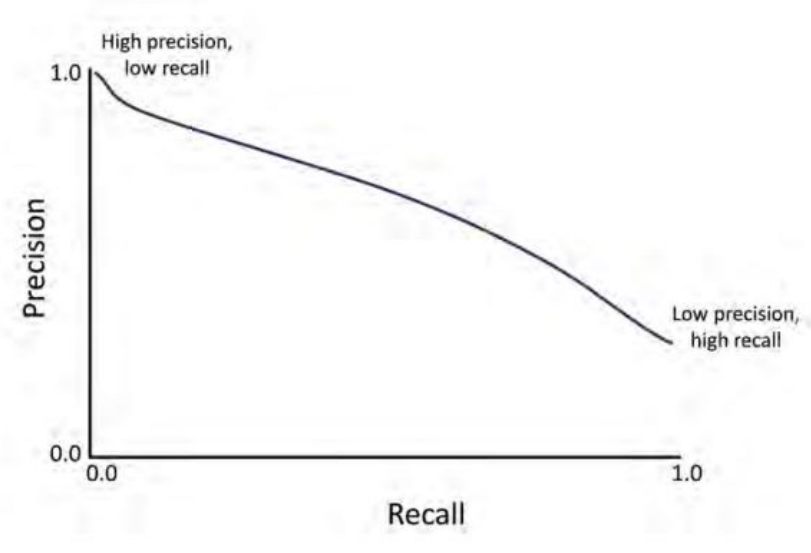
		ACTUAL	
		YES	NO
PREDICTION	YES	True Positive	False Positive
	NO	False Negative	True Negative

PRECISION–RECALL TRADEOFF

In general, there is **tradeoff** between **precision** and **recall**.

As you tune a machine learning model to **emphasize** one of these factors, you do so **at the expense** of **de-emphasizing** the other.

Which factor you choose to emphasize **depends on your business requirements**.



PRECISION–RECALL SHORTCOMINGS

As you've seen, precision and recall are more useful in **unbalanced datasets**, but they come with a tradeoff.

Sometimes, like in the bridge example, it's relatively clear which one is more useful. However, this is not always the case.

PRECISION–RECALL SHORTCOMINGS

PROBLEM: Consider a machine learning task in which you want to determine whether or not a song is in the rock genre, given a short audio sample.

A **false positive** (e.g., classifying a song as rock when it's not) is just as **undesirable** as a **false negative** (e.g., not classifying a song as rock even though it is); neither is particularly worse.

It is not always clear **which metric is more useful**. So, what's the best way to measure performance in this case?

SOLUTION: F1 SCORE

F_1 SCORE

The F_1 score helps you find the **optimal combination** of both **precision** and **recall**

FORMULA: F_1: 2 * [(Precision x Recall) / (Precision + Recall)]

F_1 SCORE

So, assume that 100 songs were correctly classified as rock, and 10 songs were incorrectly classified as rock, and 23 songs were not classified as rock even though they are

Calculate the Precision, Recall, and F_1
What do you conclude? Which one is more realistic?

F_1 SCORE

PRECISION: (TP 100 / TP 100 + FP 10) = ~91%.

RECALL: (TP 100 / TP 100 + FN 23) = ~81%

F_1 SCORE: 2 * [(0.91 x 0.81) / (0.91 + 0.81)] = 86%

RECEIVER OPERATING CHARACTERISTIC (ROC) CURVE

ROC curve makes it easy to identify the best threshold for making a decision

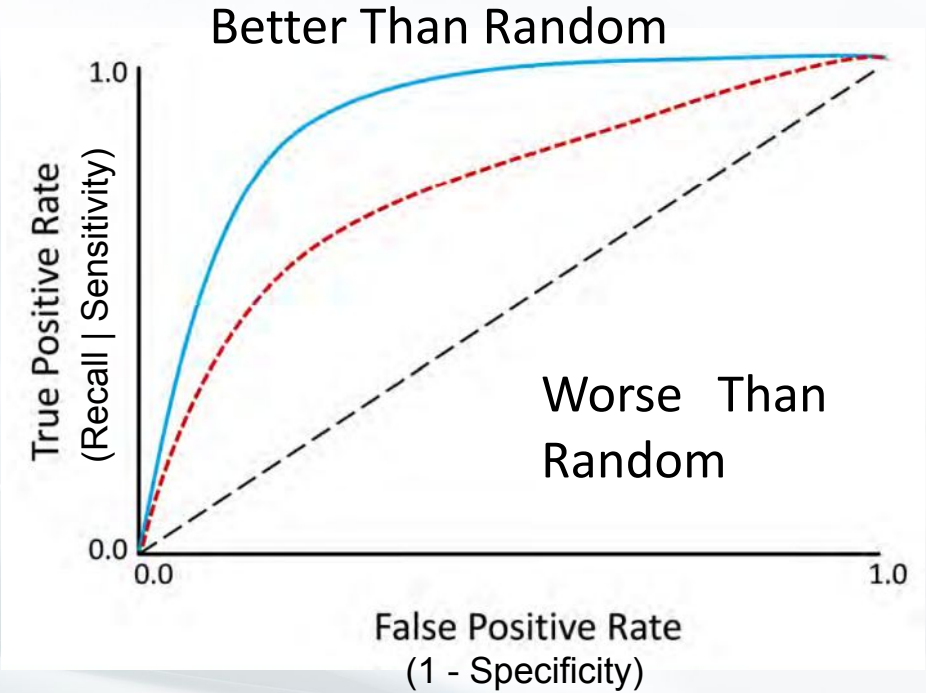
ROC is used for **balanced** binary classification problems

SPECIFICITY (TNR):

True Negatives

(True Negatives + False Positives)

SIMULATOR

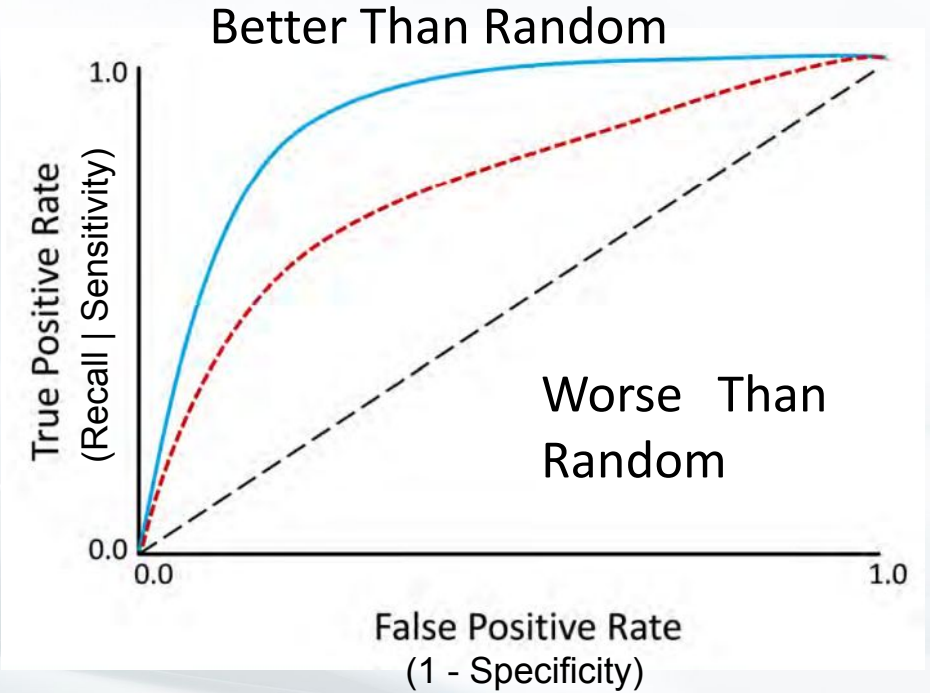


MODEL A	MODEL B	RANDOM GUESS

RECEIVER OPERATING CHARACTERISTIC (ROC) CURVE

Method of plotting the relationship between predicted "hits" (Y-axis: recall) versus false alarms (X-axis).

A perfectly predictive model would be fit from the top-left corner of the graph (0, 1). This is unrealistic, but the more a model's curve approaches this top-left corner, the better it performs.



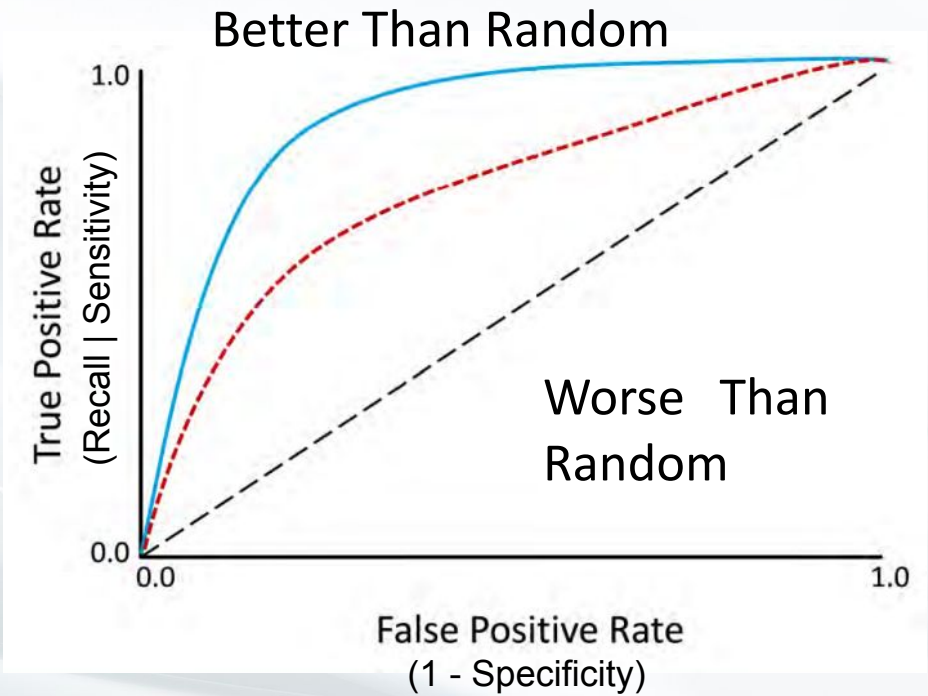
RECEIVER OPERATING CHARACTERISTIC (ROC) CURVE

TRUE POSITIVE RATE (SENSITIVITY / RECALL)

TRUE POSITIVE

TRUE POSITIVE + FALSE NEGATIVE

		ACTUAL	
		YES	NO
PREDICTION	YES	True Positive	False Positive
	NO	False Negative	True Negative



MODEL A | MODEL B | RANDOM GUESS

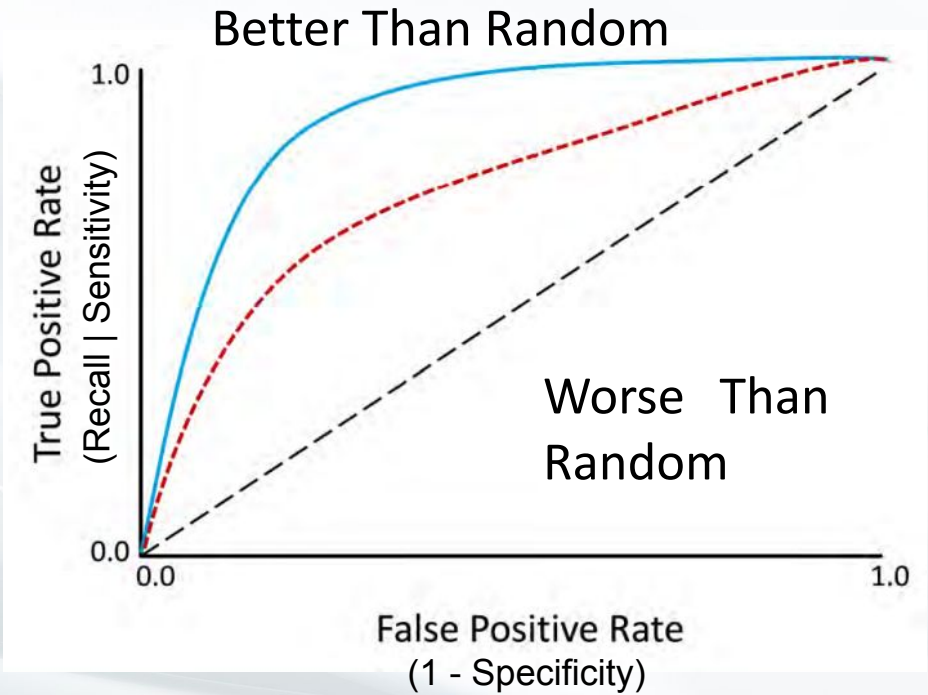
RECEIVER OPERATING CHARACTERISTIC (ROC) CURVE

FALSE POSITIVE RATE (1-SPECIFICITY)

FALSE POSITIVES

FALSE POSITIVES + TRUE NEGATIVES

		ACTUAL	
		YES	NO
PREDICTION	YES	True Positive	False Positive
	NO	False Negative	True Negative



MODEL A | MODEL B | RANDOM GUESS

THRESHOLDS

The **second way** to use a ROC curve is to evaluate a model **by itself**, not in comparison to any **other model** (**first way**).

After all, one model **may perform** better than another, but still perform poorly overall.

EXAMPLE: For the heart disease problem, I set a threshold of 0.7.
Any value greater than the threshold will be considered as 1 (heart disease)
Any value less than the threshold will be considered as 0 (No threshold)

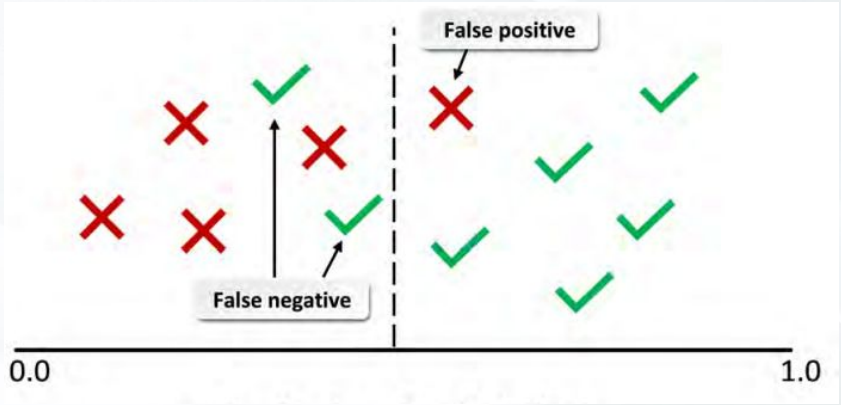
PROBLEMS OF THRESHOLDS

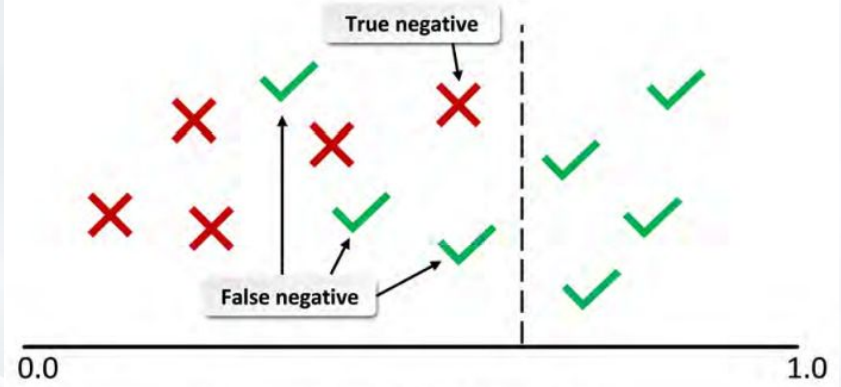
The problem with this method is that it can be very difficult to actually decide what a good threshold is for your data.

Example:
iPhone FaceReconition
What is a perfect threshold in this case?



CHANGING THRESHOLDS

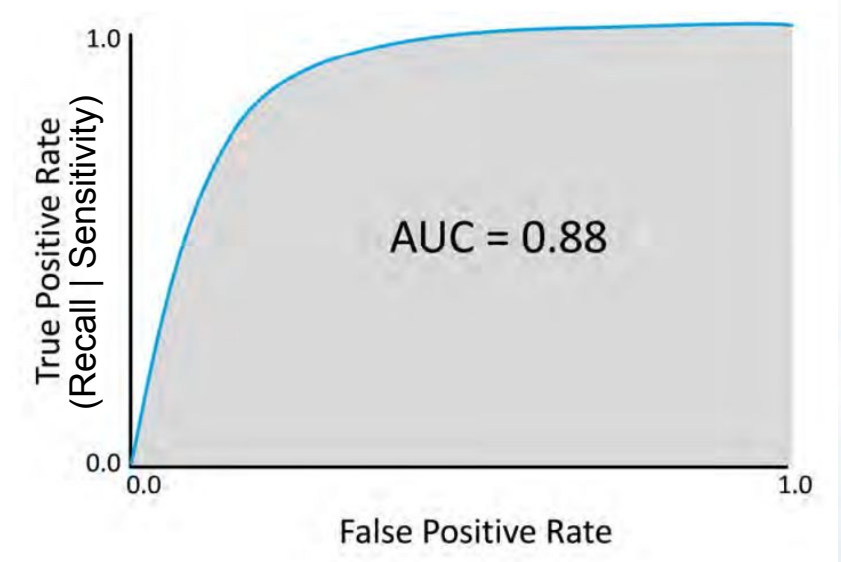




Moving the **threshold** to the **right**	TPR decreases sensitivity	TNR increases specificity
Moving the **threshold** to the **left**	TPR increases sensitivity	TNR decreases specificity

AREA UNDER CURVE (AUC)

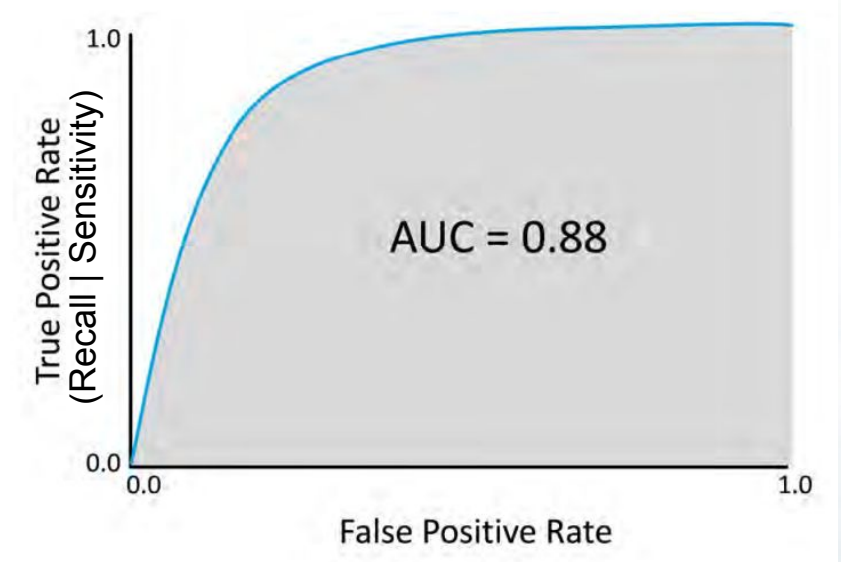
- AUC can help us decide which classification method is better
- The higher the value, the better our model is



- We use **ROC** (previous curve) to know the **best threshold.**
- **Unlike** ROC, we **don't** use AUC to choose the best threshold. On the contrary, we use AUC to **evaluate** the **overall performance** of a model

AREA UNDER CURVE (AUC)

- AUC is the total space that is under a model's ROC curve.
- The probability that the model will classify any positive instance higher than any negative instance, from 0 to 1.0.
- **Not good** for class imbalance



PRECISION–RECALL CURVE (PRC)

If your data **does** have a **class imbalance**, it is preferable to plot data on a **precision–recall curve**. A precision–recall curve (PRC) plots precision values on the y-axis and recall values on the x-axis for **different thresholds**.

Example: The bridge collapse model marks a bridge as in danger of collapse (1) or not in danger of collapse (0). The classification in this problem is **imbalanced**; very few bridges will qualify as being in danger of collapse.

PRECISION–RECALL CURVE (PRC)

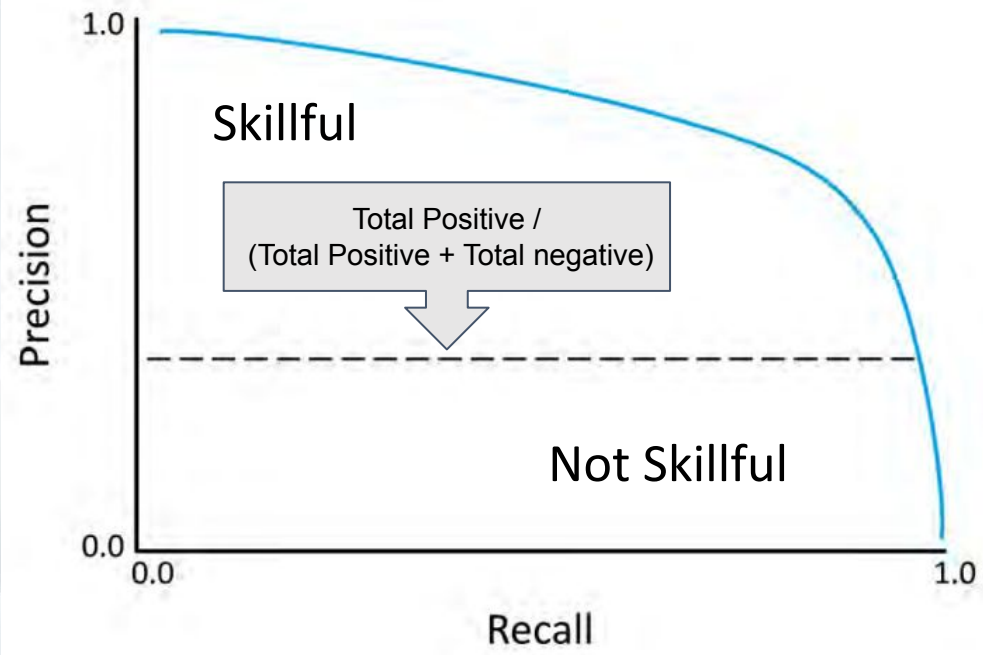
You're therefore **not all that interested** in seeing how well the model predicts true negatives (bridges will not collapse), because negatives make up the **majority** of the data.

PRCs **do not involve true negatives**. Instead, they help you evaluate how well the model predicts the class in the minority (class 1: bridge collapse) which (in this case) is the **whole point of the model**

PRECISION–RECALL CURVE (PRC)

You want your curve to be above the **NO-SKILL LINE**, otherwise the model is not useful.

A model with **perfect** skill would be drawn from the top right of the graph (1.0, 1.0). In the example figure, the curve is above the no-skill line, so the model is considered skillful.



ACTIVITY TIME

Building Classification Models Using Regression and k-NN

ACTIVITY 6-4: Evaluating a Classification Model

LESSON 6
Building Classification Models Using Regression and k-NN

Topic A	Train Binary Classification Models Using Logistic Regression
Topic B	Train Binary Classification Models Using k-Nearest Neighbor
Topic C	Train Multi-Class Classification Models
Topic D	Evaluate Classification Models
Topic E	**Tune Classification Models**

HYPERPARAMETER OPTIMIZATION

PROBLEM: I don't want to alter the hyperparameters in a manual way to check which one gives me the better combination
EXAMPLE: model_eval(0.3) ⇒ model_eval(0.1) ⇒ model_eval(0.2)

SOLUTION: HYPERPARAMETER OPTIMIZATION

Repeatedly **altering** the hyperparameters that an algorithm uses to train a model in order to determine the set of hyperparameters that lead to the best or the desired level of model performance.

NOTE: Hyperparameter optimization methods work for **any algorithm** that has hyperparameters, **not just** classification algorithms

AUTOMATED METHODS FOR HYPERPARAMETER OPTIMIZATION

1. Grid search
2. Randomized search
3. Bayesian optimization
4. Genetic algorithms

GRID SEARCH (PRACTICAL EXERCISE)

Hyperparameter optimization method that simply **takes a set** (or grid) of parameter combinations, **trains** a model using each of those combinations, and then **returns** the combination that best optimizes an evaluation metric that you specify

EXAMPLE:
```
BestFood= [{   'Burger': ['KFC','McDonalds'],
               'Fries': ['Burger King', 'McDonalds'],
               'Drink': [1, 2, 3]}]
```

I check the happiness level after every execution

Ex. I went to Subway and I tried every single combination

GRID SEARCH

```
grid = [{   'solver': ['liblinear'],
            'penalty': ['l1', 'l2'],
            'C': [0.01, 0.1, 1, 5, 10, 100]}]
```

In this case, solver, penalty, and C are all separate hyperparameters. Each hyperparameter is used in all possible combinations (Every combination will be tried)

So, grid search will first train on the dataset **1 × 2 × 6 = 12 times**

COMPLETE GRID SEARCH FORMULA

```
grid = [{   'solver': ['liblinear'],
            'penalty': ['l1', 'l2'],
            'C': [0.01, 0.1, 1, 5, 10, 100]}]
```

C is Lambda

search = GridSearchCV(model, param_grid = grid, scoring = 'f1', cv = 5)

Because cv is 5, grid search will do 5-fold cross-validation for each combination—in other words, it will train 12 × 5 = 60 times.

PROBLEMS IN GRID SEARCH

if your dataset has a **very large feature space**, grid search can take a very **long time** to exhaustively evaluate every possible combination of hyperparameters

```
BestFood= [{    'Burger': ['KFC','McDonalds','PizzaHut','BurgerKing','...']
                'Fries': ['Burger King', 'McDonalds'],
                'Drink': [1, 2, 3]}]
```

RANDOMIZED SEARCH (PRACTICAL EXERCISE)

Alternative approach in which random combinations of hyperparameters are used to train and evaluate the model.

(not all of the parameter values are tried, and instead, parameters will be sampled with fixed numbers of iterations given by n_iter.)

```
BestFood= [{      'Burger': ['KFC','McDonalds'],
                  'Fries': ['Burger King', 'McDonalds'],
                  'Drink': rand(1,10)}]
```

RANDOMIZED SEARCH

Rather than a **grid** of discrete values, hyperparameters in randomized search are typically defined in **distributions**; the algorithm will select a combination of values from these distributions for a certain **number of specified iterations**.

```
dist = [{   'solver': ['liblinear'],
            'penalty': ['l1', 'l2'],
            'C': sp_randint(1,100)}]
```

COMPLETE RANDOMIZED SEARCH FORMULA

```
dist = [{   'solver': ['liblinear'],
            'penalty': ['l1', 'l2'],
            'C': sp_randint(1,100)}]
search = RandomizedSearchCV(model, param_distributions = dist,
          n_iter = 50, scoring = 'f1', cv = 5)
```

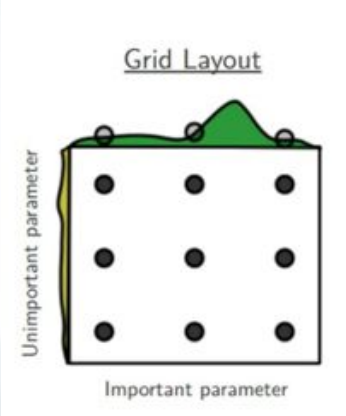
Controlling the number of iterations of random hyperparameter sampling gives you more **granular control** over the training process, and subsequently, enables you to **minimize training time** in large feature spaces.

DIFFERENCES BETWEEN THE TWO

GRID SEARCH

Exhaustive

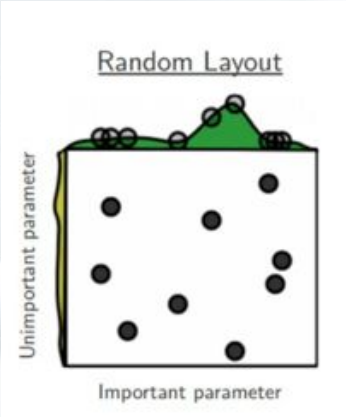
Finds truly optimal hyperparameters



RANDOMIZED SEARCH

Give Adequate / sufficient results

Better when having small number of parameters



PROBLEMS IN RANDOMIZED SEARCH

Even randomized search is susceptible to training **time issues**, especially if you need to conduct many **random sampling iterations** over **huge datasets**.

The algorithm needs to find a **combination** of random hyperparameters for each iteration, all while looking for a combination that optimizes the specified evaluation metric.

In Grid Search and Randomized Search, we **don't save** the results in a memory after each iteration (we didn't learn anything from these attempts)

BAYESIAN OPTIMIZATION

A hyperparameter optimization method that **uses past samples** to influence where **sampling is conducted** in subsequent iterations, enabling it to **determine the next optimal space** to sample from.

This makes Bayesian optimization "**smarter**" than randomized search, because it is able to get to the optimal hyperparameters much faster than truly randomized sampling

BAYESIAN OPTIMIZATION

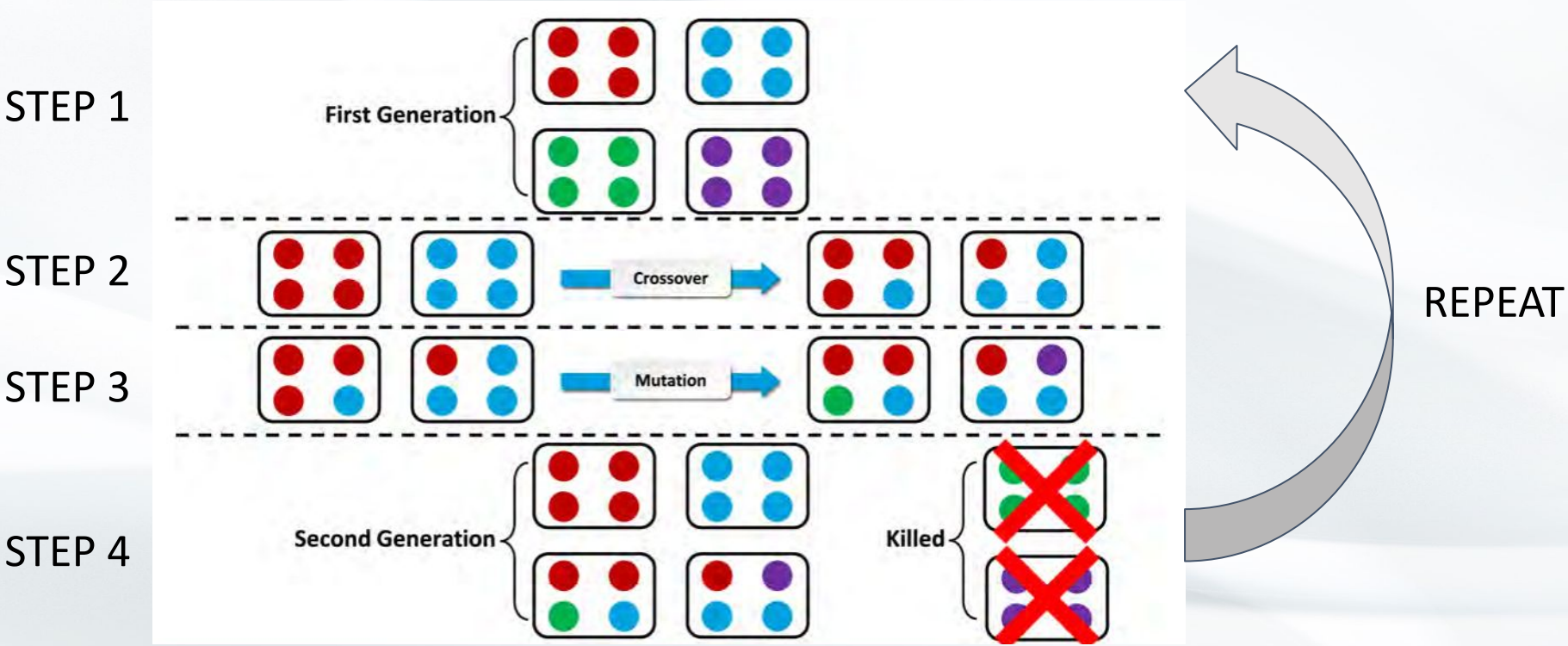
1. **Start** with an initial **random sampling** of the hyperparameter distribution space.
2. **Evaluate** the **loss function** for this space.
3. Capture the "**beliefs**" of the **prior evaluation**(s), then **update** those beliefs to account for everything that is currently known about the loss function
4. Sample a **new hyperparameter space** (best tradeoff between loss optimum prediction and high prediction uncertainty) that optimizes an acquisition function
5. **Evaluate** the **loss from** this new space (same as step 2)
6. Repeat steps 3 through 5 until some stopping criterion is met, such as a specified number of iterations or until the loss no longer improves.

PROBLEMS IN BAYESIAN OPTIMIZATION

Bayesian optimization is **more complicated** and **harder to implement** than grid search or randomized search, as it requires **more careful tuning** of the search configurations.

However, it has been shown to be **faster than grid search and randomized search** in some scenarios

GENETIC ALGORITHMS



GENETIC ALGORITHMS: PROS & CONS

Genetic algorithms are **most effective** when the number of hyperparameter combinations is **relatively low**.

With a **very large** hyperparameter space, it can **take a long time** for genetic algorithms to derive the optimal hyperparameters.
Because of these efficiency issues, they are typically **not as common as other optimization techniques**.

HYPERPARAMETER OPTIMIZATION - SUMMARY

NAME	ADVANTAGES	DISADVANTAGES
GRID SEARCH (I try every single combination)	- Finds truly optimal hyperparameters	- Exhaustive - Doesn't save the results in a memory after each iteration
RANDOMIZED SEARCH (I try random combinations)	- Can minimize training time in large feature spaces - Give Adequate / sufficient results	- Not as detailed as grid search - Exhaustive - Doesn't save the results in a memory after each iteration
BAYESIAN OPTIMIZATION (I try a new combination based on the previous one)	- Smarter - faster than grid search and randomized search in some scenarios	- Complicated - Harder to Implement
GENETIC ALGORITHM (I apply crossover and mutation)	- Most effective when the number of hyperparameter combinations is relatively low	- Takes a lot of time with a very large hyperparameter space

ACTIVITY TIME

Building Classification Models Using Regression and k-NN

ACTIVITY 6-5: Tuning a Classification Model

LESSON 6 SUMMARY ACTIVITY

What type of data in your organization do you think might be conducive to classification?

Given the datasets you're likely to use and classification problems you're trying to solve, what evaluation metrics do you think you'll find most useful when tuning a classification model?

COURSE OUTLINES

LESSON 1	Solving Business Problems Using AI and ML
LESSON 2	Preparing Data
LESSON 3	Training, Evaluating, and Tuning a Machine Learning Model
LESSON 4	Building Linear Regression Models
LESSON 5	Building Forecasting Models
LESSON 6	Building Classification Models Using Regression and k-NN
LESSON 7	**Building Clustering Models**
LESSON 8	Building Decision Trees and Random Forests
LESSON 9	Building Support-Vector Machines
LESSON 10	Building Artificial Neural Networks
LESSON 11	Operationalizing Machine Learning Models
LESSON 12	Maintaining Machine Learning Operations

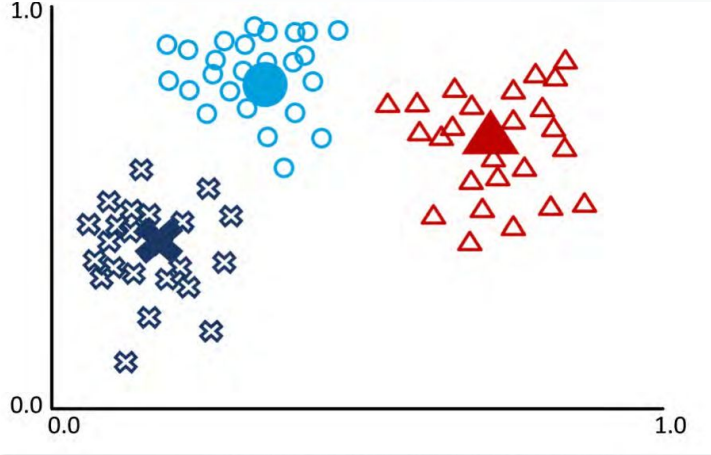
LESSON 7
Building Clustering Models

Topic A **Build k-means Clustering Models**
Topic B Build Hierarchical Clustering Models

k-Means CLUSTERING (PRACTICAL EXERCISE)

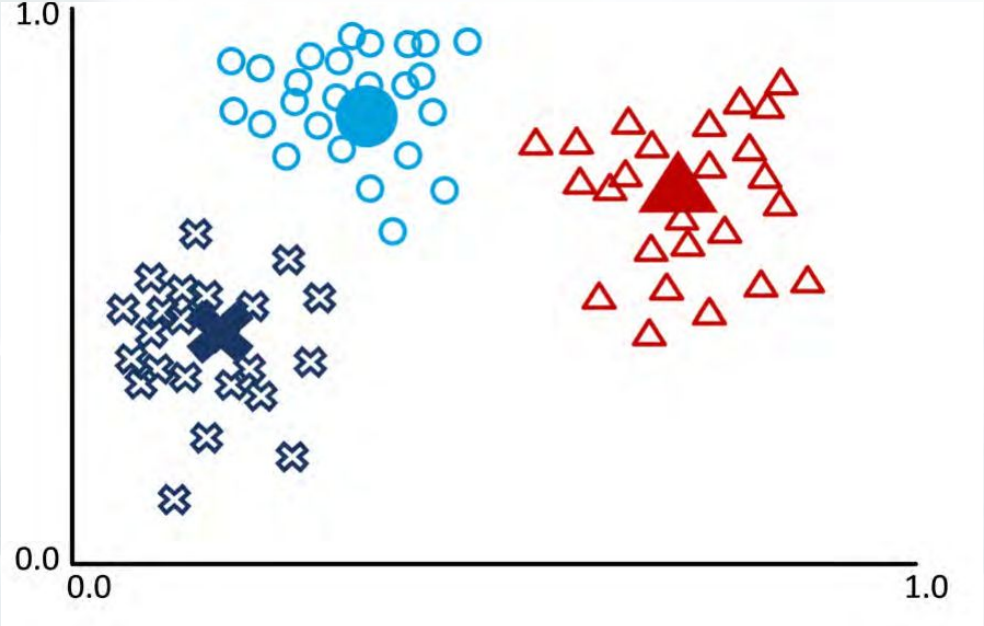
Algorithm for **unsupervised machine learning** that **groups** like data examples **together** for the purpose of revealing **patterns in the data**.

It does this by defining a **set of k groups** (clusters). Each data example is **placed within the cluster** whose center (called a centroid) is the **closest** to that data example. Closeness can be defined by a **distance** metric that is chosen during training



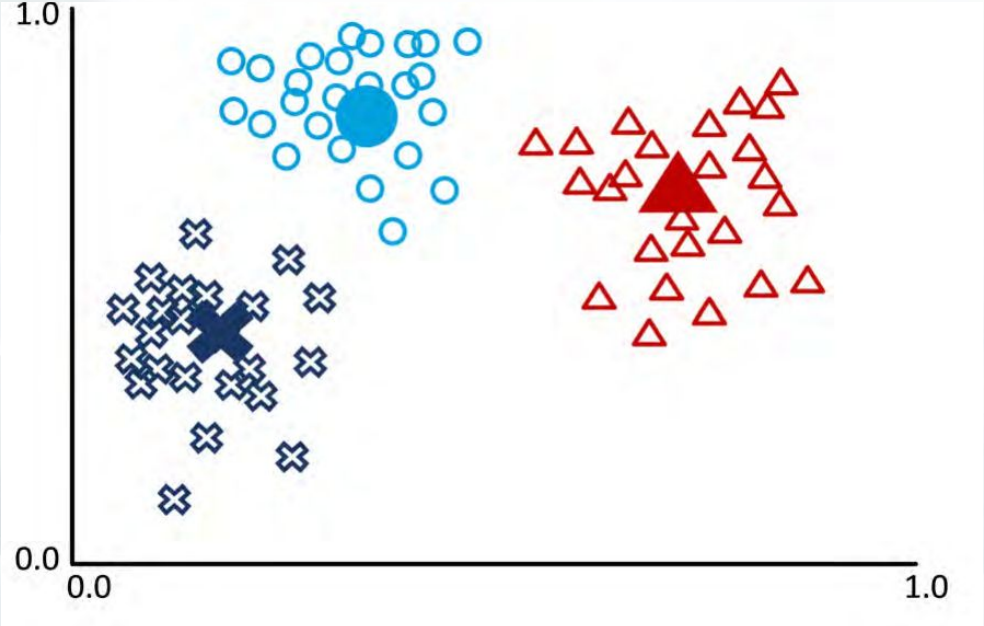
WHEN k-Means Clustering IS HELPFUL

k-means clustering is often used in the field of **customer segmentation** to group customers based on **similar characteristics**. It has also found use in categorizing sensor data from IoT devices, like images and video



WHEN k-Means Clustering IS HELPFUL

1- Select k (ex. 3)
2- Randomly select 3 distinct data points (initial clusters)
3- Measure the distance between a point and every cluster
4- Assign the point to the nearest cluster
5- Repeat



VISUAL | VIDEO 1 | VIDEO 2

k-Means COST FUNCTION

Just like any other machine learning algorithm, the ultimate goal of k-means clustering is the minimization of cost

$$J(\boldsymbol{\theta}) = \sum_{i=1}^{n} \min_{j} \left(\text{dist}(x_i, c_j) \right)$$

This cost function is calculated by default within the algorithm

k DETERMINATION

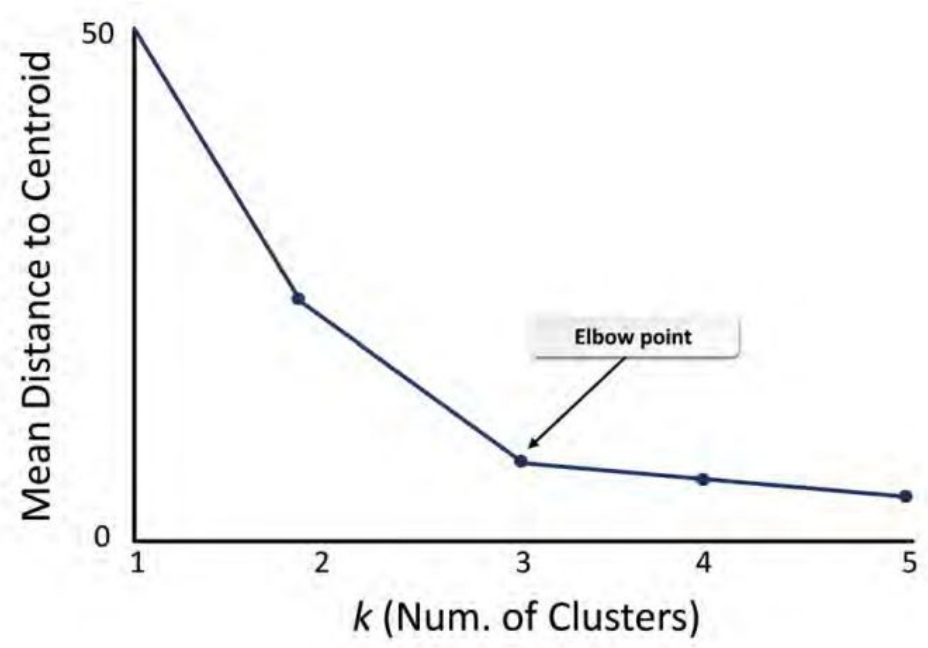
METHOD 1: Based on your **knowledge of the domain**, you may be able to place useful constraints on the data.

For example, assume you have an unlabeled dataset of **different fruits**. In this case, you know that each example is one of **three possible fruits**, and that each feature describes properties of a fruit like its shape, size, color, etc.

k DETERMINATION

METHOD 2: ELBOW POINT

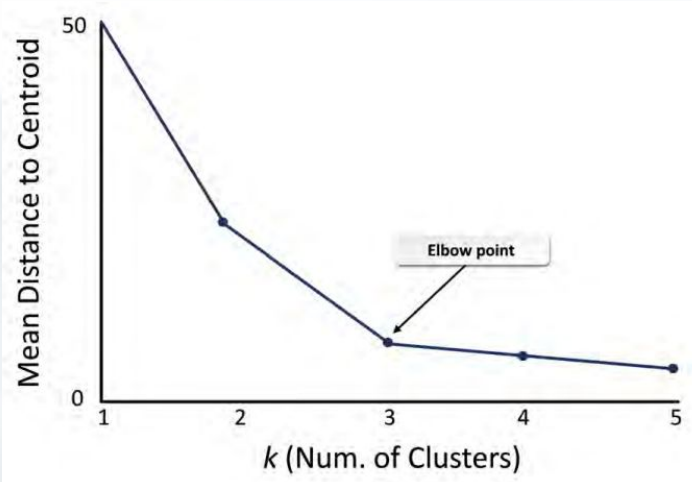
Calculate the mean distance between each data example and its associated centroid. As k increases, the mean distance necessarily decreases.



k DETERMINATION

METHOD 2: ELBOW POINT

However, at some point, increasing k becomes pointless and **doesn't reduce the mean distance in any significant way.** The point at which the mean distance no longer decreases in a significant way is called the elbow point, and it's usually a good indicator of what k should be

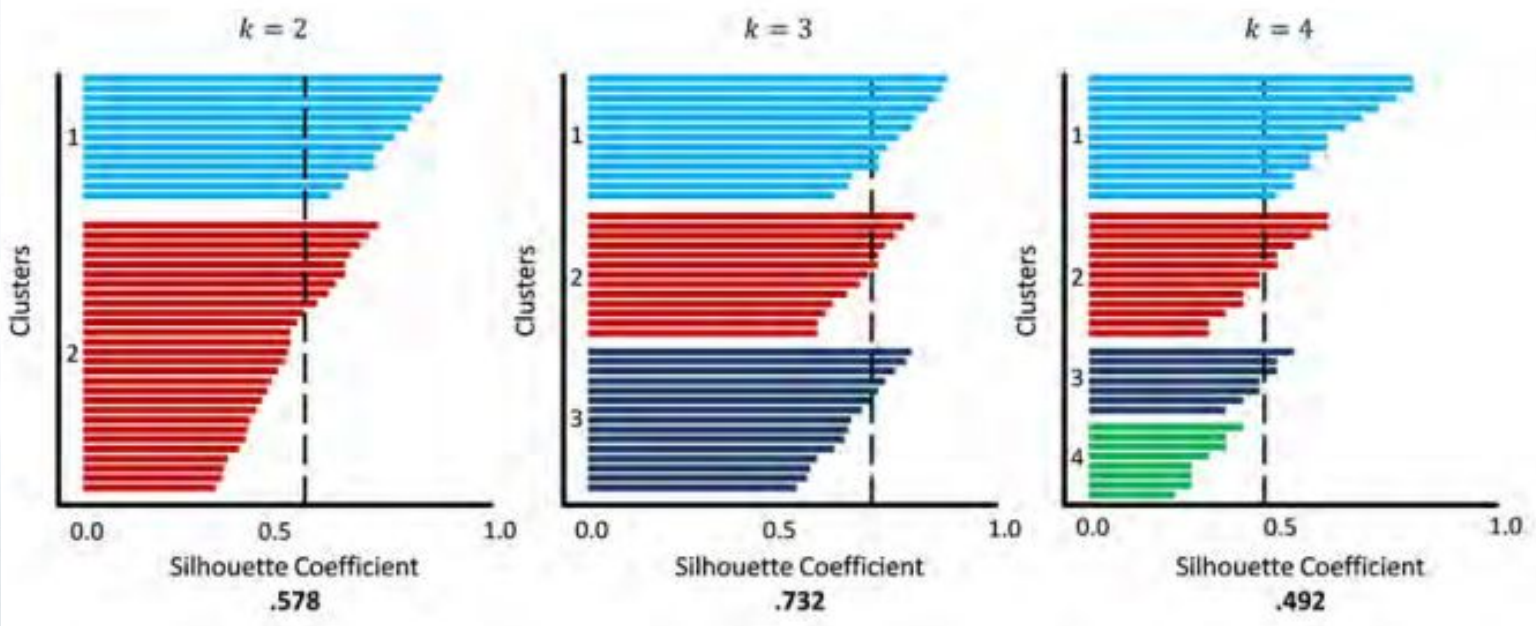


k DETERMINATION

3 Other Methods for k Determination **(SILHOUETTE ANALYSIS | DUNN INDEX | Davies–Bouldin index)**	
COMPACTNESS	**SEPERATION**
WITHIN-CLUSTER SUM OF SQUARES (WCSS)	**BETWEEN**-CLUSTER SUM OF SQUARES (BCSS)
measuring the distance between a **given data point** and **its cluster centroid**	measuring the distance between a **given centroid** and **all other centroids**
smaller values are **ideal**	**larger** values are **ideal**

k DETERMINATION

METHOD 3: SILHOUETTE ANALYSIS



k DETERMINATION

METHOD 3: SILHOUETTE ANALYSIS (It calculates Compactness and Separation)

A method of calculating how well a particular data example fits within a k-means cluster as compared to its **neighboring clusters**

k DETERMINATION

METHOD 3: SILHOUETTE ANALYSIS (It calculates Compactness and Separation)

The Coefficient value is between -1 and 1

1 Data is far from its neighbour (best case)

0 Near the decision boundary between clusters

-1 Data is closer to a neighboring cluster than its own. It's in the wrong cluster (worst case)

k DETERMINATION

METHOD 4: DUNN INDEX (It calculates Compactness and Separation)

1. Calculate smallest distance between two data points not in same cluster.
2. Calculate largest distance between two data points within same cluster.
3. Calculate ratio of values.

- Value above 0 and has no maximum value
- The higher the index, the better

k DETERMINATION

METHOD 5: Davies–Bouldin index (It calculates Compactness and Separation)

1. Calculates average ratio of **within-cluster** distance and **between-cluster** distance for each cluster as compared to its most similar cluster.

- Value above 0 and has no maximum value
- The lower the index, the better

ACTIVITY TIME

Building Clustering Models

k-Means Using Google Slides

ACTIVITY 7-1: Building a k-Means Clustering Model

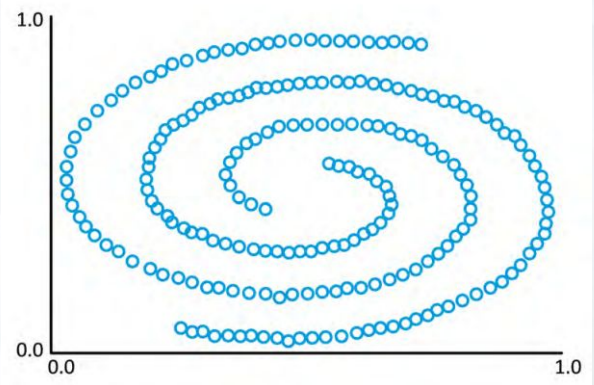
ACTIVITY 7-2: (Optional) Building a Clustering Model for Customer Segmentation

LESSON 7
Building Clustering Models

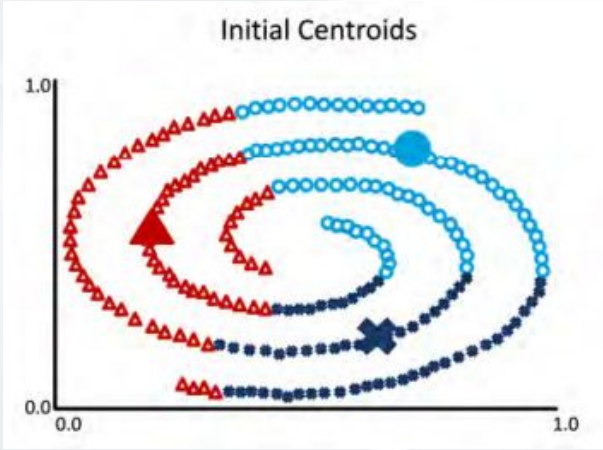
Topic A Build k-means Clustering Models

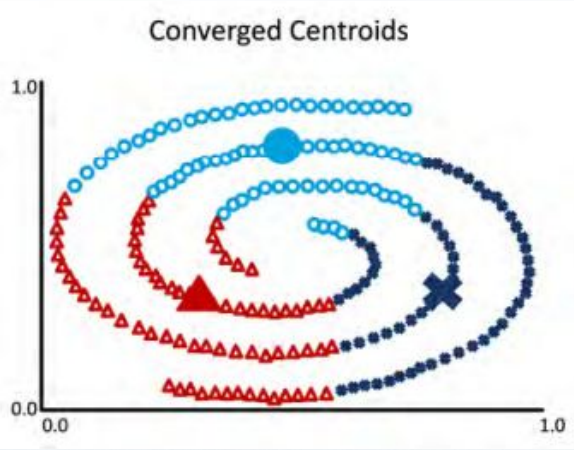
Topic B Build Hierarchical Clustering Models

k-Means CLUSTERING SHORTCOMINGS



SPIRAL DATA





HIERARCHICAL CLUSTERING

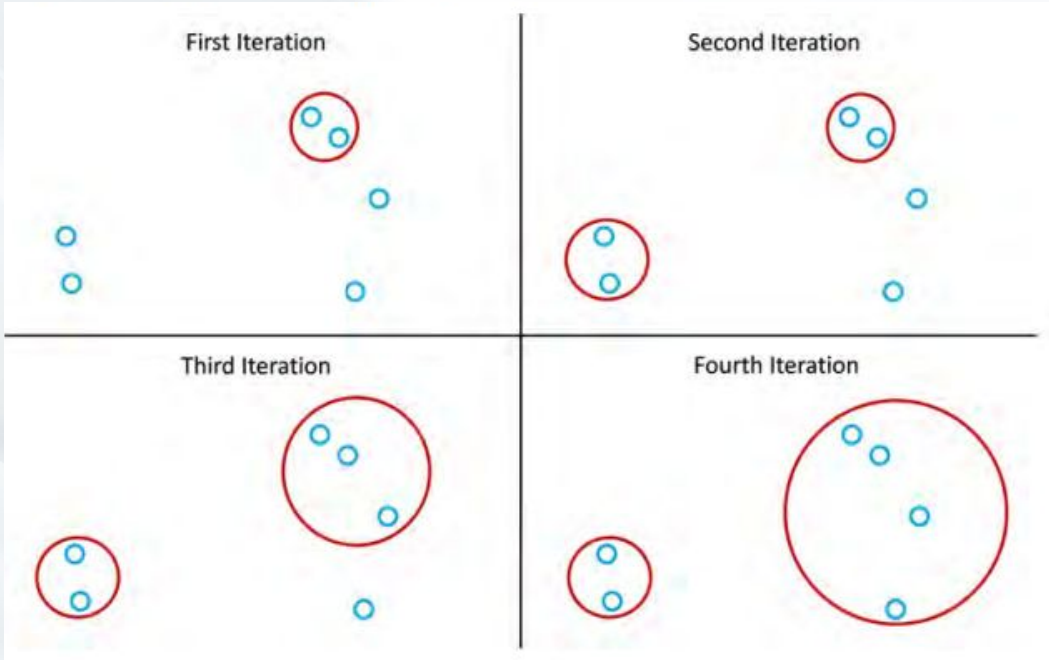
1- Hierarchical Agglomerative Clustering (HAC) (Bottom Up)

The 2 closest points

are grouped together

Example: 2 Clusters

SIMULATOR



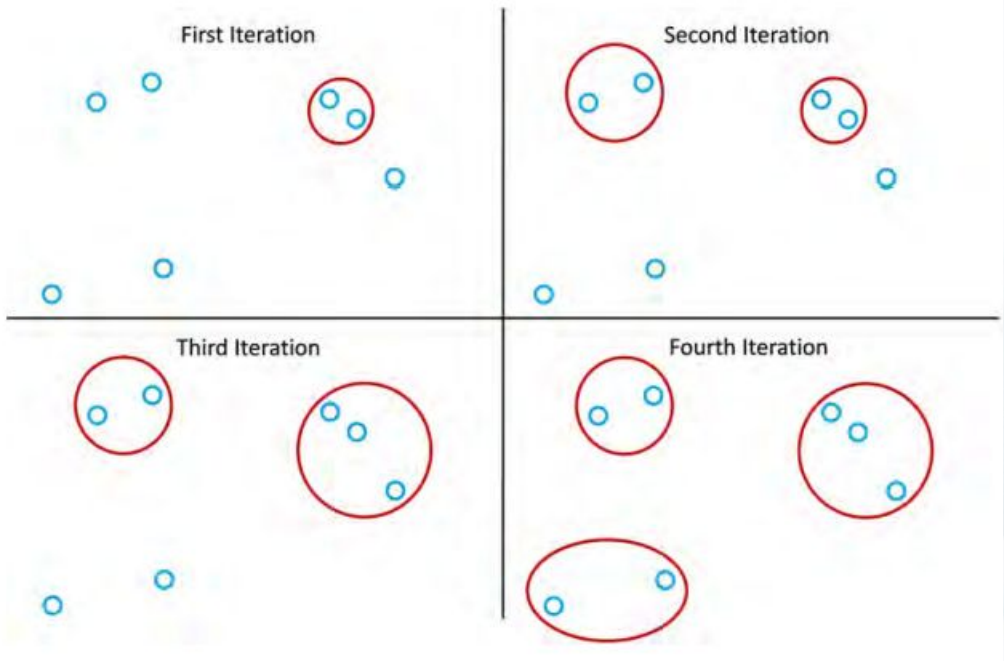
HIERARCHICAL CLUSTERING

1- Hierarchical Agglomerative Clustering (HAC) (Bottom Up)

The 2 closest points

are grouped together

Example: 3 Clusters



HIERARCHICAL CLUSTERING

2- Hierarchical divisive cluster (HDC) (Top Down)

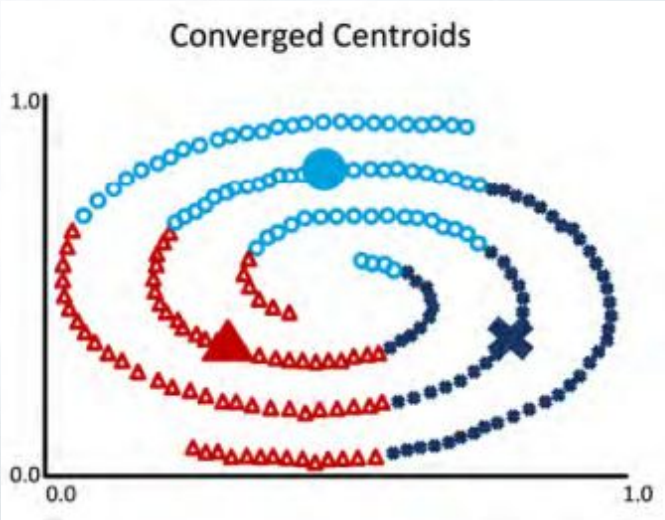
Opposite of HAC—all examples start out in the same single cluster, then this cluster is split using a particular method (It is **more complex** than HAC. There are many ways to split the cluster). The process continues until some stopping **criterion** is reached.

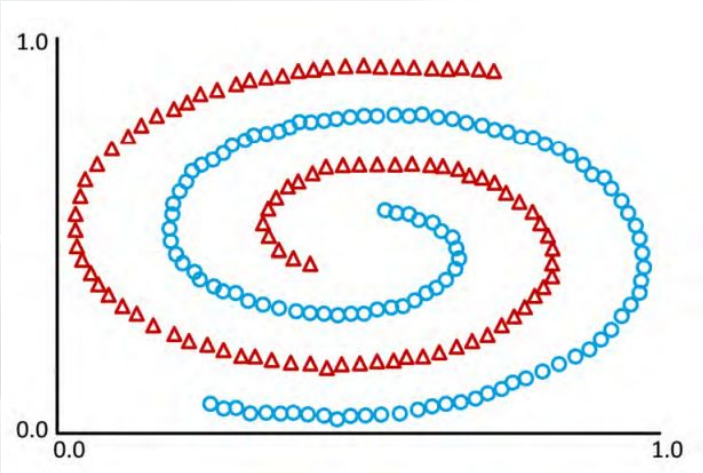
HAC VS HDC

While **HAC** has the advantage of its **simplicity**, **HDC** tends to be more **efficient** if the complete hierarchy is not created all the way to the bottom level. Also, **HDC** can be more **skillful** in its segmentation than HAC due to the **higher level of complexity** involved in its splitting decisions.

COMPARISON

BEFORE k-Means	AFTER HIERARCHICAL CLUSTERING





WHEN TO STOP HIERARCHICAL CLUSTERING

Without any stopping criterion, **HAC** will eventually **merge all data examples into a single cluster**.

Likewise, **HDC** will **keep splitting clusters until each data point is by itself**.

Neither is ideal.

So, you need to choose a point at which to stop the hierarchy from continuing in either case.

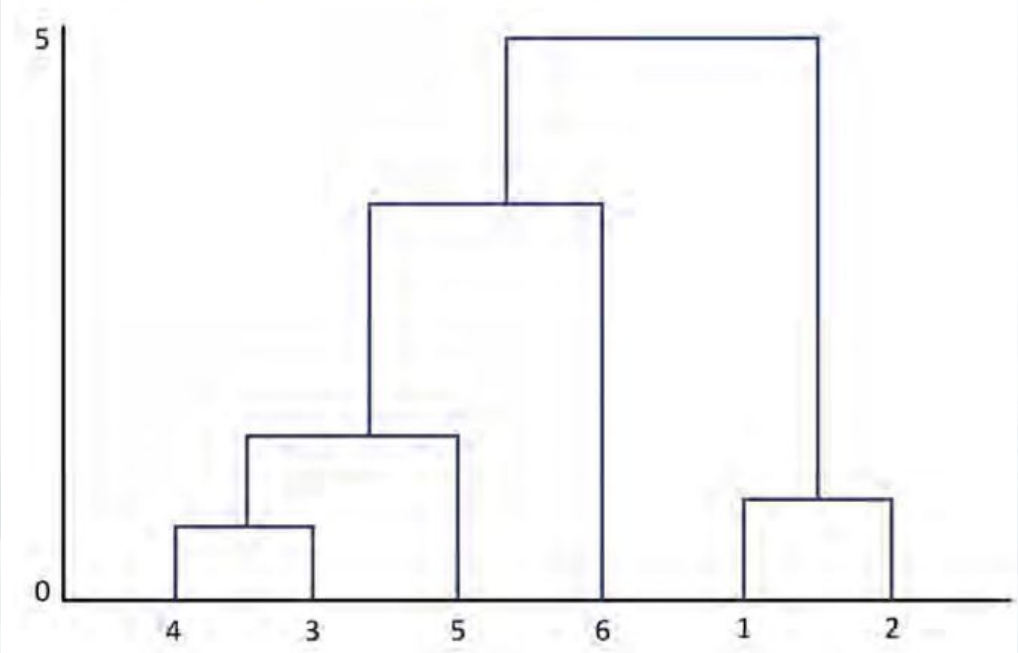
WHEN TO STOP HIERARCHICAL CLUSTERING

I can use any of the previous methods from k-Means, in addition to a **new one** (#6):

1. Manual (Domain Knowledge)
2. Elbow Point
3. Silhouette Analysis
4. Dunn Index
5. Davies–Bouldin index
6. **Dendrogram**

HIERARCHICAL CLUSTERING ANALYSIS

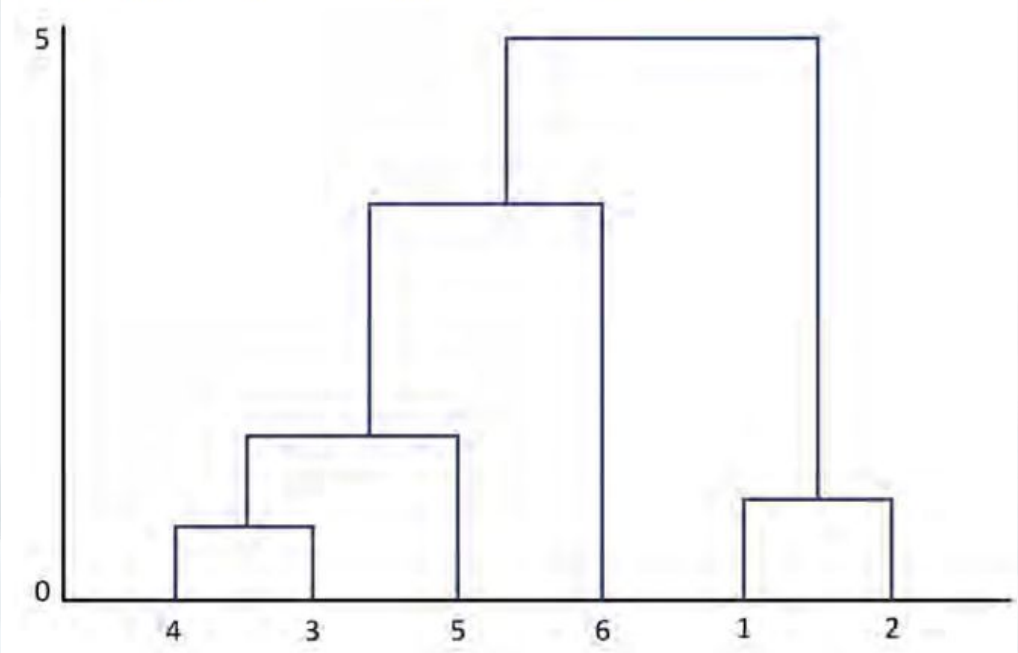
A dendrogram is a diagram that represents a **tree-like** hierarchy. It visually demonstrates which particular data points and clusters are either being **merged** (HAC) or **split** (HDC). SIMULATOR



HOW TO READ A DENDROGRAM

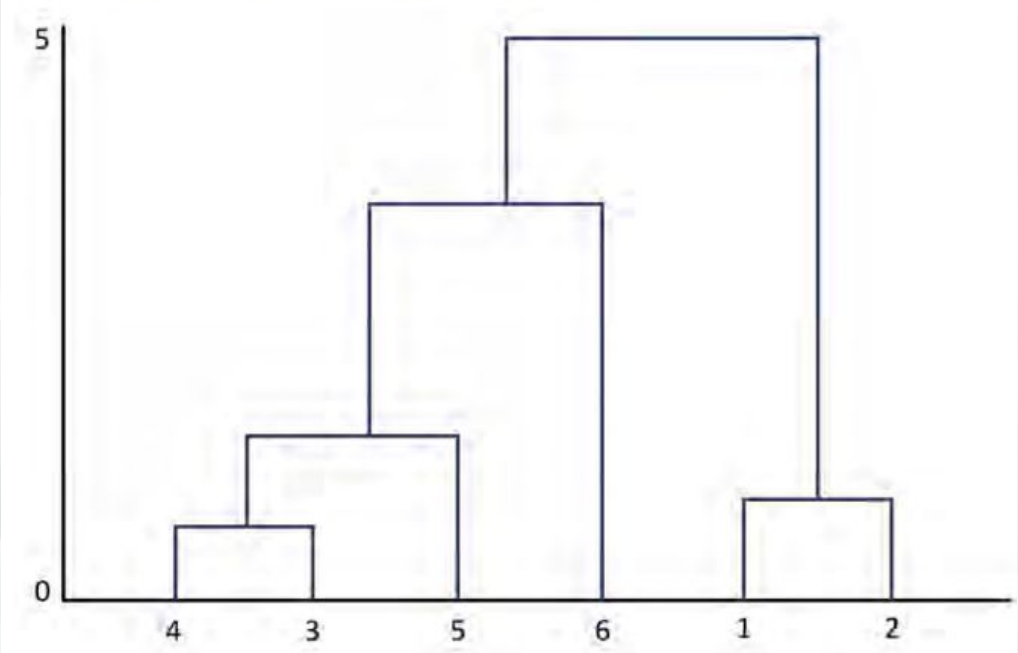
X-axis: data examples.

Y-axis: measurement of how distant the data examples and clusters are from one another



HOW TO READ A DENDROGRAM

From the x-axis, and as you go up the y-axis, each data example is placed into a cluster, and those clusters begin merging with other clusters.

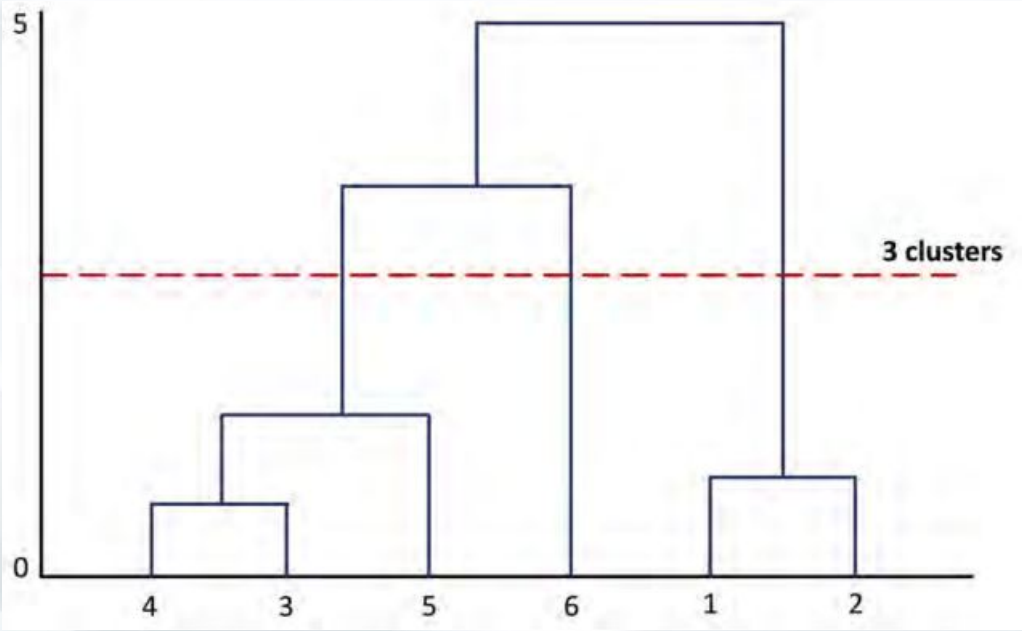


HOW TO CHOOSE A CUTOFF LINE

1- Domain Knowledge

OR

2- Calculating the distance of the longest "branch" before it begins to merge



ACTIVITY TIME

Building Clustering Models

ACTIVITY 7-3: Building a Hierarchical Clustering Model

LESSON 7 SUMMARY ACTIVITY

What type of data in your organization do you think might be conducive to clustering analysis?

What types of clustering evaluation metrics do you think you'll rely on most to determine the optimal number of clusters? Why?

COURSE OUTLINES

LESSON 1	Solving Business Problems Using AI and ML
LESSON 2	Preparing Data
LESSON 3	Training, Evaluating, and Tuning a Machine Learning Model
LESSON 4	Building Linear Regression Models
LESSON 5	Building Forecasting Models
LESSON 6	Building Classification Models Using Regression and k-NN
LESSON 7	Building Clustering Models
LESSON 8	**Building Decision Trees and Random Forests**
LESSON 9	Building Support-Vector Machines
LESSON 10	Building Artificial Neural Networks
LESSON 11	Operationalizing Machine Learning Models
LESSON 12	Maintaining Machine Learning Operations

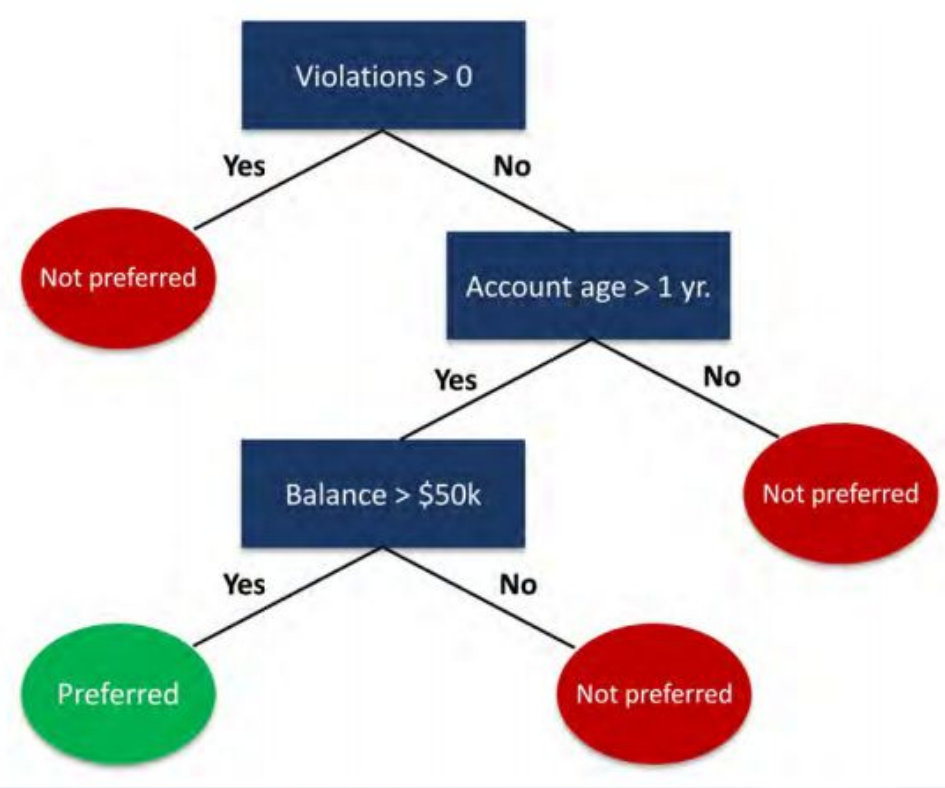
LESSON 8
Building Decision Trees and Random Forests

Topic A **Building Decision Tree Models**
Topic B Build Random Forest Models

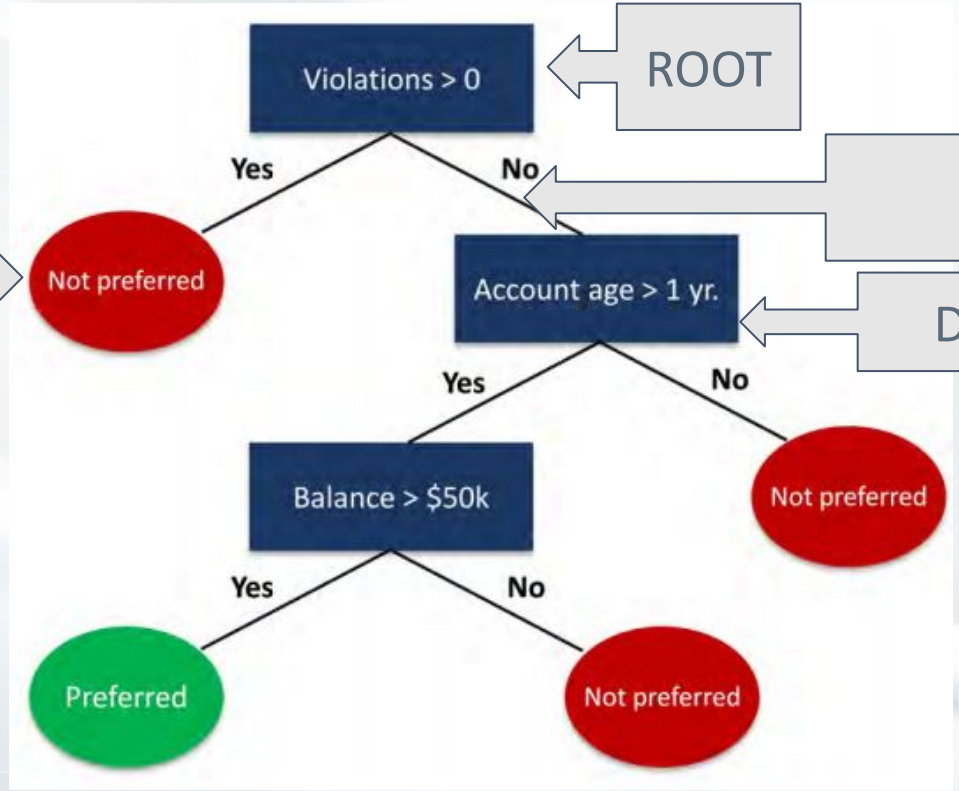
DECISION TREES

Alternative **SUPERVISED** approaches to solving **regression** and **classification** problems

Arrangement of conditional statements (true or false) and their conclusions in a branch–leaf structure

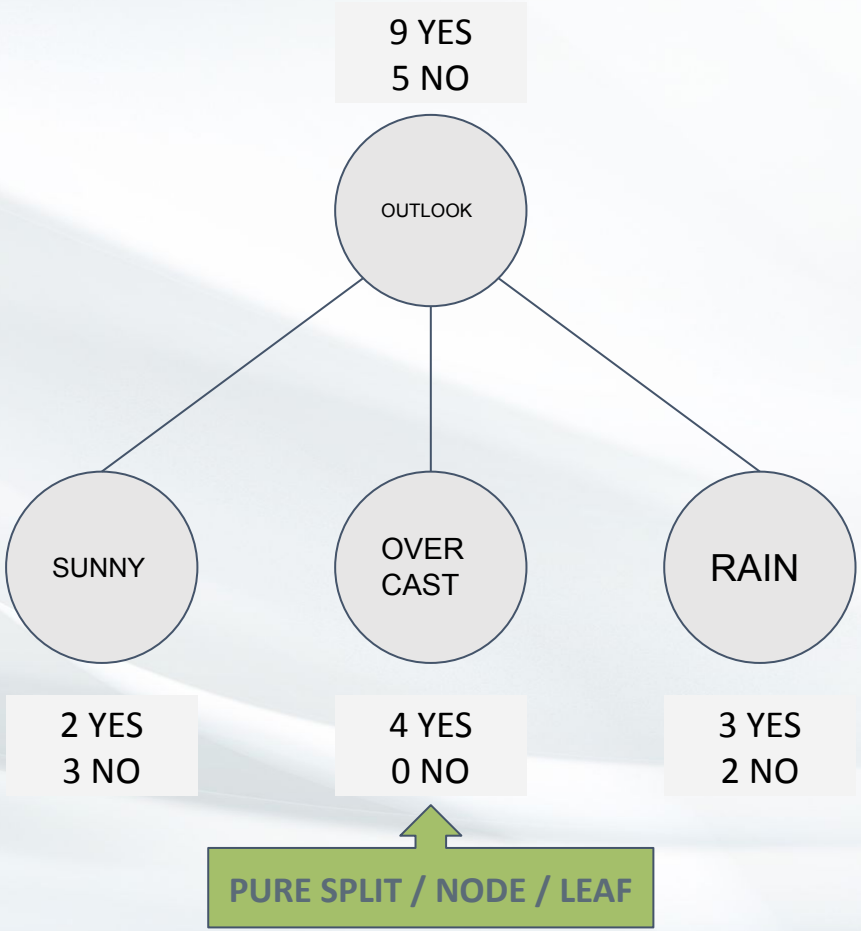


DECISION TREES



DECISION TREES

Day	outlook	temperature	humidity	wind	Decision
1	sunny	hot	high	weak	No
2	sunny	hot	high	strong	No
3	overcast	hot	high	weak	Yes
4	rainfall	mild	high	weak	Yes
5	rainfall	cool	normal	weak	Yes
6	rainfall	cool	normal	strong	No
7	overcast	cool	normal	wtrong	Yes
8	sunny	mild	high	weak	No
9	sunny	cool	normal	weak	Yes
10	rainfall	mild	normal	weak	Yes
11	sunny	mild	normal	strong	Yes
12	overcast	mild	high	strong	Yes
13	overcast	hot	normal	weak	Yes
14	rainfall	mild	high	strong	No

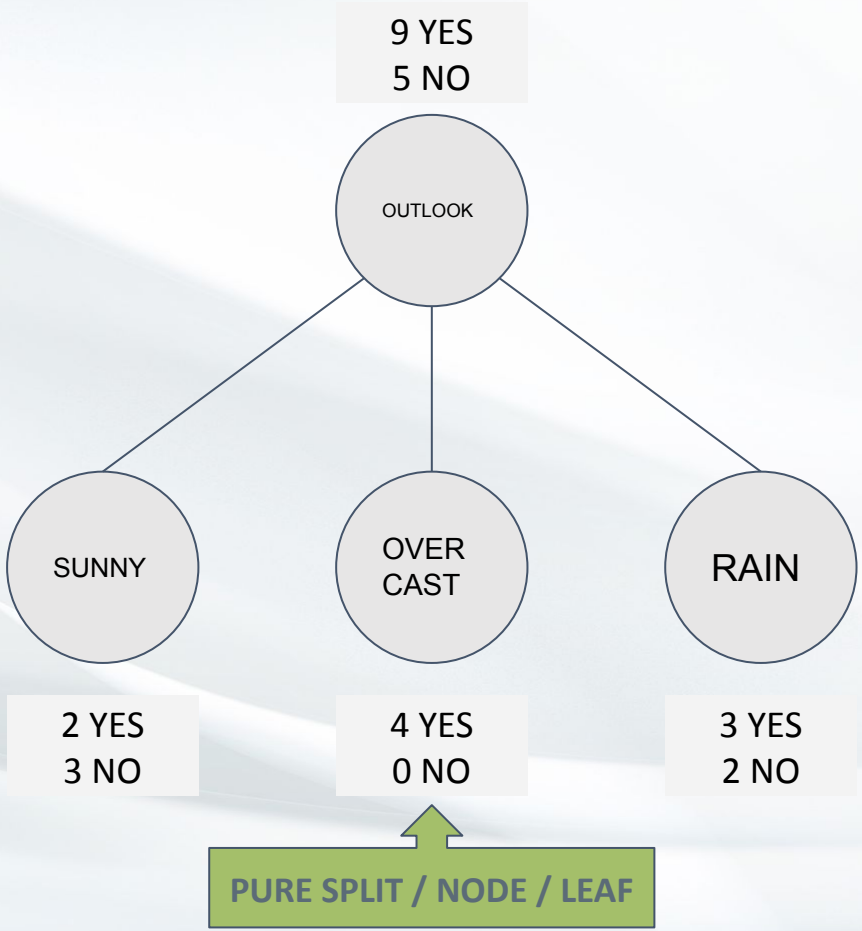


DECISION TREES

Day	outlook	temperature	humidity	wind	Decision
1	sunny				No
2	sunny				No
3	overcast				Yes
4	rainfall				Yes
5	rainfall				Yes
6	rainfall				No
7	overcast				Yes
8	sunny				No
9	sunny				Yes
10	rainfall				Yes
11	sunny				Yes
12	overcast				Yes
13	overcast				Yes
14	rainfall				No

PURE SPLIT

Regardless of all other features, whenever I use **OUTLOOK** and I choose **OVERCAST**, I directly know that this person is going to play (**no more splitting required**)



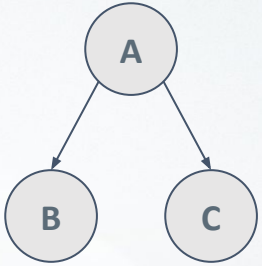
DECISION TREES

In the previous exercise, we selected **OUTLOOK** to be our **ROOT NODE**

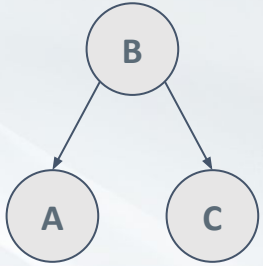
But, what who said that this feature (OUTLOOK) is the best to be chosen as a ROOT NODE?

- INFORMATION GAIN

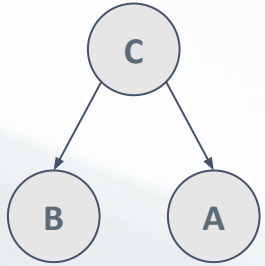
INFORMATION GAIN



INFORMATION GAIN
0.055



INFORMATION GAIN
0.049



INFORMATION GAIN
0.062

THIS ONE WILL BE **CHOSEN**

DECISION TREES

In the previous exercise, we knew that **OVERCAST** is the **purest** node by **looking** at it (manual way).

But, what if I want a **more efficient way**?
In other words, what if I need to **calculate the purity automatically**?
What if I want to know which one is a **pure split / node**?

I do a PURITY TEST

- ENTROPY
- GINI INDEX / GINI IMPURITY / GINI COEFFICIENT

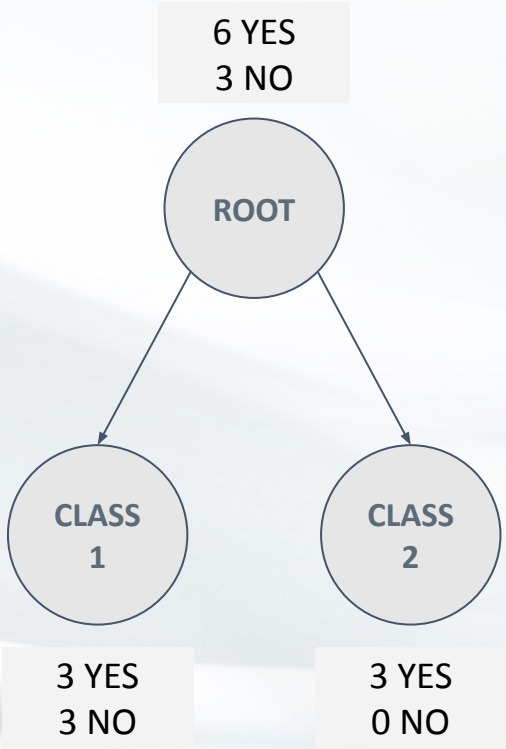
ENTROPY

Entropy helps me calculate the purity of a subset / split.
Entropy: Whether the split is pure or not
Entropy / purity ranges between 0 and 1

0 ⇒ completely **PURE** subset (100% | 0%)
1 ⇒ completely **IMPURE** subset (50% | 50%)

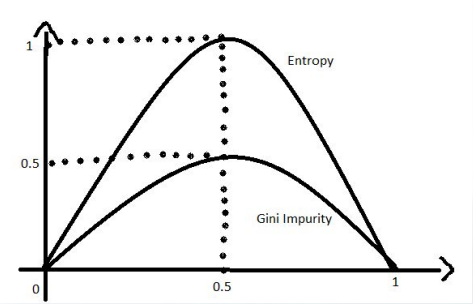
Ex. **CLASS1**: 3 YES (50%) | 3 NO (50%) ⇒ Entropy / Purity = 1 ⇒ IMPURE

Ex. **CLASS2**: 3 YES (100%) | 0 NO (0%) ⇒ Entropy / Purity = 0 ⇒ PURE (Stop splitting) ⇒ Leaf Node



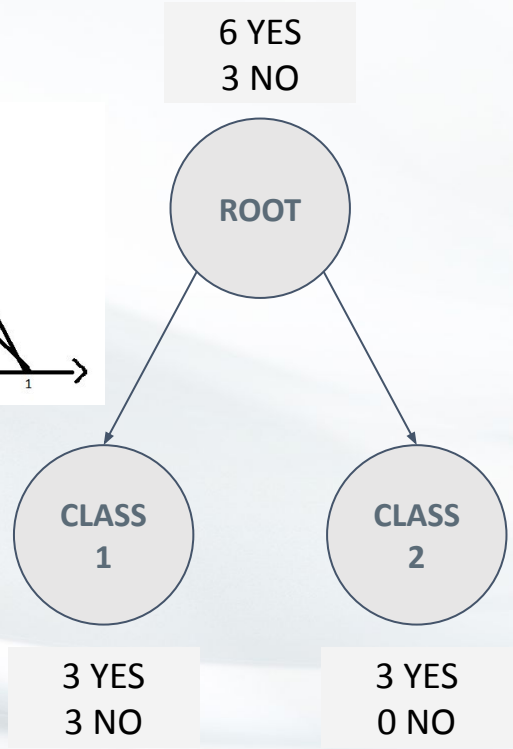
GINI INDEX / GINI IMPURITY / GINI COEFFICIENT

Similar concept to ENTROPY, but **different FORMULA**
Less value of Gini is better
Ex. **CLASS1**: 3 YES (50%) | 3 NO (50%) ⇒
Entropy / Purity = 1 ⇒ IMPURE
GINI INDEX = 0.5



Ex. **CLASS2**: 3 YES (100%) | 0 NO (0%) ⇒
Entropy / Purity = 0 ⇒ PURE (Stop splitting) ⇒ Leaf Node
GINI INDEX = 0 ⇒ PURE (Stop splitting) ⇒ Leaf Node

GINI INDEX is **faster** to compute than ENTROPY
For large datasets, GINI INDEX is better | For small datasets, ENTROPY is better



DECISION TREE ALGORITHM

CART	A decision tree algorithm that constructs trees using the **GINI** index: Which node should be the root node Gini index splits **training set into two** based on a feature with a threshold. Example: Split "Violations" based on greater than 0. we would prefer choosing the attribute/feature with the **least Gini index as the root node**
C4.5	Purity: all examples at decision node belong to only one of the classes (**lowest GINI index** = highest purity = **purest one on top**)

DECISION TREE ALGORITHM - GINI INDEX

Account age > 1	Violations > 0	Balance > $50,000	Preferred
Yes	No	Yes	No
Yes	Yes	Yes	No
No	No	Yes	No
Yes	No	Yes	Yes
Yes	No	No	No
Yes	No	Yes	Yes
No	No	Yes	Yes
No	Yes	Yes	No

DECISION TREE ALGORITHM - GINI INDEX

- Calculate Gini index for "Account age" = "Yes".

$$G = 1 - \left((2/5)^2 + (3/5)^2\right) = 0.48$$

- Calculate Gini index for "Account age" = "No".

$$G = 1 - \left((1/3)^2 + (2/3)^2\right) = 0.44$$

- Weighted sum of both:

$$G = (5/10) \cdot 0.48 + (5/10) \cdot 0.44 = 0.46$$

DECISION TREE ALGORITHM - GINI INDEX

- Process is **repeated** for other **two features**.
- When calculated, the resulted indices are:
  - Account age = 0.46
  - Violations = 0.38
  - Balance = 0.43
- "Violations" has **lowest GINI index**, so it will be root decision node.
- CART splits dataset into another subset, evaluating purity again.
  - **Repeats** process for sub-subsets, etc.
- Splitting stops when:
  - There's no more impurity to reduce.
  - Tree reaches maximum depth specified in hyperparameter.

DECISION TREE ALGORITHM

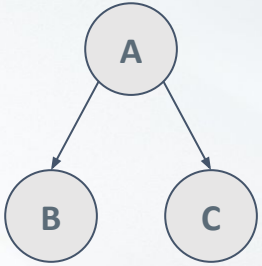
CART	It **DOES NOT** use GINI Index. It uses **Information Gain Ratio** A decision tree algorithm (for binary and multi class classification) that constructs trees with regard to **entropy** (between 0 and 1) and **information gain**. Entropy is how "ordered" data is. Similar to purity, but weights class probability using logarithm.
C4.5	Splitting is determined by information gain ratio. ● Entropy of parent node minus entropy of child nodes. ● Information is **gained** as entropy is **reduced**. ● **Feature with highest gain is root node.** ● Splitting process is repeated.

DECISION TREE ALGORITHM

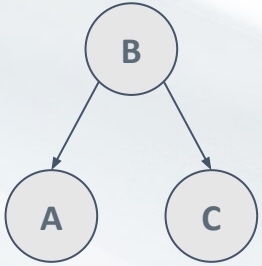
CART

C4.5

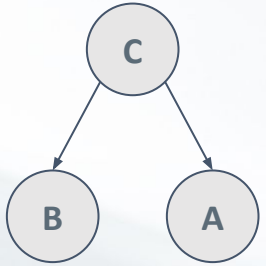
The combination that has the **HIGHEST INFORMATION GAIN** will be **chosen**



INFORMATION GAIN
0.052



INFORMATION GAIN
0.049



INFORMATION GAIN
0.062

THIS ONE WILL BE **CHOSEN**

PRE-PRUNING VS PRUNING

Note: **Similar** to **Feature Selection** in Dimensionality Reduction

PRE-PRUNING: Limiting the growth of the tree so **instead** of cutting of branches after they have been generated, we are **preventing** the branches from **growing** in the **first place**.

PRUNING: The process of **reducing the overall size** of a decision tree by **eliminating** nodes, branches, and leaves that provide **little value** for the classification or regression problem at hand.

WHY WE USE IT: Reduce overfitting, simplifying the model, and improving its overall performance, score and skill

PRUNING METHODS

REDUCED-ERROR	Converts subtree into single leaf (most common classification). Subtree with **highest reduction in error is pruned**. Simple and quick—a good first option.
MINIMUM ERROR	Error rate is weighted according to examples at branch. **If error of pruning subtree is smaller than not pruning it, it is pruned.** Doesn't do well with **class imbalance** or with many classes.
COST COMPLEXITY	"Worth" of subtree is increase in error cost of pruning, divided by number of leaves. **Subtrees with lowest "worth" are pruned.** **Slower** than other methods.

CONTINUOUS VARIABLE DISCRETIZATION

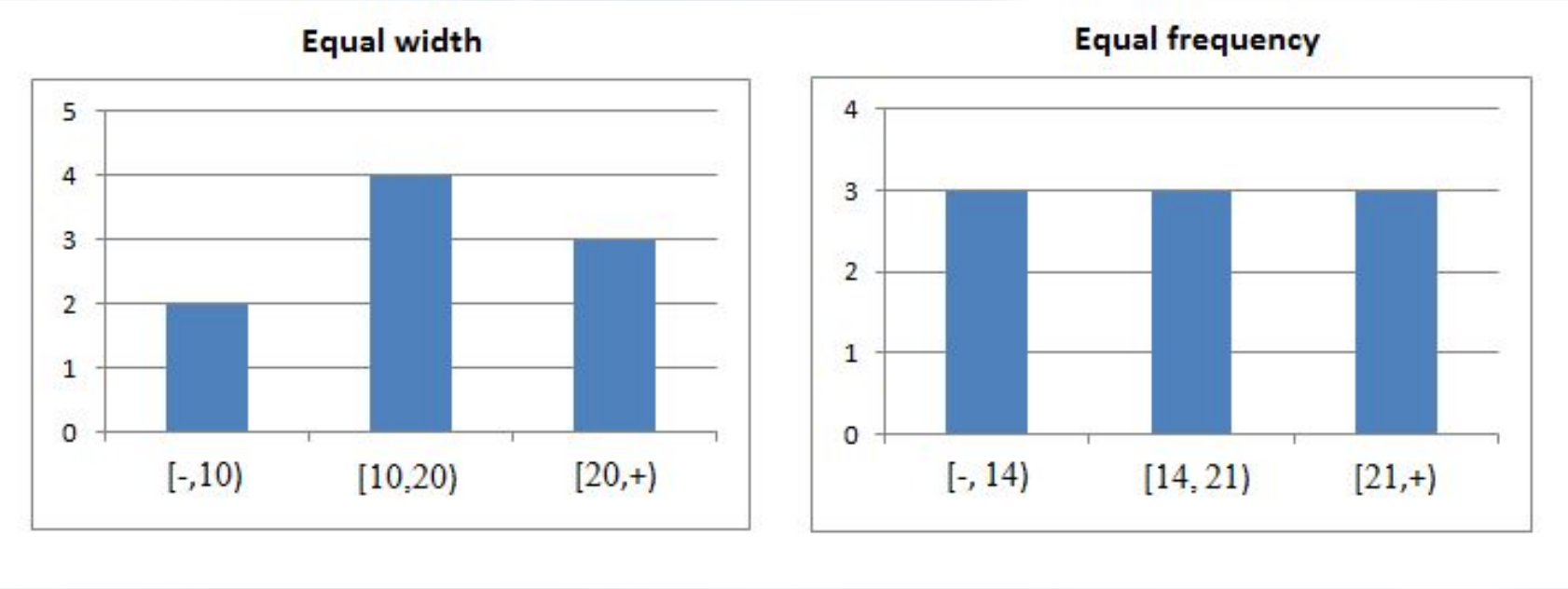
PROBLEM: Trees are very sensitive to continuous numbers
When you have a decision tree that is based on **continuous numbers** (ex. age), it is **not logical** to split your nodes for **each possible value** (ex. 15.5, 15.8, 15.9, …)

SOLUTION: That's why we do **DISCRETIZATION**: The process of converting a **continuous variable** into a **discrete variable** (prevent overfitting)

We take a continuous variable and place its values within specific, discrete intervals called **data binning**
CHALLENGE: I need to know **how many bins**, and the **range of values** for each bin (called **Width**)

BIN DETERMINATION

As for determining bin width, there are two primary approaches: **equal-width** and **equal-frequency**



ONE-HOT ENCODING

PROBLEM: scikit-learn's implementation of decision trees does not handle categorical values (Ex. Color: Red and Blue)

SOLUTION: Converting these values [Red and Blue] to [0 and 1] will make them Ordinal (Data that can be ranked - 1 better than 0)

Vehicle	isRED	isBLUE
Vehicle 1	0	1
Vehicle 2	1	0
Vehicle 3	1	0

DECISION TREES COMPARISON

	CART	C4.5
CLASSIFICATION	Binary Classification	Binary and Multiclass Classification
SPLITTING METRIC	Gini Index	Information Gain Ratio
PRUNING METHOD	Cost Complexity Pruning	Error-Based Pruning
SUPPORTS REGRESSION AND CLASSIFICATION?	Yes	Yes
SUPPORTS MULTI-CLASS SPLITTING?	No	Yes
SUPPORTS MISSING VALUES?	Yes	Yes

DECISION TREES COMPARED TO OTHER ALGORITHMS

ADVANTAGES	DISADVANTAGES
Do not require much data preparation to be effective unlike other algorithms	Overfitting (but can be a bit solved using pruning)
Simpler and easier to interpret	When it comes to regression tasks in which a dataset has many features, decision trees may not perform as well as linear regression
Faster than classification models (ex. KNN)	Decision trees are unable to derive this significance

ACTIVITY TIME

Building Decision Trees and Random Forests

ACTIVITY 8-1: Building a Decision Tree Model

LESSON 8
Building Decision Trees and Random Forests

Topic A Building Decision Tree Models

Topic B Build Random Forest Models

ENSEMBLE LEARNING

Are more models better than fewer models?

We need to aggregate (put together) models to improve accuracy / prediction and reduce cost **(More instances of the same algorithm)**
The classification that gets **the most votes (NOT the best score)** wins

Random forest is a type of Ensemble Learning

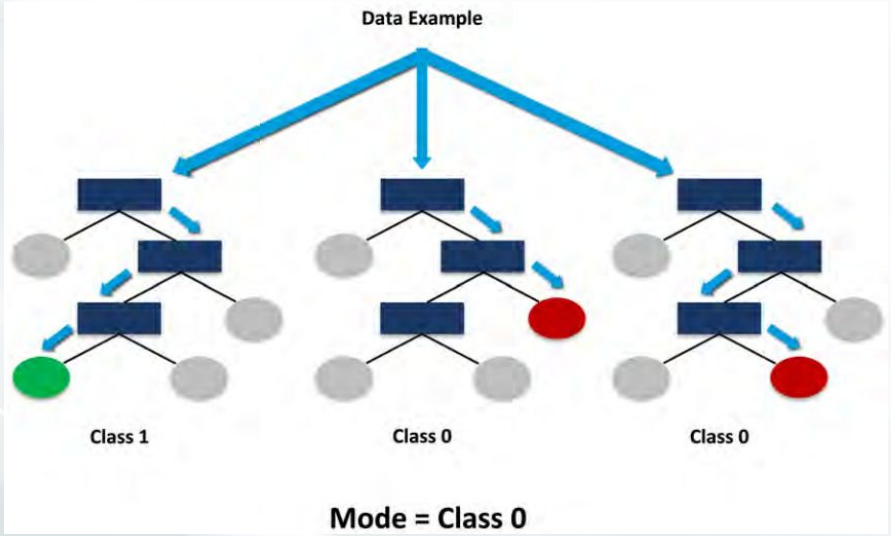
MODEL	ACCURACY	CLASS
Logistic Regression	75%	**Class 1**
SVMs	83%	Class 0
CART	80%	**Class 1**

RANDOM FOREST

Collection of Decision Trees

Method that aggregates multiple decision tree models together (ensemble learning) and selects either:

- The **mode** (**voting**) of the classifiers between all decision trees (**classification**)
- The **mean** (**average**) of the predictors between all decision trees (**regression**)



SAMPLING DATA

Before, we used Holdout Method or Cross Validation

PROBLEM: What if all trees predicted the same thing? Isn't it equivalent to using only one model instead?
That's why we need our trees to be different from each other

SOLUTION: we use 1- BAGGING (Bootstrap Aggregating) or 2- **Feature Randomness (Feature Selection)**

SAMPLING DATA

1- BAGGING (Bootstrap Aggregating)

Taking **samples of our datasets** in order to create **completely new datasets**
Example: The computer takes 2/3 and the remaining 1/3 is called **out of bag**

Similar to CROSS VALIDATION

SAMPLE		OUT OF BAG
FOLD 1	FOLD 2	FOLD 3

SAMPLING DATA

2- **Feature Randomness (Feature Selection)**

We limit the number of features that each tree can even consider splitting on

Example: [Name, Age, Sex]

TREE1: Name & Age
TREE2: Name & Sex

NOTE: Similar to the concept of SoftMax

OUT OF BAG ERROR

Even though **no one tree** sees all of the training data, the **entire forest does**. You can therefore **evaluate** the **performance** of **each tree** in the forest based on the out-of-bag data it hasn't seen, outputting an error value. Then, you'd take the **average** of each **error value** to determine the out-of-bag error for the entire ensemble.

ADVANTAGE: This helps the random forest to avoid overfitting, **much like** how k-fold cross validation helps non-ensemble methods avoid overfitting.

OUT OF BAG ERROR

EXAMPLE: An out-of-bag error of 0.19 indicates that 19% of the predictions were incorrect

The LESS the BETTER

The **advantage** of the OOB method over cross validation is that it requires **less computation** and allows us to **test the model <u>while it is being trained</u>**.

DECISION TREE VS RANDOM FOREST

Decision trees are simpler and faster but prone to overfitting to the training data

Random forests usually give better results and reduce variance and therefore help reduce overfitting issues.

ACTIVITY TIME

Building Decision Trees and Random Forests

ACTIVITY 8-2: Building a Random Forest Model

Note: n_estimators is the number of trees in the forest

LESSON 8 SUMMARY ACTIVITY

In your own environment, how might you use decision trees to solve business problems?

In your own environment, how might you use random forests to solve business problems?

COURSE OUTLINES

LESSON 1	Solving Business Problems Using AI and ML
LESSON 2	Preparing Data
LESSON 3	Training, Evaluating, and Tuning a Machine Learning Model
LESSON 4	Building Linear Regression Models
LESSON 5	Building Forecasting Models
LESSON 6	Building Classification Models Using Regression and k-NN
LESSON 7	Building Clustering Models
LESSON 8	Building Decision Trees and Random Forests
LESSON 9	**Building Support-Vector Machines**
LESSON 10	Building Artificial Neural Networks
LESSON 11	Operationalizing Machine Learning Models
LESSON 12	Maintaining Machine Learning Operations

LESSON 9
Building Support-Vector Machines

Topic A **Build SVM Models for Classification**
Topic B Build SVM Models for Regression

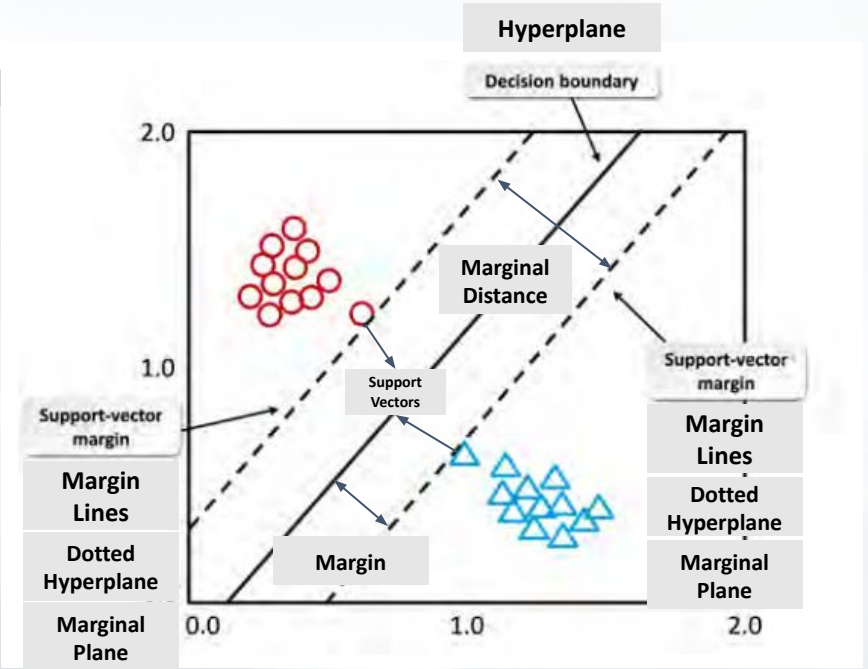
SVM: SUPPORT VECTOR MACHINE

Supervised learning algorithms used to solve both **classification** and **regression** problems.

AIM / OBJECTIVE

Easily separate 2 classes using a hyperplane and the biggest margin distance of the margin lines

Margin lines should be passing by at least 1 nearest positive **and** negative point (red or blue irrespectively)



SIMULATOR

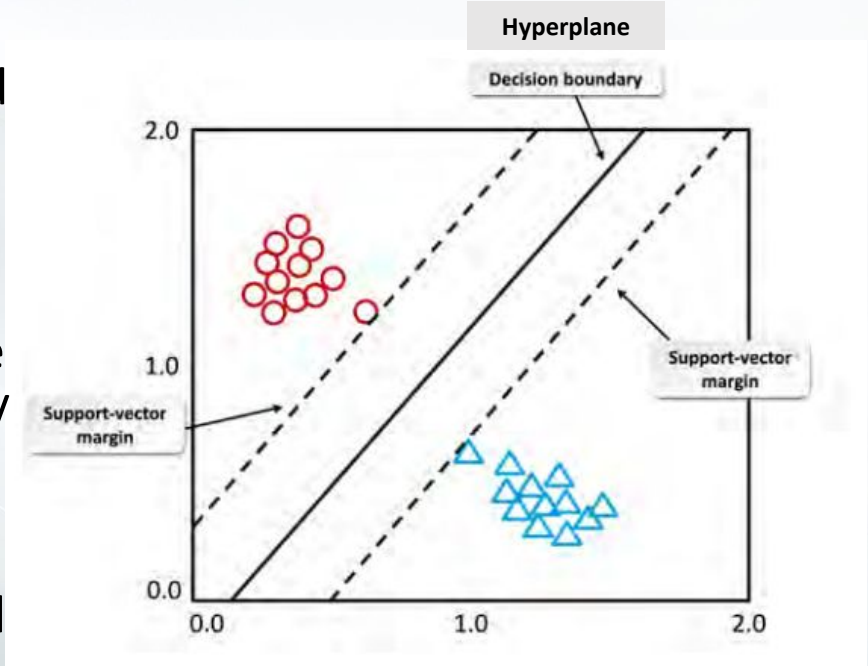
SIMULATOR2

SVM: SUPPORT VECTOR MACHINE

Supervised learning algorithms used to solve both **classification** and **regression** problems.

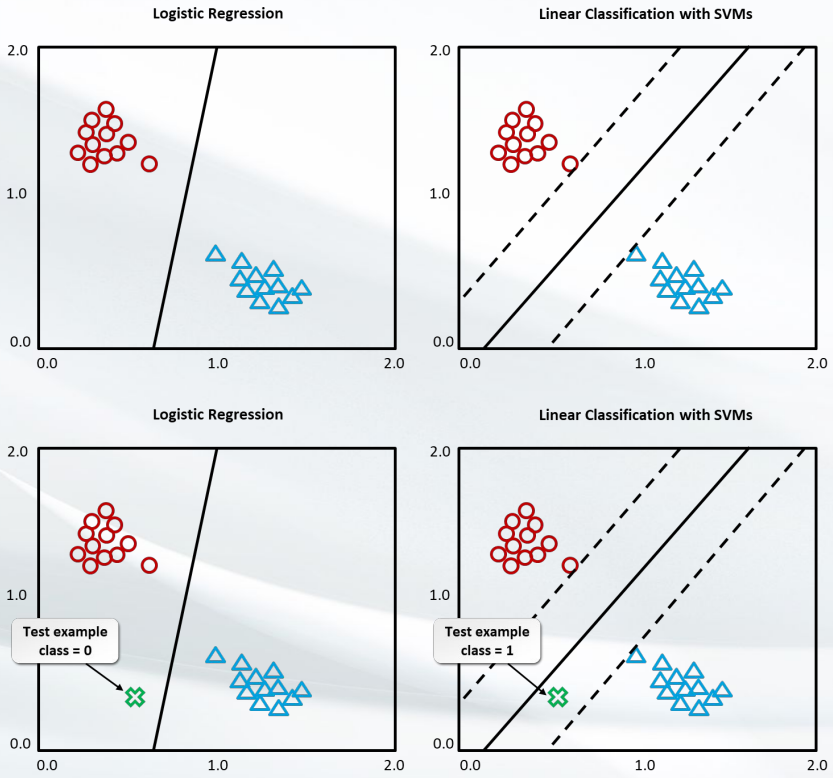
Frontier which best segregates the two classes (decision boundary / hyperplane / line)

Extreme Data Points (also called Support Vectors) are only important whereas other points are ignored

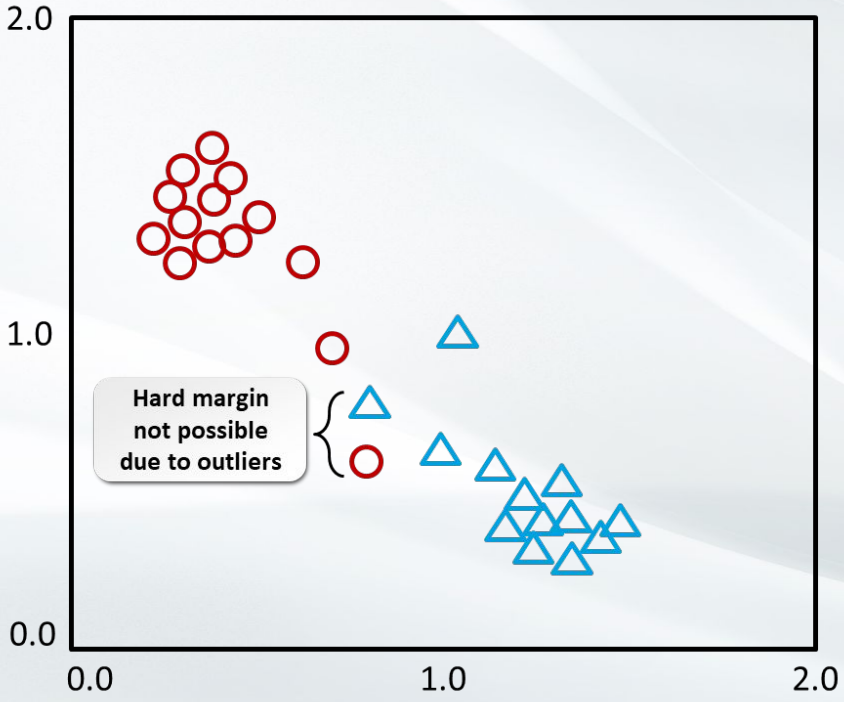


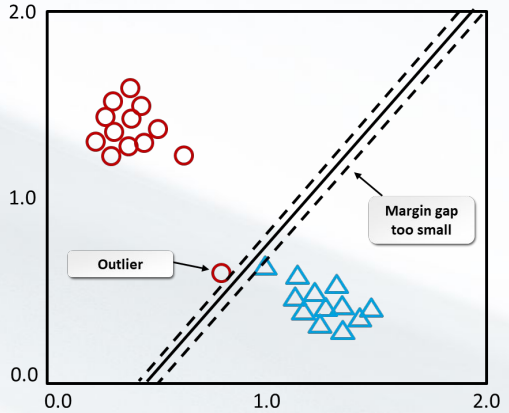
LOGISTIC REGRESSION VS SVM

SVM is better than Logistic Regression whenever we are classifying data with **outliers**



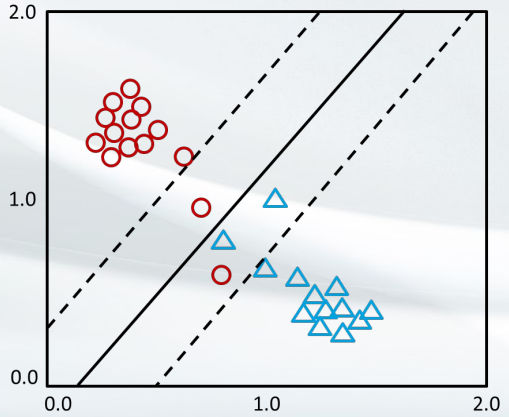
HARD VS SOFT MARGIN CLASSIFICATION





HARD MARGIN

MODEL **LESS** TOLERANT TO DATA WITHIN THE MARGIN = **OVERFITTING**



SOFT MARGIN

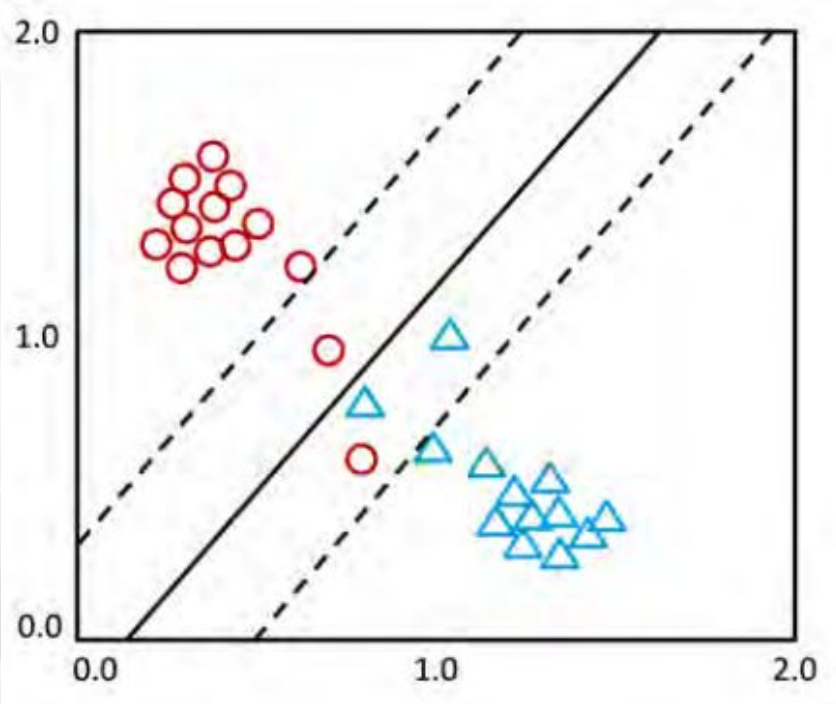
MODEL **MORE** TOLERANT TO DATA WITHIN THE MARGIN

AVOID OVERFITTING

SOFT-MARGIN CLASSIFICATION

Balance between keeping the distance between the margins as large as possible and minimizing the number of examples that end up inside the margins (the support vectors)

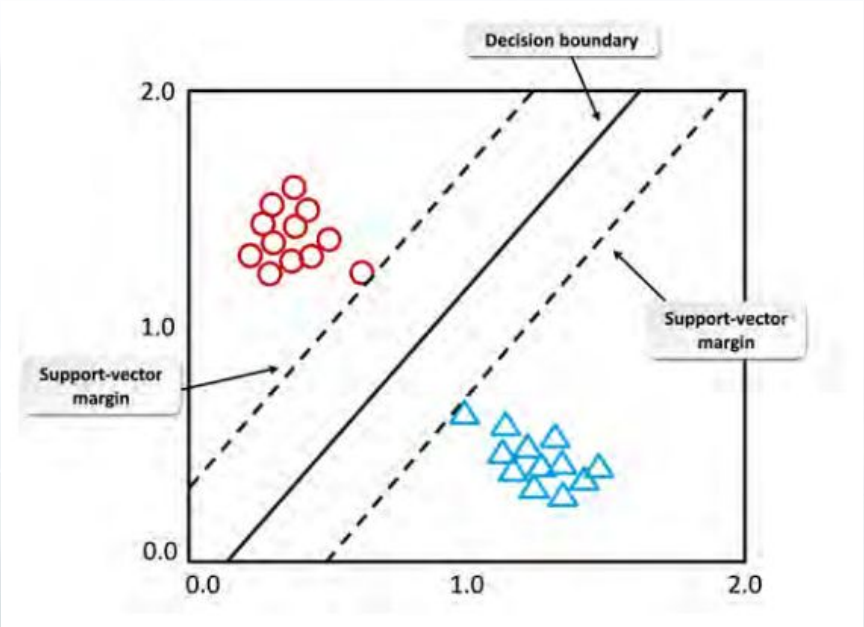
It is not perfect, but it is a tradeoff



SVM: SUPPORT VECTOR MACHINE

The **goal** of SVM Classifier is to ideally have a modal where there are **no data points** within the margins.

BUT, if that leads to **overfitting**, then we will **soften** the margins to generalize the model

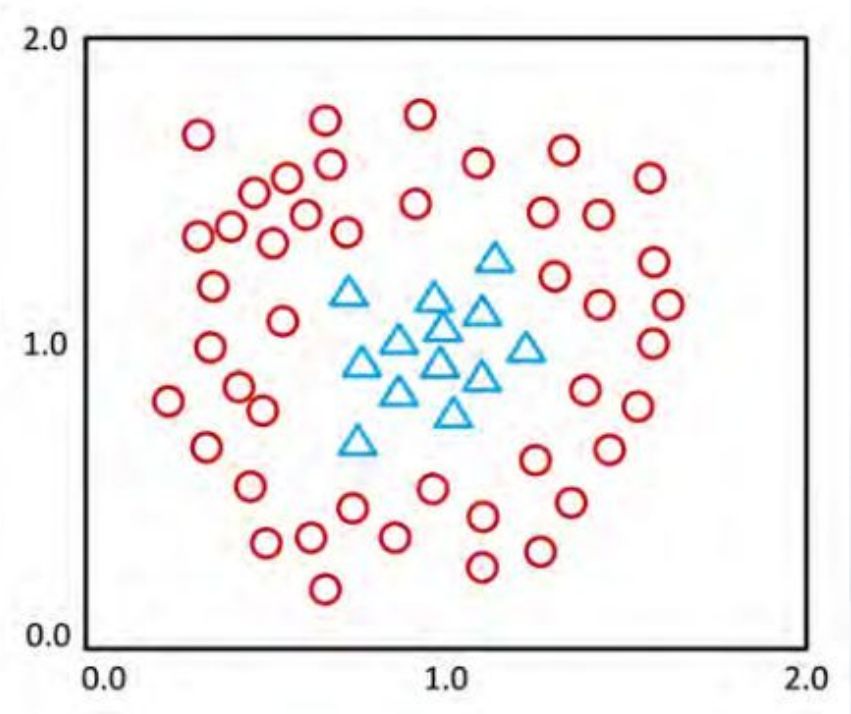


SVMS FOR NON-LINEAR CLASSIFICATION

The previous examples were successfully done using SVMS For Linear Classification.

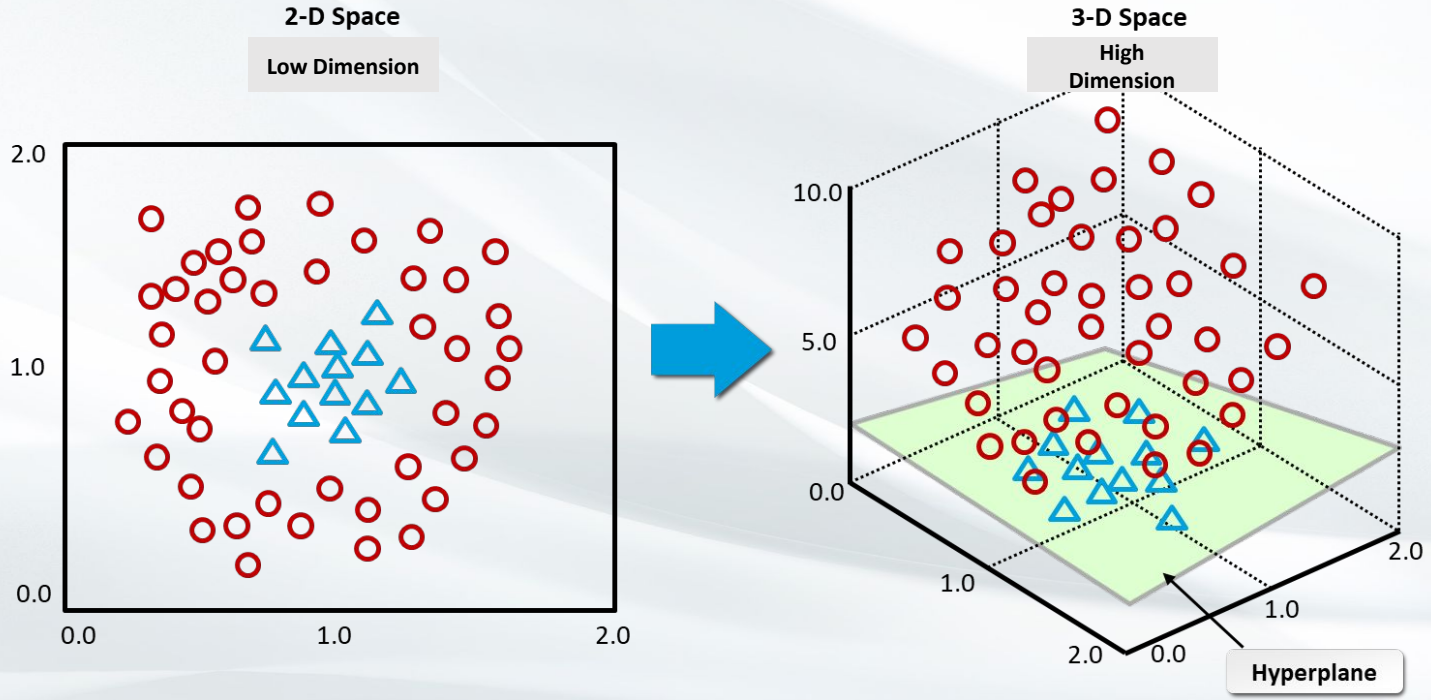
Do you think it is doable using this dataset?

How are we able to split the data?

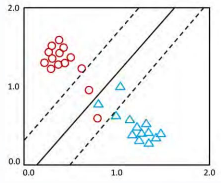
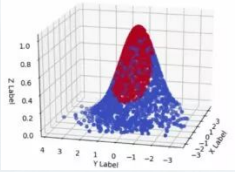
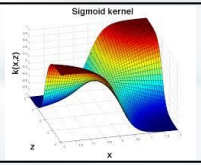


KERNEL TRICK

Representing nonlinearly separable data in higher-dimensional space



KERNEL METHODS

KERNEL METHOD	DESCRIPTION
Linear kernel	•Simple, only applies to linearly separable data. •Useful when feature space is large. •Popular in text classification. •Good first choice. 
Gaussian RBF kernel	•**Gauss function** projects new feature space in higher dimensions. •Measures distance between examples and centers. •Most effective when examples **greatly outnumber** features. •May be prone to **overfitting** unless regularized. 
Polynomial kernel	•Uses polynomial values in feature space projection. •High polynomial degree can lead to **overfitting**; low can lead to **underfitting**. ANIMATION •Less popular than RBF.
Sigmoid kernel	•Uses hyperbolic tangent function (tanh). SIMULATOR •Similar to a perceptron neural network. •Used in object detection/image classification. 

ACTIVITY TIME

Building Support-Vector Machines

ACTIVITY 9-1: Building an SVM Model for Classification

NOTE: SVC('C')			
High Value of C	Hard Margin	Narrow Lines	C = 100
Low Value of C	Soft Margin	Wide Lines	C = 0.1

LESSON 9
Building Support-Vector Machines

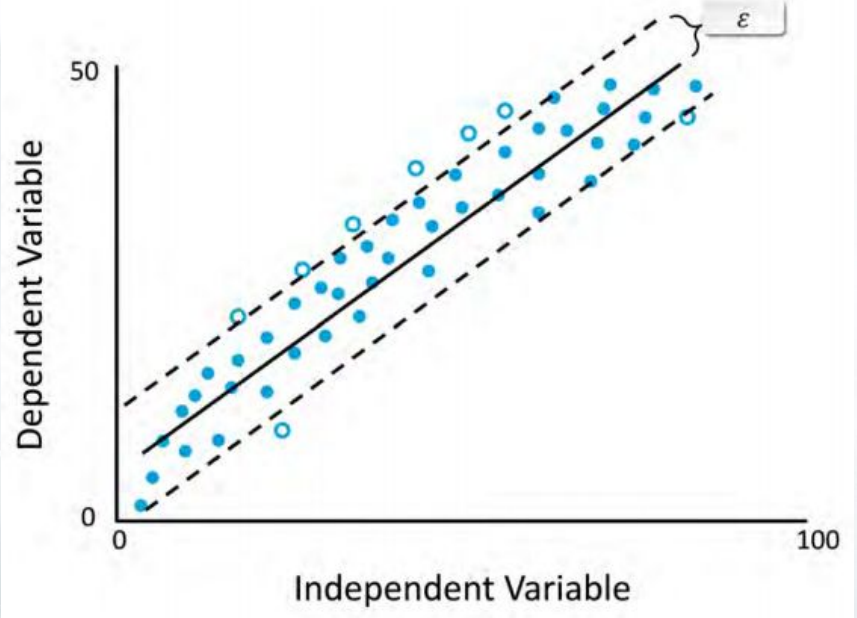
Topic A Build SVM Models for Classification

Topic B Build SVM Models for Regression

SVMs FOR REGRESSION

In addition to classification, **SVMs** can also take on **regression** tasks.

Like with classification, the ideal hyperplane (decision boundary) is one that creates the **largest possible space** between the support-vector margins.

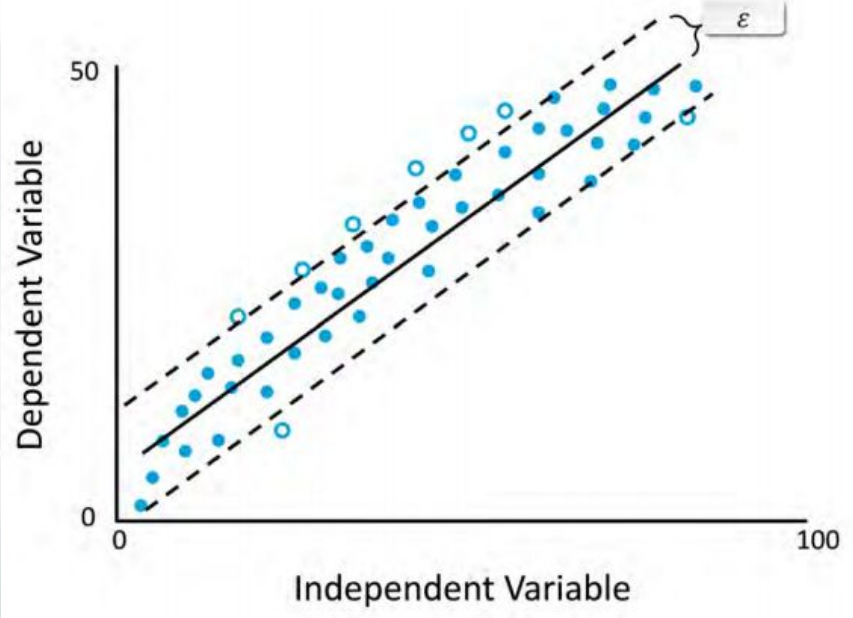


SVMs FOR REGRESSION

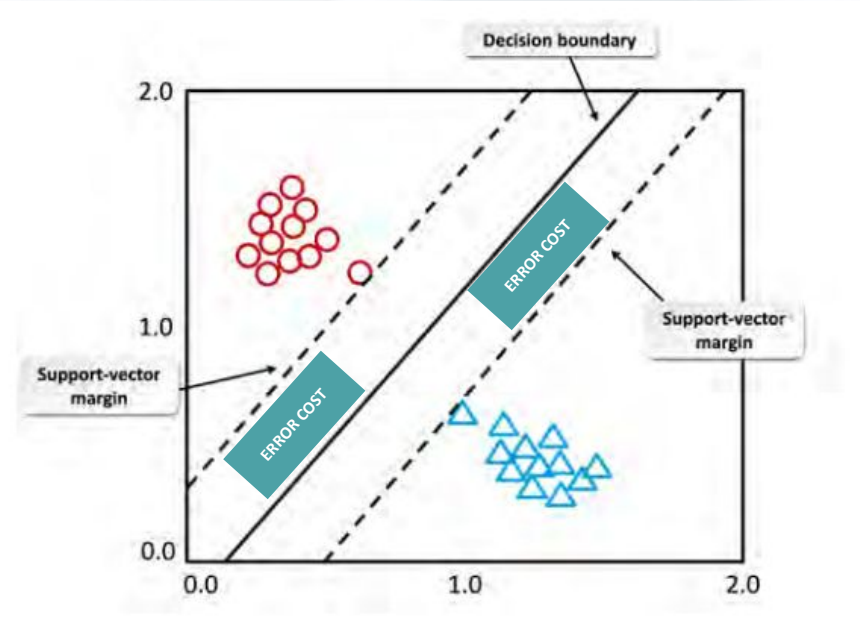
However, **unlike** with classification, you actually want to fit as many examples **within** the margins as possible.

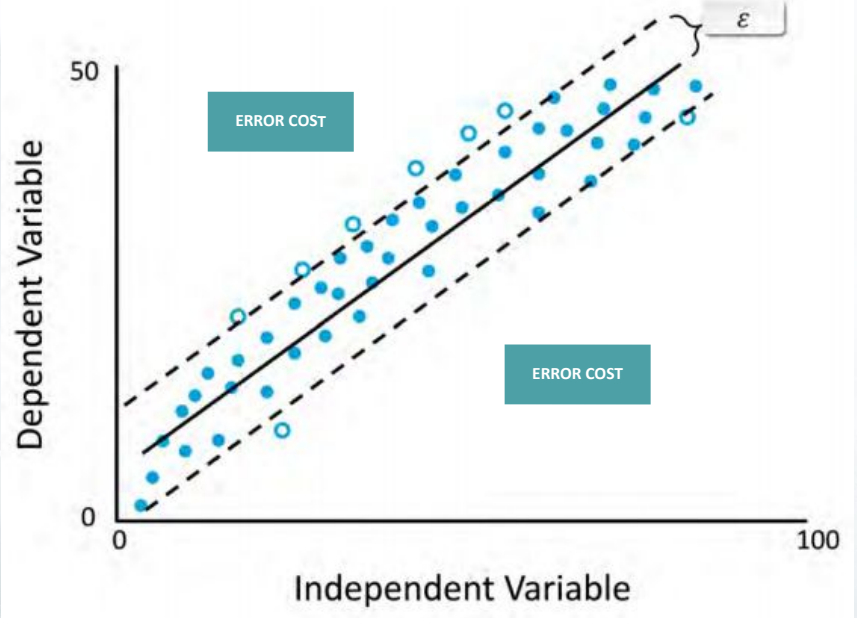
Only examples **outside** of the margins—the support vectors themselves—are incorporated in the **cost function**'s calculations.

Data points **outside** the margins are **high error values** (far from model)



COMPARISON: CLASSIFICATION VS REGRESSION





SVM CLASSIFICATION	SVM REGRESSION

ACTIVITY TIME

Building Support-Vector Machines

ACTIVITY 9-2: Building an SVM Model for Regression

NOTE: SVC is Support Vector Classifier | SVR is Support Vector Regression

In **SVC**, we use the **C Value** to harden or soften the margins
In **SVR**, we use the **Epsilon Value** to choose the maximum error

LESSON 9 SUMMARY ACTIVITY

In your own environment, how might you use SVMs to solve classification problems?

In your own business environment, how might you use SVMs to solve regression problems?

COURSE OUTLINES

LESSON 1	Solving Business Problems Using AI and ML
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LESSON 10	**Building Artificial Neural Networks**
LESSON 11	Operationalizing Machine Learning Models
LESSON 12	Maintaining Machine Learning Operations

LESSON 10
Building Artificial Neural Networks

Topic A	**Build Multi-Layer Perceptrons**
Topic B	Build Convolutional Neural Networks
Topic C	Build Recurrent Neural Networks

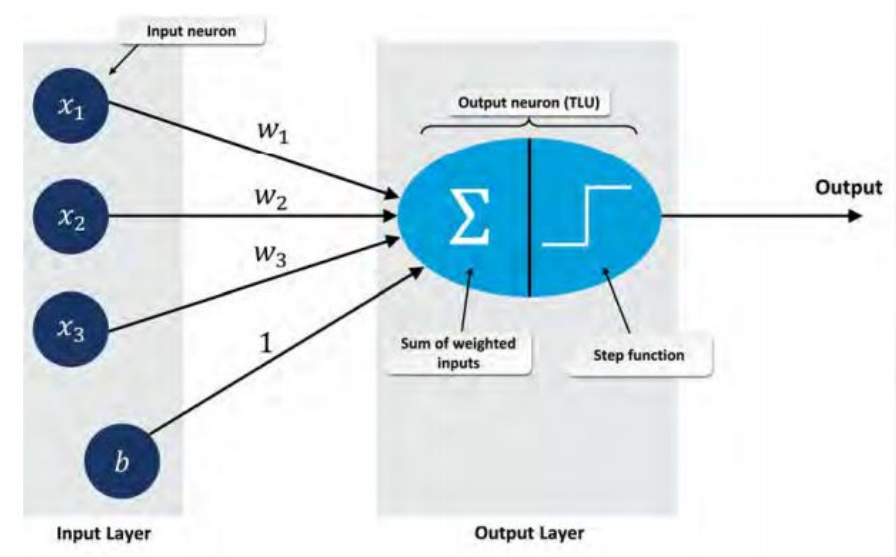
TERMINOLOGIES

Artificial Neural Network - ANN: machine approximation of biological neural networks, like the connective structure of the human brain, for the purpose of learning

INPUT NEURON: Nodes that transmit information

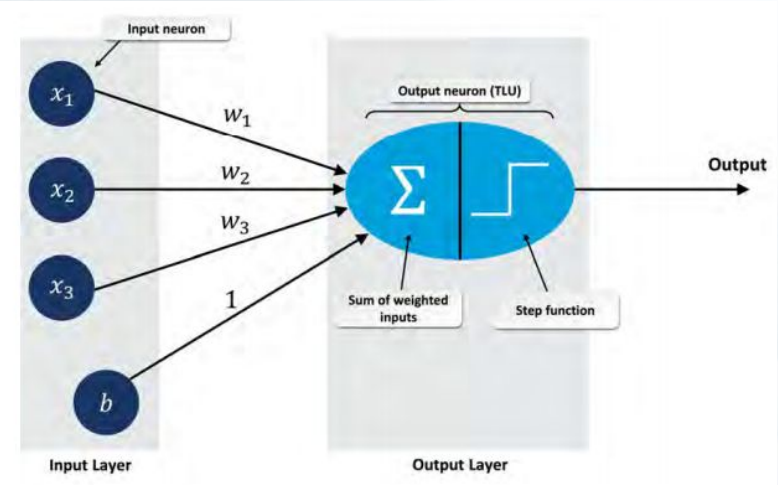
INPUT LAYER: Place for grouping Neurons

PERCEPTRON: algorithm used in ANN to solve **binary classification** problems



TERMINOLOGIES

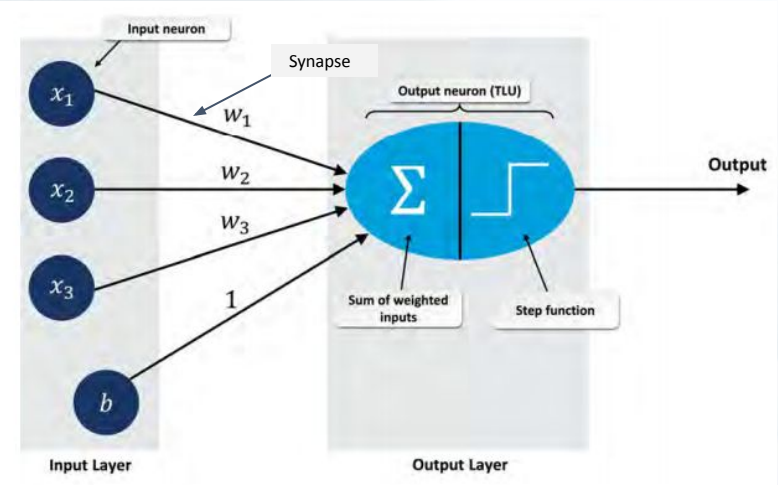
Perceptron is composed of **INPUT LAYER** and **OUTPUT LAYER**
Input Layer is composed of one or more **INPUT NEURON**
Output Layer is composed of one or more **OUTPUT NEURON (TLU)**
Output Neuron is composed of **SUM FUNCTION** and **STEP FUNCTION**



TERMINOLOGIES

Each **synapse** will have a weight proportional to their influence on the outcome

If we find that this synapse is leading to a **correct outcome**, we **higher** its weight. If **not**, we **reduce** its weight.



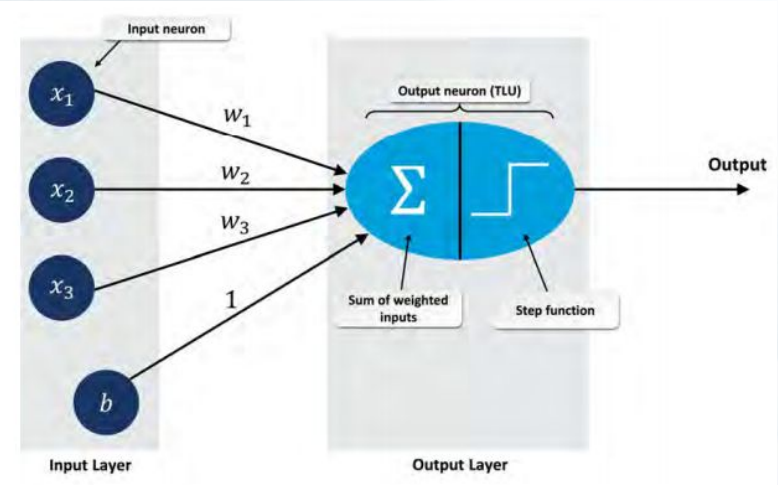
TERMINOLOGIES

Decisions are taken using the **weighted sum**: SUM((input*weight) + bias)

TLU: output neuron that calculates this weighted sum and then pass it through a step function

b (bias term):
similar to b in linear formula y=ax+b

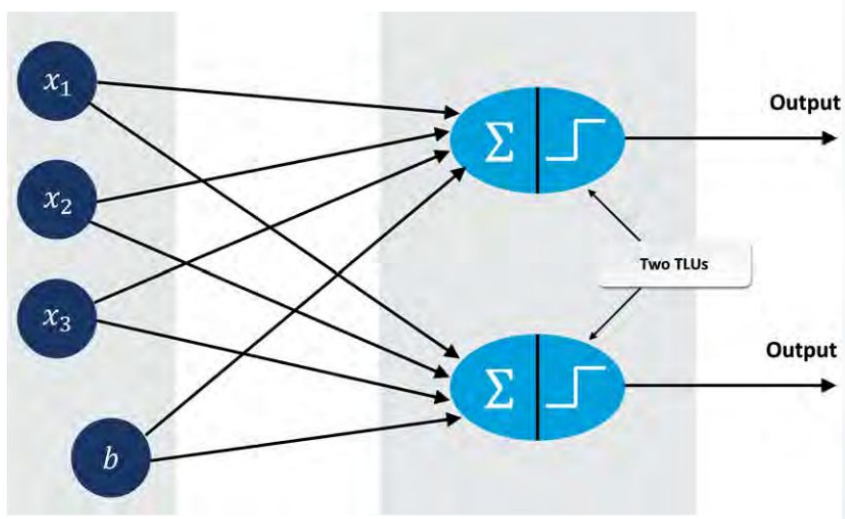
Step Function = Heaviside Function
Step function will output 0 or 1



MULTI-LABEL BINARY CLASSIFICATION PERCEPTRON

neural network algorithm that has multiple output neurons

1ST OUTPUT NEURON: CAR | PLANE
2ND OUTPUT NEURON: RED | BLUE



PERCEPTRON SHORTCOMINGS

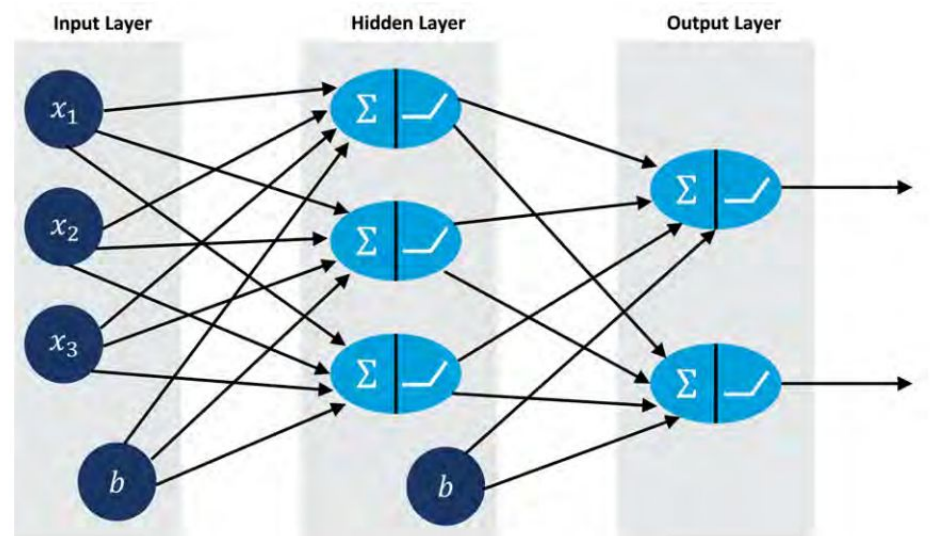
PROBLEM: A single perceptron **cannot** model non linearly separable concept / logic

SOLUTION: we use a MULTI LAYER PERCEPTRON

MULTI-LAYER PERCEPTRON (MLP)

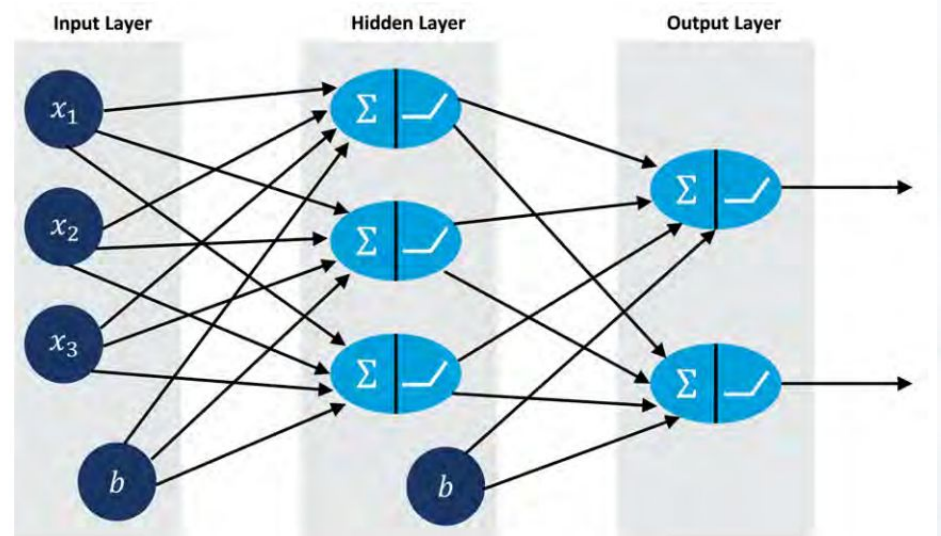
neural network algorithm that has multiple distinct layers of Output Neuron (TLU)s, rather than just one layer.

If the hidden layers are more than 1, it is called DNN (Deep Neural Network)



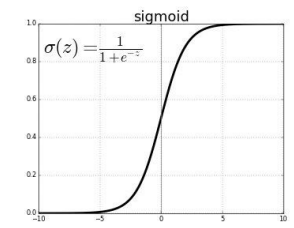
MULTI-LAYER PERCEPTRON (MLP)

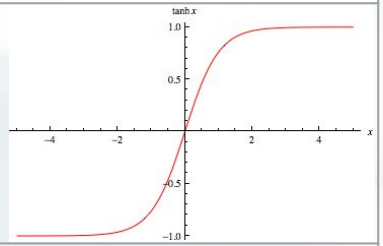
SINGLE PERCEPTRON	Step Function (Heaviside Function)	Can **only** solve **linearly** separable problems
MULTI LAYER PERCEPTRON	Activation Function	Can solve **non linearly** separable problems

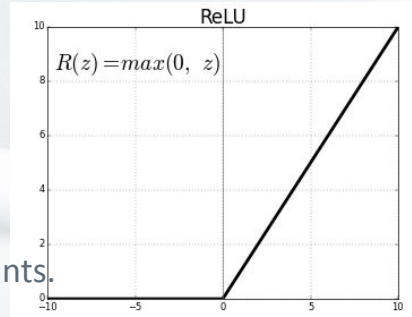


ACTIVATION FUNCTION

Sigmoid	• Outputs S-shaped curve. • Outputs range from **0 to 1** to constrain values. • Good choice for **output layer** in classifiers. • Backpropagation may be **too slow** if applied to hidden layers.
tanh	• **Scaled** version of sigmoid function. • Constraints output from **−1 to 1**. • More normalized output; **can center on 0**. • Better than sigmoid at **reducing bias** in error gradients.
ReLU	• **Rectified Linear Unit** • Calculates **linear function** of inputs. • If result is positive, it outputs; if result is negative, output is 0. • No maximum value constraint. • **Makes network sparse.** • Capable of vanishing gradients; addressed by leaky ReLU and other variants.







BACKPROPAGATION

Method by which an MLP neural network is trained.

the error calculations start at the end of the network and then work backward. This enables the algorithm to more efficiently calculate the error gradients, as the calculations of one layer are used in the calculations of the layer before it.

EPOCH: Training the neural network with all the training data for **one cycle (forward and backward)**

ACTIVITY TIME

Building Artificial Neural Networks

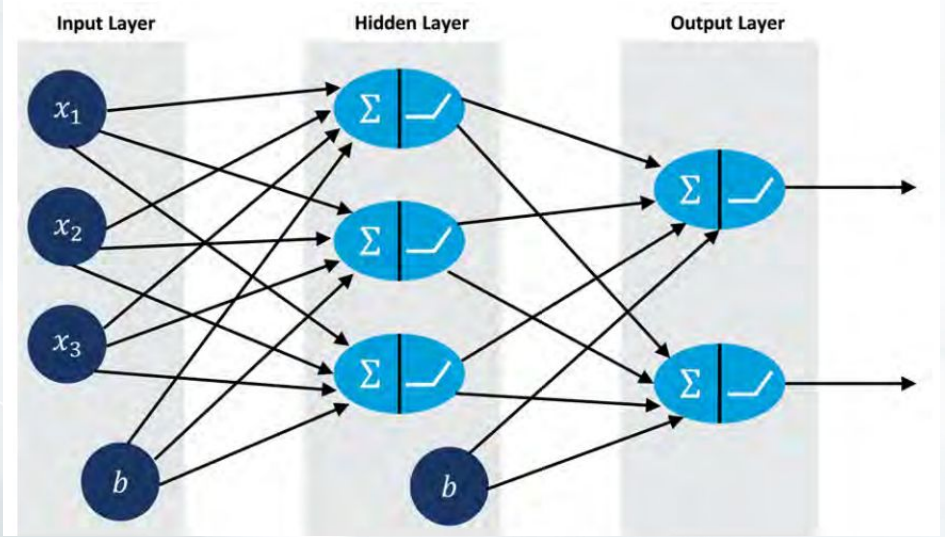
ACTIVITY 10-1: Building an MLP
Detect the presence of people in a room
OCCUPANCY: Room Not Occupied (0) OR Room Occupied (1)

LESSON 10
Building Artificial Neural Networks

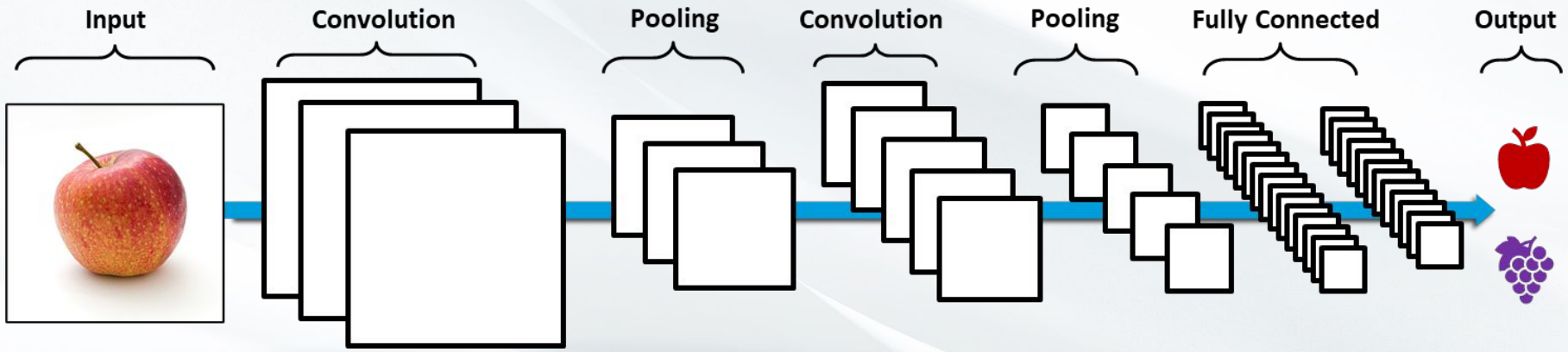
Topic A	Build Multi-Layer Perceptrons
Topic B	**Build Convolutional Neural Networks**
Topic C	Build Recurrent Neural Networks

ANN SHORTCOMINGS

- Fully connected MLPs **struggle** when it comes to **processing images**.
- Each image is represented by pixels, and even small images have many pixels.
- Requires a **very high number** of neuron connections.
- Increases **network density** and reduces performance.
- Requires flattening **2-D image to 1-D** before input.
- Can remove crucial information.



CONVOLUTIONAL NEURAL NETWORKS



CONVOLUTIONAL NEURAL NETWORKS

- **Convolutional layer (convolution) comes from research into human vision.**
  - Each neuron in brain only reacts to **small portion** of stimuli in entire **receptive field.**
  - **Example: One reacts to simple lines, another to more complex shapes.**
  - Each neuron builds on neurons near it.
  - Neurons combine to form entire receptive field.
- **Same basic idea behind a convolution.**
  - First layer only connected to input neurons whose pixels are in relevant field.
  - Second layer only connected to first layer neurons in relevant field.
  - And so on.
- **CNNs are partially connected.**
  - **Ideal for processing pixel-dense images. (Computer Vision)**

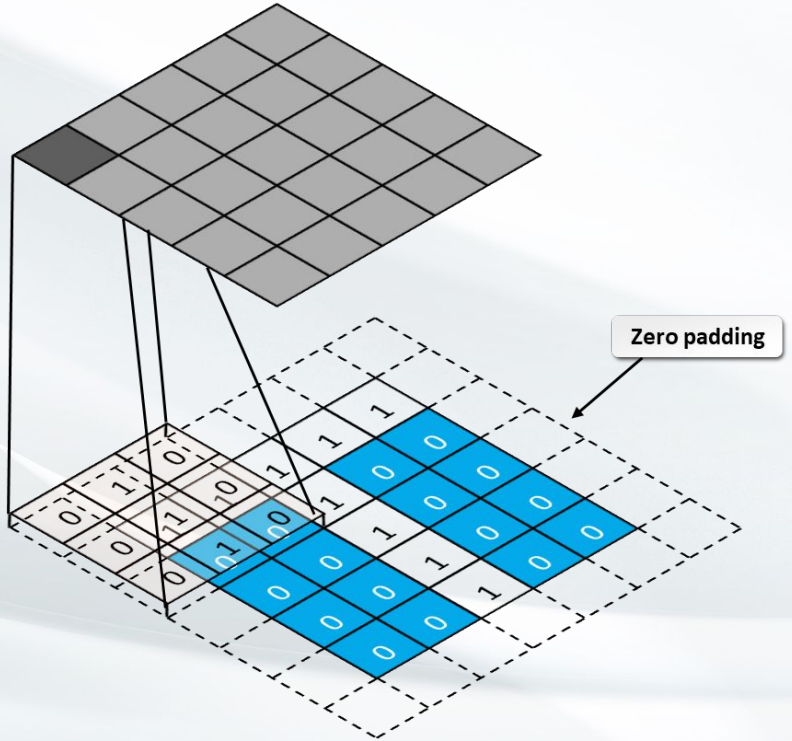
ANN VS CNN

ANN	CNN
FULLY CONNECTED LAYER	PARTIALLY CONNECTED LAYER
every neuron in one layer will be connected to **every** neuron in the next layer and that layer will be connected to **every** neuron in the next layer	**Only** the neurons that are **relevant** to the other neurons in the other layer will be connected. **Only** Pooling (reducing) layer and the output layer will be **fully** connected

PADDING

PROBLEM: when we started from the corner, we will miss measuring the very first column of pixels

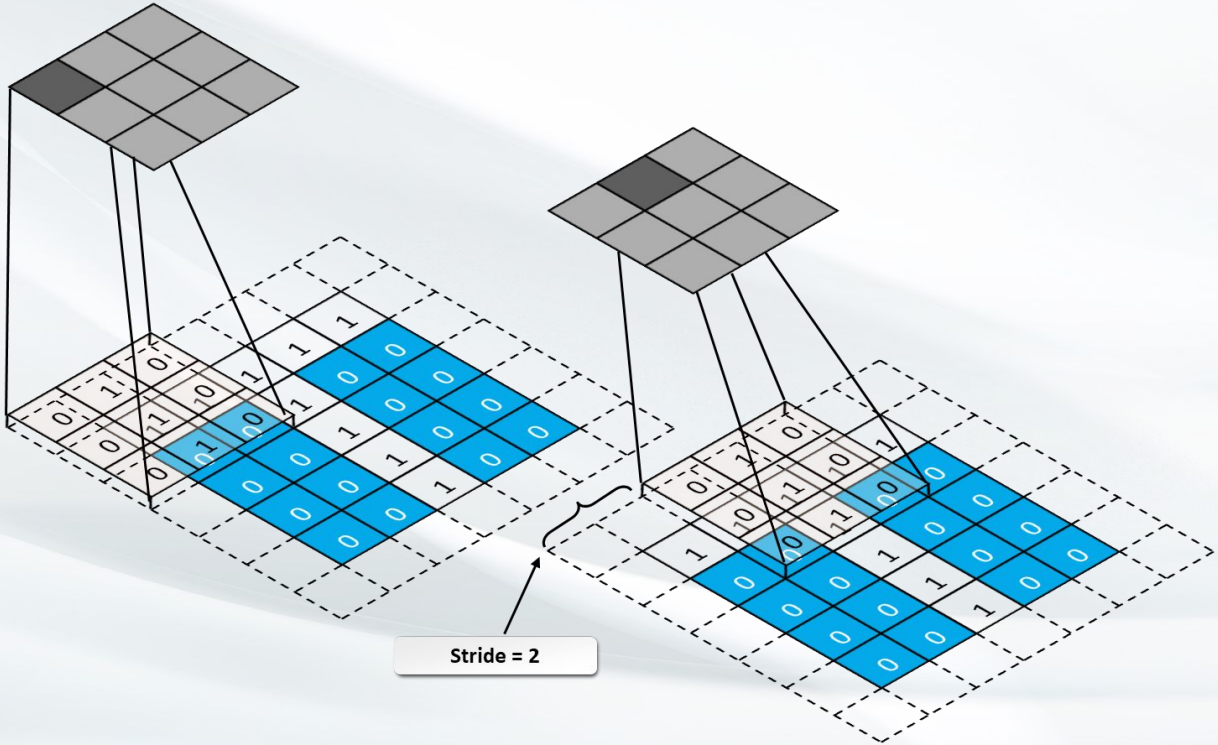
SOLUTION: PADDING
Adding bulk space placed around the image that will allow the filter to fully scan the image.



STRIDE

Number of steps to move

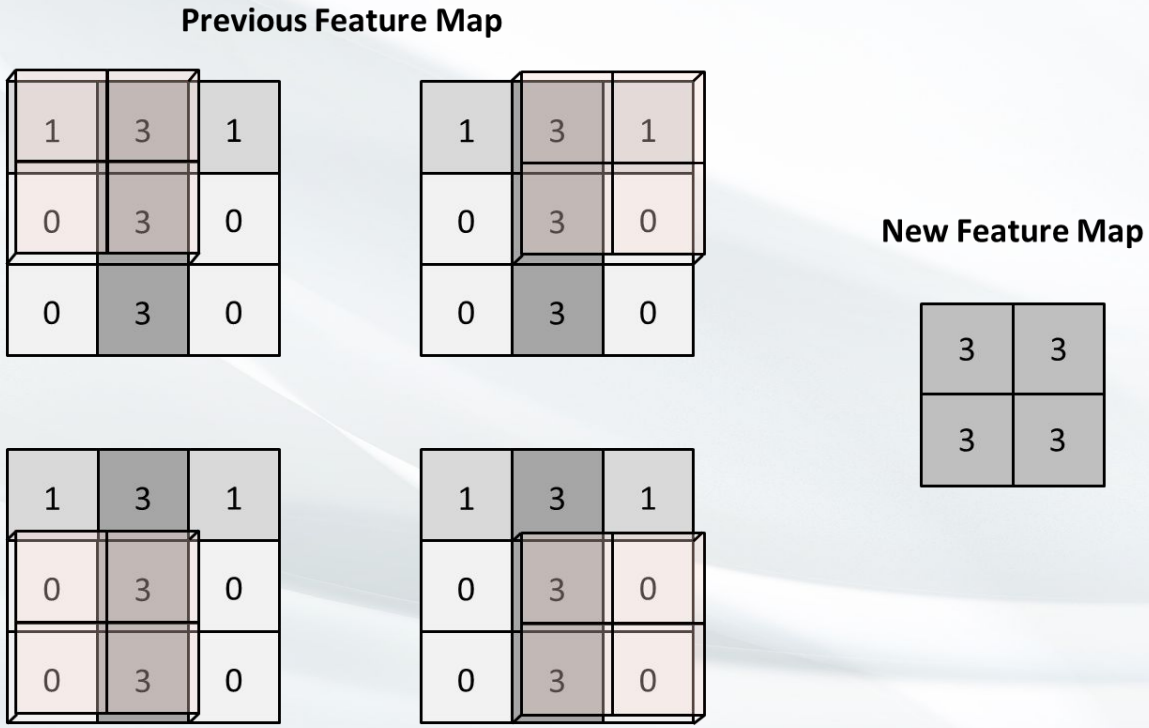
How quickly to scan the image (skipping some information) but we get the overall view



POOLING LAYER

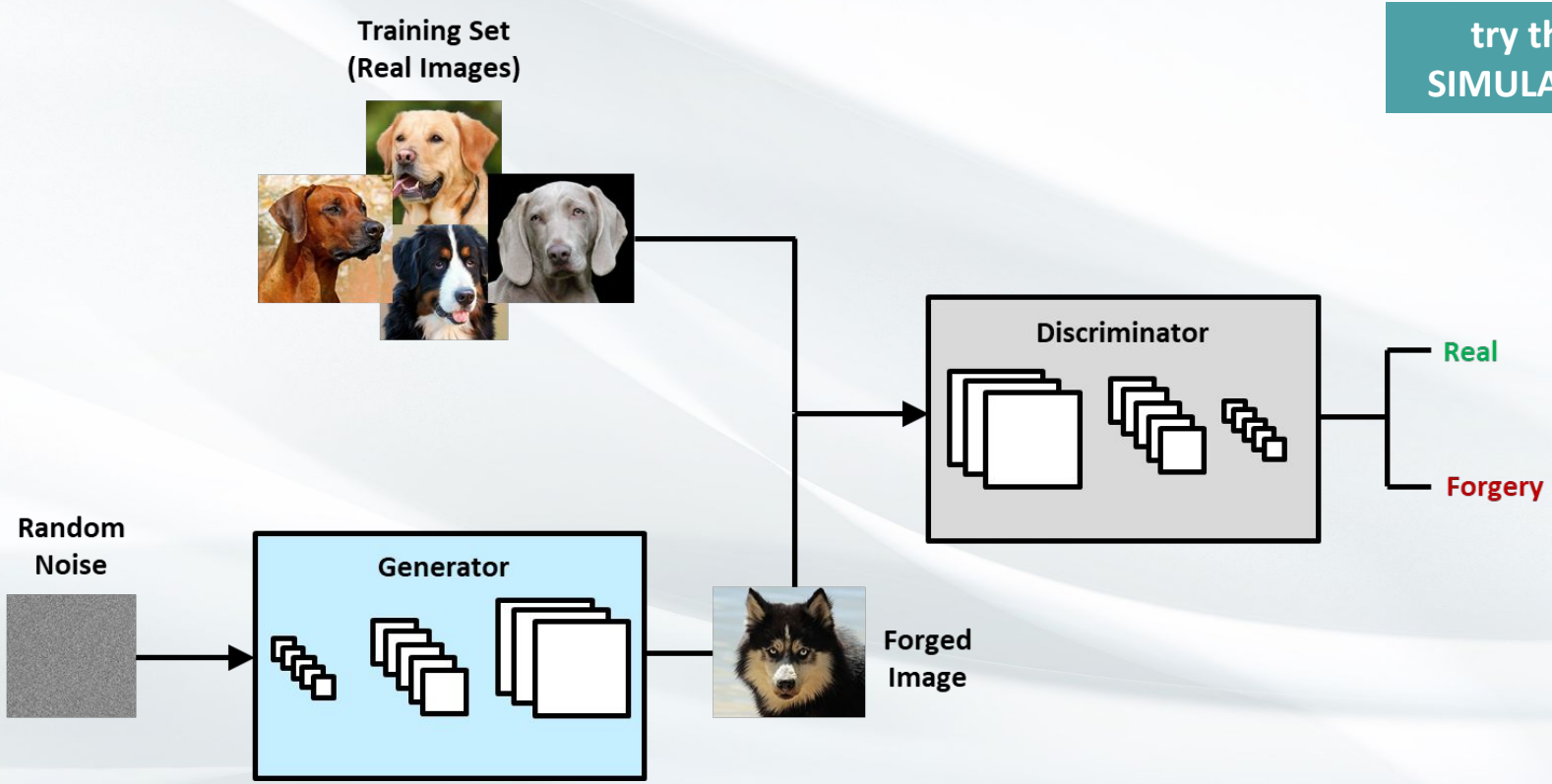
Improve performance by only sampling a portion of the input

This saves time and can reduce overfitting



GENERATIVE ADVERSARIAL NETWORK (GAN)

try the SIMULATOR



GENERATIVE ADVERSARIAL NETWORK (GAN)

- One network is discriminative, the other is generative.
- Training networks in **opposition,** leads to more **skillful** models.
- **Discriminator**:
  - Similar to networks mentioned already.
  - **We give a set of features ⇒ It predicts a label**
  - Example: Looks for correlations between features to predict heart disease.
- **Generator**:
  - Does the opposite.
  - **We give a label ⇒ It predicts features**
  - Maps distribution of feature values and generates new values that fit in distribution.
  - Example: For a patient with heart disease, determines what features play a role.
- **Both are in a constant state of conflict.**
  - Generator creates new data to send to discriminator.
  - Generator wants to "fool" the discriminator, and discriminator wants to detect this.
- Tug-of-war strengthens network's ability to generate convincing data.

ACTIVITY TIME

Building Artificial Neural Networks

ACTIVITY 10-2: Building a CNN

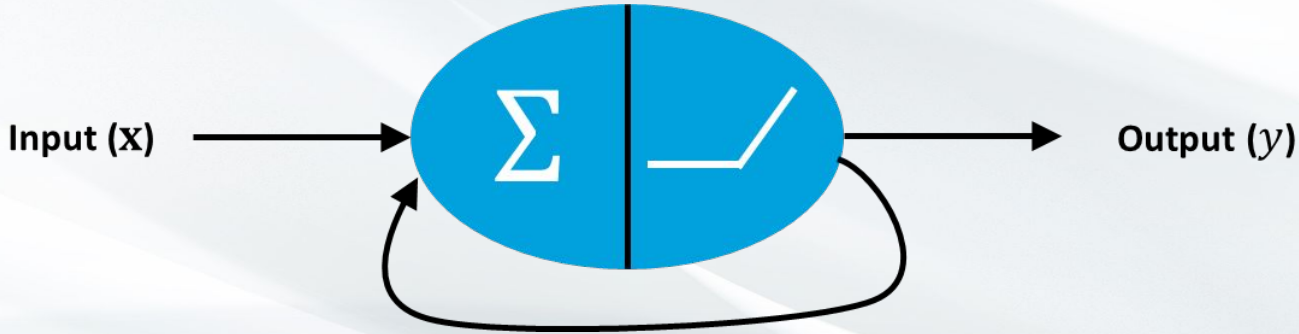
ACTIVITY 10-3 (OPTIONAL): Building a CNN to Classify Handwritten Characters

LESSON 10
Building Artificial Neural Networks

Topic A	Build Multi-Layer Perceptrons
Topic B	Build Convolutional Neural Networks
Topic C	**Build Recurrent Neural Networks**

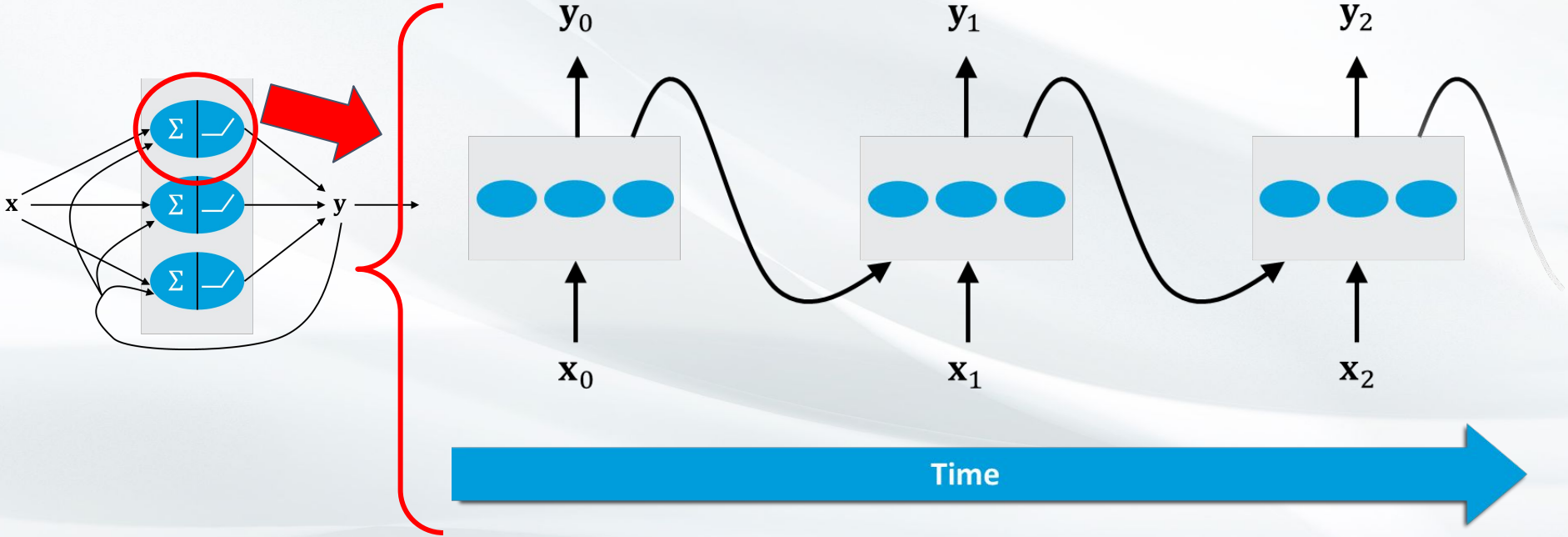
RECURRENT NEURAL NETWORKS

Because it uses a **feedback memory** of what was done earlier, it is mostly used for **Natural Language Processing** (Because a sentence and its meaning is often based on the word before)

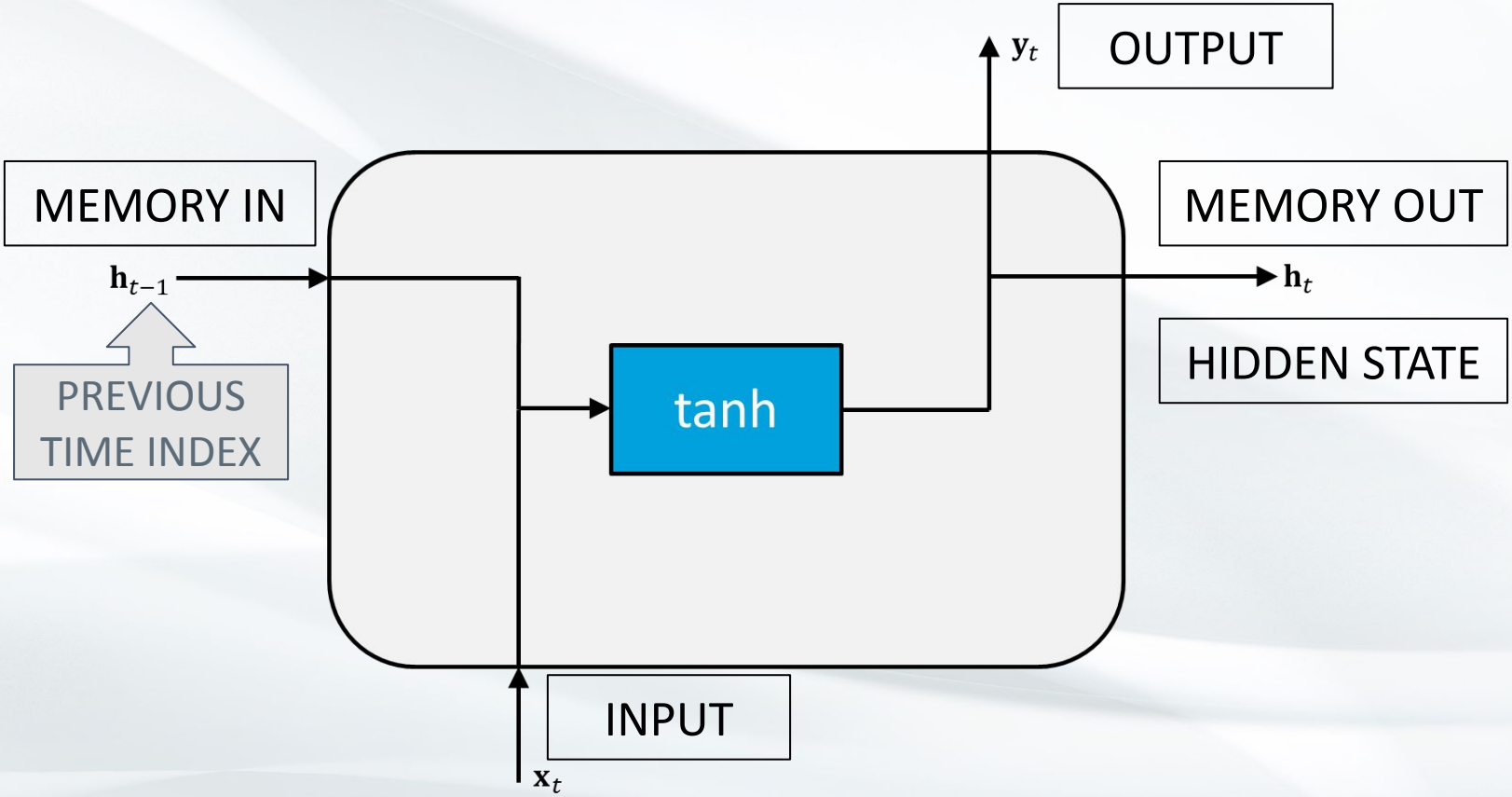


RECURRENT NEURAL NETWORKS

This is actually the **SAME NEURON** but it is duplicated for representation purposes only



RECURRENT NEURAL NETWORKS



RNN TRAINING

Backpropagation through time (BPTT): The process of training an RNN.

1. Time sequence of RNN layers is **unrolled**.
2. **Backpropagation** is performed just **as it is for a traditional ANN**.
3. Error gradients are identified starting at last time step.
4. Gradients are propagated backward for next layer's error calculation.
5. Repeats until reaching first time step.
6. Weights between neurons are updated.

RNN SHORTCOMINGS

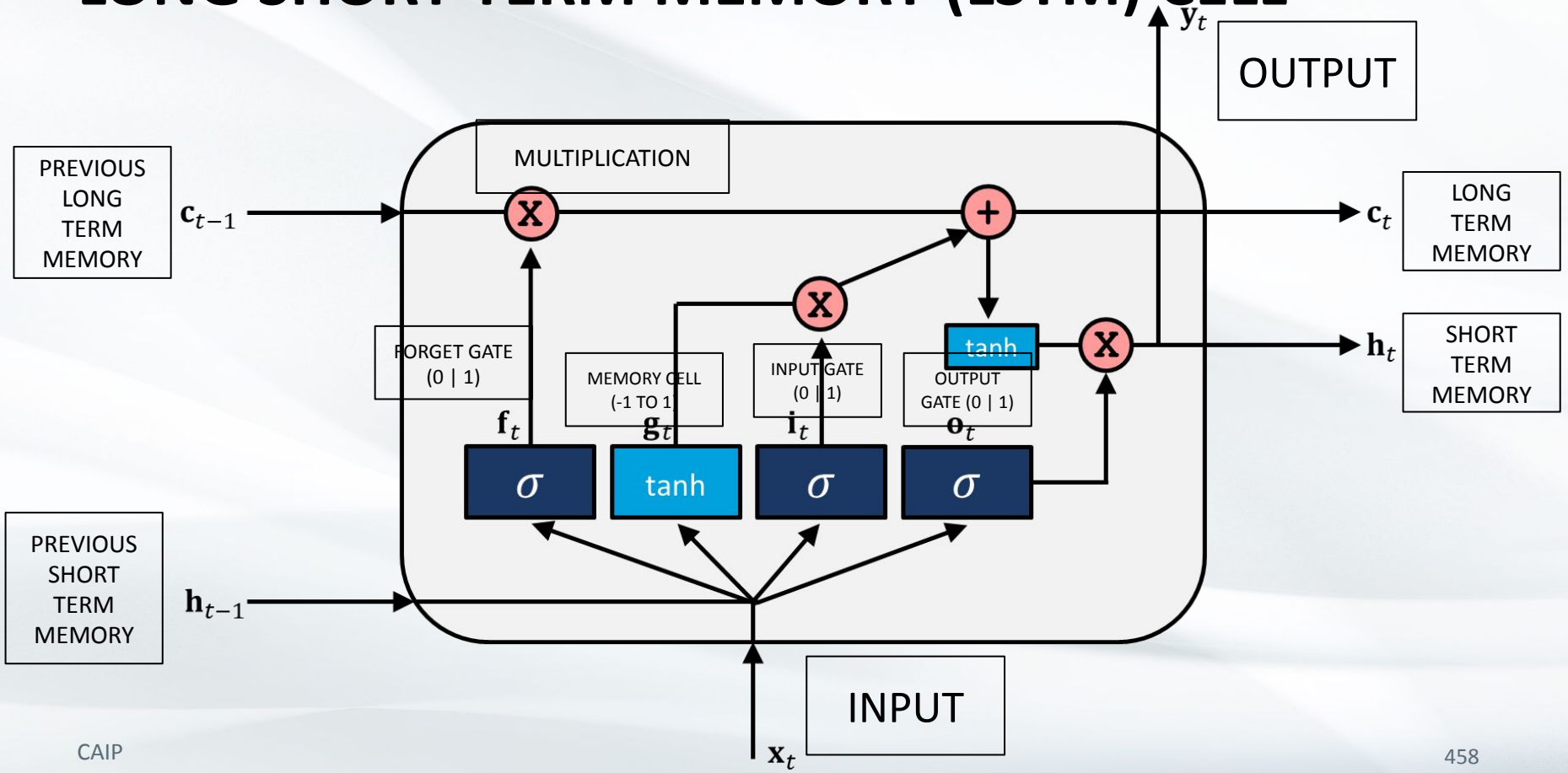
- RNNs can **suffer performance issues** on long input sequences.
- Can reduce time steps used for unrolling to reduce computational needs.
  - Model may not learn patterns that manifest over a long time.
  - **Can lead to reduction in model skill**.
- Memory cell issue:
  - Some data, through calculations, is **lost in the network**.
  - Later memory cells start missing key pieces of info.
  - Example: **Review** might start, "I really like this place, but …"
  - **Review is positive, but network might "forget" this initial sentence.**
- More complex and effective memory cell is required.

LONG SHORT-TERM MEMORY (LSTM) CELL

Long Short-Term Memory (LSTM) Cell: A memory cell in which the hidden state comprises two vectors: a short-term state and a long-term state.

- Able to **preserve** input significant to training process.
- Able to "**forget**" input that is not significant.
  - Speeds up training.
- **Used in NLP.**
  - Suitable with sequential processing of input, like text.
- Also useful in **forecasting time series**.
  - Can predict trends based on past information, like expected sales over time.

LONG SHORT-TERM MEMORY (LSTM) CELL



GATED RECURRENT UNIT (GRU) CELL

- simplified version of the LSTM cell
- both short-term and long-term states are merged into one
- The forget and input gates are also merged into one
- There is a reset gate rather than an output gate

EMBEDDING

Embedding: The process of condensing a language vocabulary into small-dimensional vectors. **(Used for speech recognition)**

- These vectors are **more efficient than traditional encoding schemes**.
  - Example: One-hot encoding a word into 40,000 possible words.
    - Would have 39,999 values of 0—**very inefficient**.
- Embedding represents a word as its own vector.
  - Usually no more than a few hundred dimensions.
  - Vectors constitute overall embedded space.
  - Vector occupies a point in this space.
- Network trains vectors on some language input to **find similarities** between words.
  - Similar words placed close together in embedded space.
  - Network can group together words in meaningful ways.
- Consider leveraging pre-trained embeddings.
- If training your own embeddings, consider the following tools:
  - Word2vec
  - Doc2vec
  - fastText

BAG OF WORDS

Bag of Words: A list of each individual word in a text without respect to grammar, punctuation, etc. **(used for text analysis)**

- Used as a way to represent text for training, before being embedded as vectors.
- Example: "I really enjoyed this movie—more than I enjoyed the first one."
  - List: ['I', 'really', 'enjoyed', 'this', 'movie', 'more', 'than', 'I', 'enjoyed', 'the', 'first', 'one']
  - Dictionary: {'I': 2, 'really': 1, 'enjoyed': 2, 'this': 1, 'movie': 1, 'more': 1, 'than': 1, 'the': 1, 'first': 1, 'one': 1}
- Similar approach: bag-of-n-grams.
  - Each element is an n-gram rather than a word.
  - Commonly used to represent features for language classification.

ACTIVITY TIME

Building Artificial Neural Networks

ACTIVITY 10-4: Building an RNN

COURSE OUTLINES

LESSON 1	Solving Business Problems Using AI and ML
LESSON 2	Preparing Data
LESSON 3	Training, Evaluating, and Tuning a Machine Learning Model
LESSON 4	Building Linear Regression Models
LESSON 5	Building Forecasting Models
LESSON 6	Building Classification Models Using Regression and k-NN
LESSON 7	Building Clustering Models
LESSON 8	Building Decision Trees and Random Forests
LESSON 9	Building Support-Vector Machines
LESSON 10	Building Artificial Neural Networks
LESSON 11	**Operationalizing Machine Learning Models**
LESSON 12	Maintaining Machine Learning Operations

LESSON 11
Operationalizing Machine Learning Models

Topic A	**Deploy Machine Learning Models**
Topic B	Automate the Machine Learning Process with MLOps
Topic C	Integrate Models into Machine Learning Systems

DEPLOYMENT OF MACHINE LEARNING MODELS

In the field of machine learning, deployment refers to **taking a model and putting it into a production environment** so that **users** or services can **feed** the model input, as well as **receive output** from the model.

In other words, they are able to **acquire resources** produced by the model in a streamlined way, **without** having to interface directly with the underlying model code.

OFFLINE VS. ONLINE MODELS

Offline models are trained every so often on batches of new data. The new data is input to the model, which then learns new parameters. The model is updated on the new data all at once.

Online models, on the other hand, are **continuously** trained on new data. The model learns from **each new training** example as it flows in. So, it is constantly updating its parameters.

OFFLINE VS. ONLINE MODELS

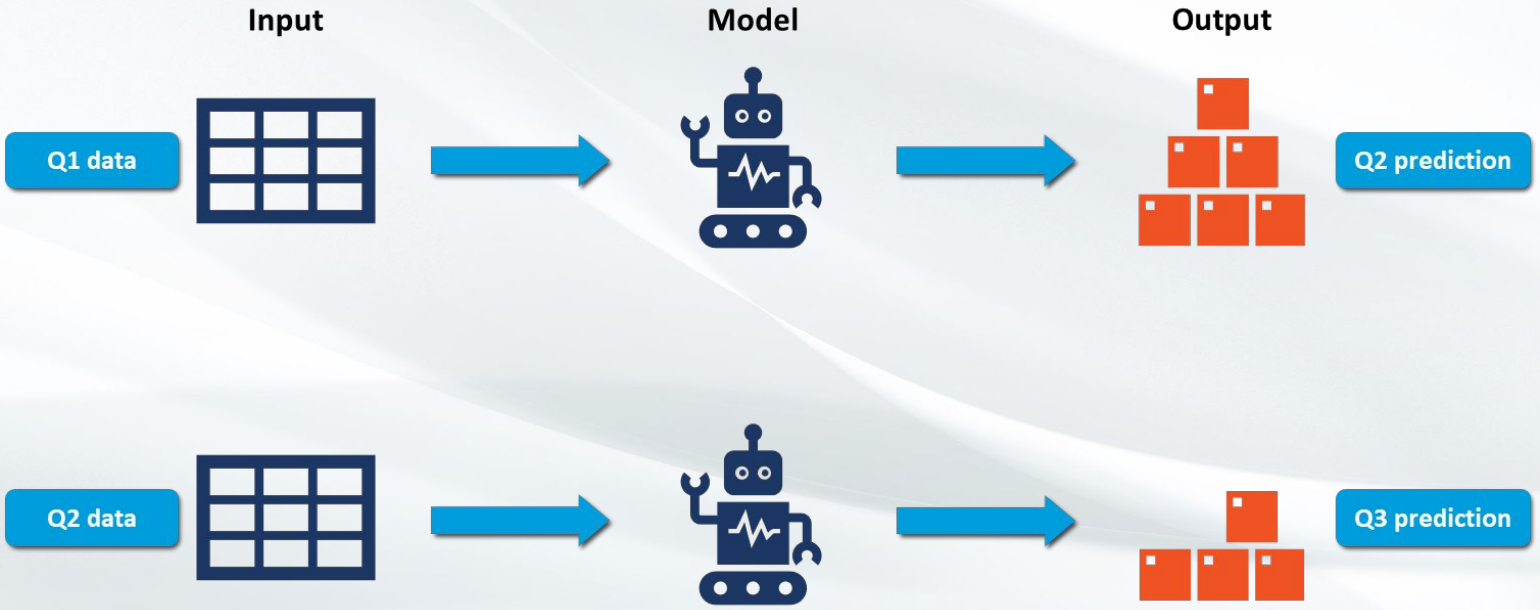
Aspect	Online Models (Advantages / Disadvantages)	Offline Models (Advantages / Disadvantages)
Keeping Model Up-to-Date	✅ Continuously update parameters with latest data → stay fresh and relevant.	❌ Trained only once per cycle → assumptions may get outdated before next training.
Performance Over Time	✅ May outperform offline models because they adapt immediately to new patterns.	❌ May lag in performance if environment changes rapidly between training cycles.
Training Speed	❌ Training is continuous → requires constant updates and computation.	✅ Training happens once per cycle → usually faster to complete.
Handling Large Datasets	✅ Processes data in small batches → scalable even for large data streams.	❌ Very large training sets can slow down training since they must be processed all at once.

OFFLINE VS. ONLINE MODELS

Aspect	Online Models (Advantages / Disadvantages)	Offline Models (Advantages / Disadvantages)
Computational Power Needs	❌ Constantly consumes computing resources to keep model updated.	✅ Requires less computation since training happens periodically.
Data Review & Quality Control	❌ Hard to manually inspect new data before it affects the model.	✅ Engineers can review and clean data before retraining.
Risk of Data Poisoning	❌ More vulnerable to corrupted/malicious data (e.g., fake traffic reports).	✅ Safer, since only curated datasets are used during retraining.
Best Use Cases	✅ Ideal for scenarios with real-time, streaming data (e.g., weather forecasting, navigation apps).	✅ Best for tasks without constant new data or labeled examples (e.g., sentiment analysis on text).

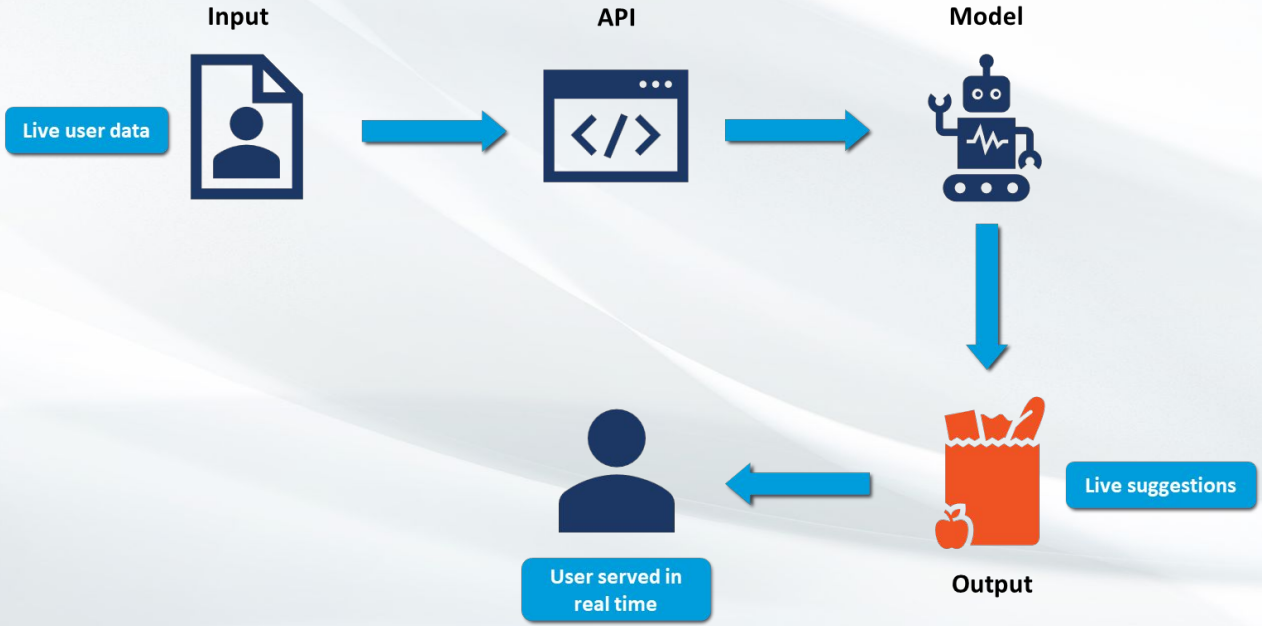
DEPLOYMENT METHODS FOR MODEL OUTPUT

Batch Deployment	Real-Time Serving	Streaming



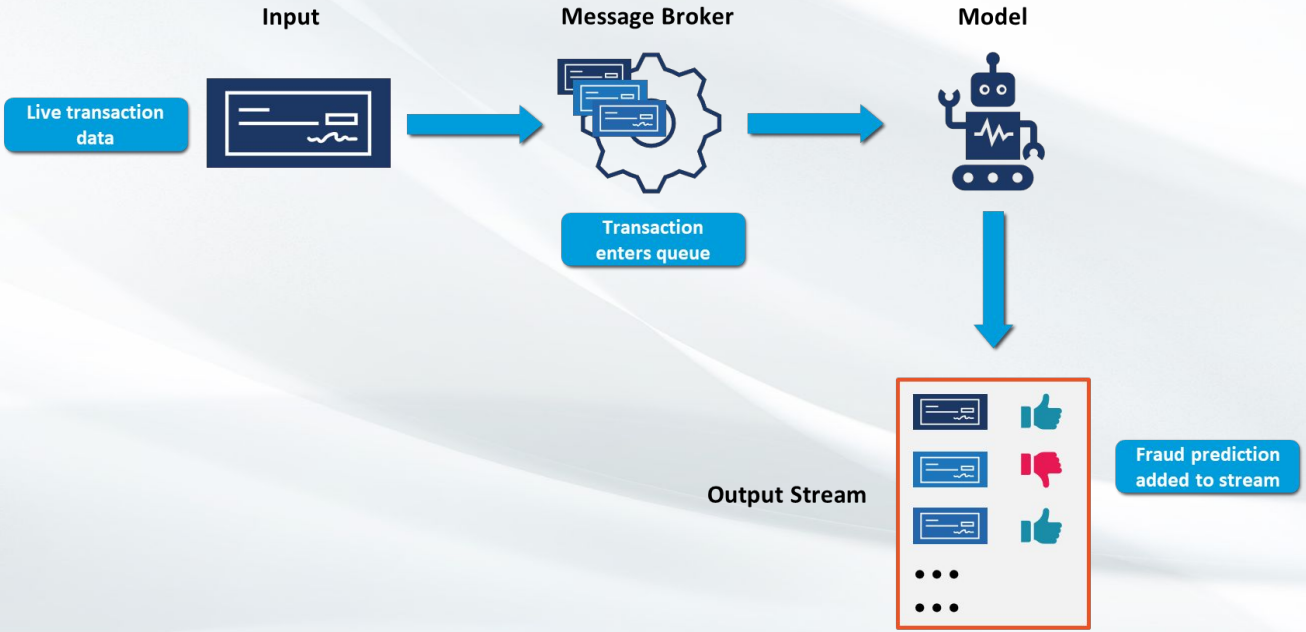
DEPLOYMENT METHODS FOR MODEL OUTPUT

Batch Deployment	**Real-Time Serving**	Streaming



DEPLOYMENT METHODS FOR MODEL OUTPUT

Batch Deployment	Real-Time Serving	**Streaming**



STAKEHOLDER REQUIREMENTS

Consumers/end users:

- Expect to receive timely and accurate results.

Sponsors/champions:

- Verify deployment is on time and within budget.

Program/project managers:

- Ensure engineers have access to resources.

Team members:

- Must have access to correct platform/tools to do the job.

STAKEHOLDER REQUIREMENTS

Business partners:

- Shoulder financial/operational burden.

Governments:

- Impose laws/regulations.

Societies:

- Expect ethical products.

LESSON 11
Operationalizing Machine Learning Models

Topic A	Deploy Machine Learning Models
Topic B	**Automate the Machine Learning Process with MLOps**
Topic C	Integrate Models into Machine Learning Systems

PRECONCEIVED NOTIONS

OBSERVER BIAS

A problem when we treat **preconceived notions as experience** and allow them to **lead us** to see what we expect to see or want to see.

THE BLACK BOX CHALLENGE

A mechanism to which you **provide** one or more **inputs** to produce one or more **outputs**, **without knowledge or visibility** into its inner workings.

In the world of Ethical Practices, it can be very difficult to **determine** how the machine **actually accomplishes its evaluation** or the extent to which it **uses various data points**, since these things were not directly programmed

PREJUDICE BIAS

It can be very difficult to **determine** how the machine actually **accomplishes its evaluation** or the extent to which it uses various data points, since these things were **not directly programmed**

It is **fair** to use the **Income** as an attribute, but not fair to use the ZIP code. Although the location of the person may indicate his salary, but he may have **debts and loans** to pay for

ETHICS IN NLP

- We should consider people who speak a **different language**
- We should consider people who are **unable to hear or speak**

For example, developers of voice activated home automation devices might occasionally "**listen** in" on conversations between users and the application to assess how well the device is responding to voice commands.

This may **violate** the user's privacy

GUIDELINES FOR PROMOTING ETHICAL PRACTICE

- **APPLICATION PROCESSING**: Model unfairly rejects an application for a job, loan, educational program, medical benefits, and so forth due to prejudice bias.
- **CRIME (FRAUD, TERRORISM DETECTION)**: Model incorrectly classifies the subject as a potential offender, possibly leading to pain and suffering, detention/loss of time, legal expenses, embarrassment, and possible loss of reputation.
- **GOVERNMENT BENEFITS**: Model incorrectly classifies potential recipients of government benefits, resulting in eligible people not receiving benefits and ineligible people receiving benefits.

GUIDELINES FOR PROMOTING ETHICAL PRACTICE

- **HOUSING**: Model makes an unfair assessment of a property based on prejudice bias (related to location, community, geography, and so forth), resulting in unfair high tax assessments or unfair low sale prices.
- **INSURANCE**: Model assesses unfair, high premiums for a customer based on prejudice bias such as race or gender.

LESSON 11
Operationalizing Machine Learning Models

Topic A	Deploy Machine Learning Models
Topic B	Automate the Machine Learning Process with MLOps
Topic C	**Integrate Models into Machine Learning Systems**

PRIVACY AND DATA GOVERNANCE FOR AI AND ML

The various functions of the organization should work together to **identify applicable laws and standards** and to translate them into policies, practices, procedures, and tools that will ensure the entire organization is handling data privacy in a **unified manner**.

INTELLECTUAL PROPERTY

DURATION OF IP PROTECTIONS	Some laws protecting IP rights are in effect for a certain period, based on the lifetime of the author. But, if the author is an immortal machine, then it may not be clear how long the copyright protection lasts.
IP RIGHTS VIOLATED BY AI	If an AI system produces creative works that violate someone else's IP rights, it may not be clear who is liable for infringement
PROTECTING THE AI OR ML ITSELF	Protecting the approach as a trade secret (keeping its inner workings hidden and protected) may violate principles of data privacy, since it conflicts with the idea of transparency.

HUMANITARIAN PRINCIPLES

Transparency	Because of the "black box" nature of many AI algorithms, it can be difficult to hold developers of an AI system accountable
Privacy	Individuals can be located using their smartphone and other connected devices, and located visually using facial recognition features.
Job Elimination	Industrial robotics and workforce automation have already resulted in job elimination in many industries
Disinformation	Bots can inject misleading traffic and communications into social media sites. This can undermine democracy, incite terror and social unrest, and diminish the benefits of a free press

HUMANITARIAN PRINCIPLES

Cybersecurity	AI can be leveraged to automate attacks, perform them faster, with greater precision, and on a greater scale.
Disparity	Dilemma between safety and ethics (Self driving car)
Weaponization	Remote controlled robotic weapons and drones are already in significant use.
Unintended Consequences	Many have raised the concern that artificial intelligence can eventually surpass human intelligence, making it hard to contain and control, eventually leading to the "rise of the robots" scenario played out in many science fiction movies.

FINAL ASSESSMENT

https://docs.google.com/forms/d/e/1FAIpQLSfW285Ea9J6jZSBE-ixFalLvKceb1If_-cNVCvr7-El1vSXXQ/viewform

NOTE: Before you go to the exam, make sure to do a revision using the exam blueprint. Click HERE

EXAM SUMMARY / NOTES

Thank You

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